



UNIVERSITÀ DI TRENTO

Department of Information Engineering and Computer Science

Master's Degree in
Computer Science

FINAL DISSERTATION

MACHINE LEARNING FOR RACE PERFORMANCE PREDICTION IN PROFESSIONAL CYCLING *The Norwegian Team Coop - Repsol Case Study*

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Academic year 2024/2025

”Non leggiamo e scriviamo poesie perché è carino. Noi leggiamo e scriviamo poesie perché siamo membri della razza umana. E la razza umana è piena di passione. Medicina, legge, economia, ingegneria sono nobili professioni, necessarie al nostro sostentamento. Ma la poesia, la bellezza, il romanticismo, l’amore, sono queste le cose che ci tengono in vita.”

— John Keating - (Robin Williams) —

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List of Acronyms

ACWR Acute:Chronic Workload Ratio

AI Artificial Intelligence

API Application Programming Interface

ATL Acute Training Load

AUC Area Under the Curve

AUC-PR Area Under the Precision-Recall Curve

CTL Chronic Training Load

DNF Did Not Finish

DNS Did Not Start

EWMA Exponentially Weighted Moving Average

FIT Flexible and Interoperable Data Transfer

FN False Negative

FP False Positive

FTP Functional Threshold Power

GFL German Football League

GPS Global Positioning System

HR heart rate

HRV Heart Rate Variability

IF Intensity Factor

kJ kilojoules

KPI Key Performance Indicator

LuTRIMP Lucia's Training Impulse

ML Machine Learning

MLB Major League Baseball

MMP Maximum Mean Power

NFL National Football League

NLP Natural Language Processing

NP Normalized Power

OTL Over the Time Limit

PCS ProCyclingStats

PR Precision-Recall

RI Readiness Index

ROC Receiver Operating Characteristic

ROC-AUC Receiver Operating Characteristic Area Under the Curve

SMOTE Synthetic Minority Over-sampling Technique

TL Training Load

TN True Negative

TP True Positive

TRIMP Training Impulse

TSB Training Stress Balance

TSS Training Stress Score

†TSS TRIMPS Training Stress Score

UCI Union Cycliste Internationale

VAM Vertical Ascent in Meters

VO₂max Maximum Oxygen Consumption

XGBoost Extreme Gradient Boosting

Abstract

In professional road cycling, performance outcomes emerge from complex interactions between training load, recovery, physiological adaptation, and individual responses accumulated over long periods of time. While modern wearable sensors and training platforms generate large volumes of high-resolution data, transforming this information into actionable and interpretable performance indicators remains a challenging task. This thesis investigates the use of machine learning techniques to predict race performance in professional cycling, with a specific focus on individualized modelling and performance readiness estimation.

The study is conducted as a real-world case study on a Norwegian Union Cycliste Internationale (UCI) Continental team and is based on two years of daily training data and race results from five professional athletes. Raw data from *TrainingPeaks* are integrated with race metadata and external results from ProCyclingStats (PCS), then processed through a complete data engineering pipeline including cleaning, alignment, imputation, and feature engineering. A training-block framework is introduced to capture meaningful temporal structures in athletes' preparation cycles, from which domain-specific workload, fitness-fatigue, and tapering features are derived.

Multiple modelling strategies are explored, starting from a logistic regression baseline and progressing to gradient-boosted decision trees (Extreme Gradient Boosting (XGBoost)). To address strong class imbalance and inter-individual variability, rider-specific models and personalized decision thresholds are employed. Predicted probabilities are further transformed into a Readiness Index (RI), a synthetic and interpretable score that quantifies an athlete's likelihood of achieving a strong race performance.

Results indicate that the proposed readiness-based XGBoost model significantly enhances the identification of strong performances, using Top-20 race finishes as an informational proxy, compared to baseline approaches, highlighting the effectiveness of individualized modelling and structured feature engineering. The thesis concludes with the implementation of an interactive dashboard that translates model outputs into a practical decision-support tool for coaching staff, highlighting the potential of machine learning to support performance analysis in real-world settings.

Chapter 1

Introduction

1.1 Motivation

In recent years, professional sports have experienced a rapid transformation driven by the increasing availability of digital data, wearable sensors, and analytical tools. This trend is particularly evident in Sport Sciences, where Artificial Intelligence (AI) and data-driven approaches have started to play a relevant role. A lot of major sport organizations - including the National Football League (NFL), Major League Baseball (MLB), and German Football League (GFL) - have already promoted open AI competitions to explore new analytical solutions to solve real-world problems [1].

Cycling represents one of the most promising domains for such innovation. As a sport grounded in continuous mechanical work, cycling benefits from uniquely rich and high-resolution data streams: power output, cadence, heart rate, Global Positioning System (GPS), environmental conditions, time-series from training sessions, and detailed metadata from daily training logs [2]. Unlike many team sports, cycling allows the continuous measurement of mechanical work, producing quantifiable load metrics that are ideal for computational analysis. When accumulated over months or years, these data provide valuable insight for performance prediction, injury risk monitoring, and strategic decision-making [3, 4, 5].

AI and Machine Learning (ML) methods offer natural solutions for capturing these complex, individualized relationships. ML algorithms can identify patterns without explicit programming, learn from experience, and adapt to the heterogeneous nature of sports data [1]. In cycling, an AI system can analyze a rider's workload history, physiological responses, and race schedule to highlight strengths, detect weaknesses, and suggest adjustments to maximize performance and minimize risks such as fatigue, injury, or illness. As ML techniques evolve, they provide new opportunities for processing physiological and biomechanical signals that historically have been difficult to interpret using traditional statistical frameworks [2].

Despite their potential, AI applications in endurance sports - and cycling in particular - remain under-explored compared to other disciplines such as soccer, basketball, or American football. The existing literature shows a significant focus on injury prediction and performance forecasting in team sports, whereas cycling research has only recently begun to adopt modern machine-learning approaches [1]. Given the sport's global growth, its increasing economic impact, and the availability of large datasets, there is a strong motivation to investigate how ML can contribute to performance modeling in professional road cycling.

This thesis is positioned in this context: exploring how machine learning and training-load analytics can be used to understand, quantify, and ultimately predict race performance for a Norwegian professional cycling team.

1.2 Research Gap

Existing research in cycling and endurance sports follows two methodological directions. First, physiological modelling approaches aim to describe performance capacity through power-duration relationships, fatigue estimation, or fitness-fatigue dynamics. These models focus on characterizing underlying physiological states, but they typically remain descriptive and are not explicitly designed to estimate the probabilistic likelihood of race performance outcomes at the individual level.

Second, race outcome prediction models often rely on contextual or competition-level variables, such as race characteristics, team strength, or environmental conditions. While effective for forecasting results, these approaches capture external determinants of performance and do not explicitly model the rider's accumulated training adaptation as a latent readiness state.

Because of this, a methodological gap emerges at the intersection of training-response modelling and individualized race performance prediction. In particular, probabilistic modelling of race-day performance

based on structured, longitudinal workload history at the single-rider level remains limited in the literature.

This thesis addresses this gap by formulating race performance prediction as a supervised learning problem based on structured representations of accumulated training adaptation.

1.3 Problem

Building on the research gap identified in the previous section, predicting athletic performance is very complex, especially in endurance sports where outcomes depend on a mix of psychophysics factors, accumulated workload, recovery status, tactical choices, weather conditions, and team strategies. Cycling amplifies this complexity: performance results emerge from months of preparation, high variability in training stimuli, and inter-individual differences in how riders respond to load [2].

This thesis focuses on a specific problem:

Given the training sessions from the past n weeks, can we estimate the probability that a cyclist will achieve a strong race performance, using historical training data and pre-race workload features?

Formally, the problem is framed as a supervised binary classification task. Let $X_{i,t}$ denote the vector of pre-race workload features for rider i in race instance t , and let $Y_{i,t} \in \{0, 1\}$ indicate whether the rider achieved a strong race performance (Top-20 finish). The objective is to estimate

$$P(Y_{i,t} = 1 \mid X_{i,t})$$

using only information available prior to the race, to prevent temporal leakage. The predicted probability is interpreted as a quantitative proxy for the latent performance readiness of the athlete. The framework is reported in Figure 1.1.

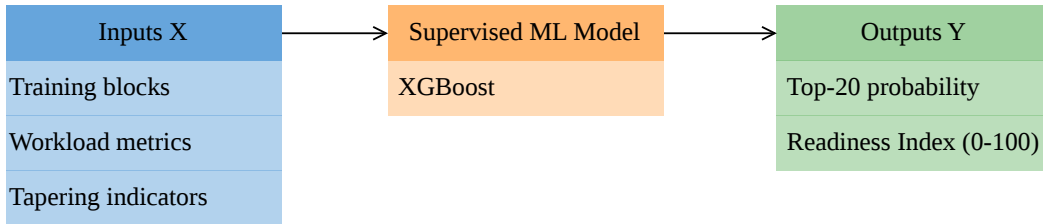


Figure 1.1: Supervised Learning Framework.

This goal introduces several challenges:

- Data imbalance: Top-20 results, used as an operational proxy for strong performance and model supervision, are relatively rare compared to non-Top-20 finishes.
- Heterogeneity across riders: each rider has unique training history, physiological profile, and role in the team.
- Temporal dependencies: performance is shaped by long-term load, short-term tapering, and acute fluctuations.
- Feature ambiguity: metrics are commonly used but may not capture all aspects of performance.
- Absence of a direct readiness indicator: teams often rely on experience-based assessment, lacking a quantitative index that summarizes overall form before a certain event.

So, the problem addressed in this thesis is twofold:

1. Developing a performance prediction model tailored to individual riders, capable of learning from their longitudinal data.
2. Constructing a Readiness Index, defined as a synthetic and interpretable score reflecting the model-estimated probability of a strong race performance.

The choice of using a Top-20 finish as the target label is motivated by its practical relevance and statistical feasibility. More restrictive thresholds, such as Top-10 results, would be too rare given the strong class imbalance, while broader thresholds would be less informative. As such, the Top-20 outcome is treated as an observable proxy to infer an underlying, latent state of performance readiness, which can not be directly measured.

1.4 Approach

To address the problem, this work uses a multi-stage approach combining data engineering, feature construction, and machine learning modeling.

The dataset includes two years of daily training files of five athletes, time-series metrics, and race results from a professional Norwegian UCI Continental team. The raw *TrainingPeaks* exports, a widely used training management platform for endurance sports, are cleaned, standardized, and merged with race metadata and the scraped PCS information, a public database providing detailed professional cycling results and race statistics. A concept of training blocks is introduced to segment the season according to natural cycles of preparation and recovery. From these blocks, numerous features are created, covering:

1. training load metrics;
2. fitness-fatigue dynamics;
3. tapering indicators;
4. rolling statistics over multiple windows (7, 28, 45 days);

Multiple modeling strategies are tested. A logistic regression serves as a baseline, followed by a more expressive gradient-boosted tree approach (XGBoost). Techniques such as Synthetic Minority Over-sampling Technique (SMOTE) are applied to handle class imbalance, and personalized thresholds are computed using the Youden index for each rider. Feature importance analyses are conducted to identify the most important predictors.

Additionally, predicted probabilities are mapped into a RI (0-100), customized for each athlete based on their personal distribution of predicted performance likelihood.

Finally, the results are integrated into an interactive dashboard to provide coaches and staff with a practical tool for interpreting readiness before competitions.

1.5 Results

The baseline logistic regression metrics-based model provides a first indication of the relationship between training load and race outcomes, but its predictive ability remains limited. In particular, it correctly identifies only 5 out of 22 Top-20 (recall 0.227), and its precision is modest due to the strong class imbalance. While overall accuracy appears acceptable, it is driven by the high proportion of non-Top-20 results, revealing that the baseline is not sufficiently discriminative.

The XGBoost-based readiness model demonstrated a clear improvement in identifying positive race outcomes. It successfully detects 16 out of 22 Top-20 results (recall 0.727), more than 50% compared to the baseline model. Although precision remains moderate - a typical challenge when predicting rare positive outcomes - the readiness model provides a more balanced trade-off between identifying strong race performances and limiting false alarms.

Overall, these results indicate that the proposed readiness approach offers a good predictive signal. It is important to say that the RI should be interpreted as an *estimate of an athlete's performance potential* prior to a race, rather than as a deterministic prediction of the final ranking.

1.6 Contributions

This thesis provides the following scientific contributions:

- It formalizes race-day performance prediction as an individualized supervised learning problem based on longitudinal training-load history, explicitly linking accumulated adaptation to probabilistic performance outcomes.

- It introduces a structured training-block representation as a modelling abstraction of periodized adaptation dynamics, enabling the transformation of high-frequency session data into physiologically coherent predictive units.
- It provides empirical evidence, based on multi-season professional data, that interactions between acute load, chronic load, and tapering metrics contribute significantly to race performance likelihood at the individual level.
- It proposes a probabilistic Readiness Index (RI) as an interpretable proxy for latent performance readiness, bridging predictive modelling and applied decision-making in professional cycling.

Together, these contributions advance the methodological foundations of individualized performance modelling in endurance sports and provide a formal bridge between adaptation theory and probabilistic outcome prediction.

The remainder of this thesis is organized as follows.

- **Chapter 2** presents the literature review, summarizing previous work on training-load modelling, machine learning in endurance sports, and performance prediction in cycling. It highlights the methodological gaps that motivate the present study.
- **Chapter 3** describes the case study, the data sources, and the pre-processing pipeline. It details the extraction of *TrainingPeaks* files, the integration of race metadata, and the construction of training blocks used as the structural backbone of the dataset.
- **Chapter 4** introduces the modelling framework. It first discusses preliminary attempts and their limitations, then formalizes the two families of models used in this thesis: the metrics-based models and the readiness-based model. The chapter also presents the full modelling pipeline, covering the handling of class imbalance, hyperparameter tuning, and personalized decision thresholds.
- **Chapter 5** reports the experimental results obtained for each model family. It compares predictive performances, discusses strengths and weaknesses, and analyses the interpretability of the proposed readiness framework.
- **Chapter 6** discusses the methodological limitations of the proposed framework and it also highlights potential improvements for future work.
- **Chapter 7** summarizes the main findings of the thesis and how it can support a professional cycling team.

This structure guides the reader from background concepts to methodological development, experimental validation, and practical applications of the proposed readiness framework.

Chapter 2

Literature Review

Machine Learning in Sport Performance Analysis

The application of AI and ML in sports has grown in recent decades, driven by the proliferation of digital data. Early foundational work already recognized the potential of AI to support decision-making in performance analysis. One of the earliest contributions showed that artificial neural networks could be used to analyze biomechanical patterns in human gait, anticipating a shift from observational assessment toward more data-driven computational methods [6].

More recent research has highlighted how ML can automatically identify performance-related patterns and adapt training decisions. Some studies have demonstrated that AI-based systems can analyze training history, detect when an athlete has trained too intensely, and autonomously adjust future sessions [7]. This represents a transition from general training guidelines toward athlete-specific optimization enabled by ML.

In sport analytics, different research directions have emerged depending on the modelling objective. Broadly, existing studies can be grouped into four main categories: physiological performance modelling, match outcome prediction, recommendation systems, and training-response modelling. Each of these approaches focuses on different prediction targets, input variables, and assumptions regarding the determinants of performance.

Physiological Performance Modelling

A substantial portion of the literature focuses on modelling the physiological determinants of athletic performance. These studies typically rely on physiological metrics such as power output, energy expenditure, or fatigue-related variables to characterize an athlete's performance capabilities.

In cycling, several works have investigated the relationship between power production and race demands. For example, Pinot and Grappe analyzed elite cyclists through their power-duration profiles, demonstrating how these curves reflect the physiological demands of different race scenarios [8]. Similarly, other research has shown that the ability to sustain power despite accumulating work represents a key determinant of success in professional road cycling [9].

Recent studies have also explored predictive models based on physiological metrics. One framework proposed the prediction of Maximum Mean Power (MMP) in upcoming races based on recent Training Load (TL), showing that hybrid models such as XGBoost and CatBoost outperform simpler baselines [10].

While these studies successfully capture physiological capabilities and fatigue-related dynamics, their prediction targets remain primarily limited to laboratory-derived indicators or physiological metrics rather than actual race performance outcomes.

Match Outcome and Performance Prediction

Another major research direction focuses on predicting competitive outcomes using machine learning techniques. In this context, the prediction target typically corresponds to match results, rankings, or finishing times. Applications of these approaches have been widely explored across multiple sports. For example, systematic reviews in tennis illustrate how ML models are used to predict match outcomes, analyze performance, and optimize tactics [11]. Similar advancements have been documented in basketball, where big-data analytics and ML algorithms support training management and performance forecasting [12]. In team handball, ML has been used to classify performance levels and predict match outcomes based on physiological and technical variables [13].

In cycling specifically, several works have attempted to predict race outcomes directly. Early studies focused on stage-level characteristics rather than rider rankings, predicting expected finishing times in Grand Tour stages using physical models [14] or estimating average stage velocity using gradient boosting [15].

More recent contributions have considered rider-level performance. One approach applied gradient boosting

to predict relative finishing times in one-day classics such as the Tour of Flanders, correctly identifying several Top-10 finishers using historical performance, weather, and team features [16].

Despite these advances, most models in this category focus primarily on competition-level predictors or historical race outcomes, while training dynamics and long-term physiological preparation are often not explicitly incorporated into the predictive framework.

Recommendation Systems for Athlete Support

A different line of research focuses on recommendation systems designed to support coaching decisions or athlete-specific strategies. Originally developed in domains such as e-commerce or media platforms, recommendation systems can be adapted to suggest personalized actions based on historical data and user preferences.

Several works have explored their application in endurance sports. For example, systems have been proposed for personalized running recommendations [17] and marathon pacing strategies based on historical performance and physiological characteristics [18].

In cycling, the RaceFit system is a representative example: based on historical coaching decisions and performance patterns, it recommends which riders are best suited for specific race stages. Although focused on tactical allocation rather than performance prediction, this work illustrates how ML can model expert-driven decisions in a professional cycling environment [19].

Although these systems demonstrate the feasibility of machine learning for supporting coaching decisions, they typically focus on tactical allocation or strategy optimization rather than estimating athlete readiness for upcoming races.

Training Load and Training-Response Modelling

Another research area concerns the modelling of the relationship between training load, fatigue, and performance adaptation. These approaches aim to quantify how training stimuli influence physiological responses and subsequent performance outcomes.

Foundational frameworks introduced the concept of internal training load, linking perceived exertion and training duration to overall workload [20]. Subsequent research expanded these concepts to include additional indicators such as acute–chronic workload ratios, monotony indices, tapering dynamics, and recovery processes.

Empirical studies across multiple sports have demonstrated that training load patterns influence performance outcomes. For example, training load monitoring has been associated with performance fluctuations in Australian Football [21] and beach volleyball [22]. These findings support the idea that both the magnitude and temporal structure of training stimuli play an important role in determining athletic performance.

However, despite the maturity of training load research, many of these models remain descriptive or retrospective. The integration of longitudinal training data into predictive machine learning frameworks for competition performance remains relatively limited.

Summary and Research Gap

The existing literature demonstrates that machine learning methods are increasingly applied across sport science to model physiological capabilities, predict competitive outcomes, support coaching decisions, and analyze training adaptation. However, these research directions remain largely separated in terms of modelling objectives and data sources.

Physiological performance models primarily focus on laboratory-derived metrics such as power-duration profiles or maximum power outputs. Match outcome prediction models rely mainly on competition data, historical results, or contextual race variables. Recommendation systems aim to support tactical decisions but do not typically estimate the likelihood of future performance. Finally, training-load research studies adaptation processes but is often not integrated into predictive models of race performance.

A key gap remains in the literature: the absence of individualized frameworks capable of estimating probabilistic readiness for race performance by integrating longitudinal training load data with machine learning models. In particular, few studies combine multi-season training histories, athlete-specific physiological indicators, and race outcomes to produce interpretable predictions that can support coaching decisions.

This gap motivates the present thesis, which proposes a machine-learning framework for personalized race performance prediction in professional cycling. By integrating multi-year training and competition data, the study aims to estimate the probability of strong performance for individual riders and derive a rider-specific RI designed to support decision-making in a professional cycling team environment.

Chapter 3

The Case Study: Team Coop-Repsol

In the Chapter 1, the general motivation, problem definition, and objectives of this thesis were introduced, followed by an overview of the current state of the art in Chapter 2. In this Chapter 3, the focus is on the description of the data workflow developed during the project, as reported in the figure 3.1. In particular, after the introduction of the context, the Section 3.2 details the data collection process, describing the sources. The following Section 3.3 presents the data pre-processing, then the data cleaning phase in Section 3.4, concluding with the feature engineering in Section 3.5, including the creation of derived metrics with the logic used to define the workout blocks in Section 3.5.2. To conclude, Section 3.6 discusses the identification of the target variable and the set of candidate features, while Section 3.7 presents the exploratory data analysis conducted to gain an initial understanding of patterns, distributions, and relationships within the dataset.

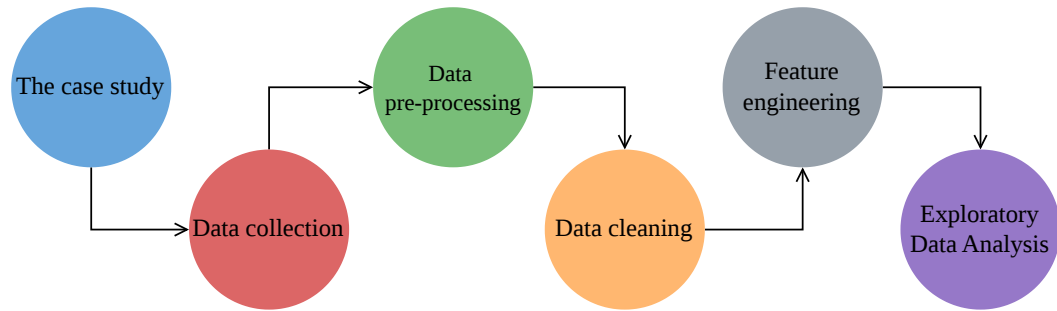


Figure 3.1: The workflow.

3.1 Context: the Norwegian Continental cycling team

The study presented in this thesis is based on data from *Team Coop-Repsol*, a Norwegian UCI Continental cycling team competing at the professional level. The team consists of approximately twenty riders distributed across two squads (men and women) and competes across a wide range of European races during the season.

From a research perspective, this setting offers a valuable opportunity to study performance dynamics in a real professional environment. While operating at the professional level, Continental teams typically have more limited financial and analytical resources compared to World Tour teams. As a consequence, advanced analytical infrastructures and dedicated data science departments are rarely available. This creates a context in which data-driven approaches can potentially provide meaningful support to coaching and performance analysis.

An initial exploratory phase identified several areas in which analytical methods could provide practical value, including organizational planning, nutritional monitoring, and performance analysis. Among these possibilities, the present study focuses on the performance and training management domain.

This choice is primarily driven by data availability. Organizational and financial planning would require operational and economic data not directly related to athletes' training processes. Similarly, nutrition-related analyses would require systematic records of caloric intake and macronutrient distribution, which are not consistently tracked by the team. In contrast, the team maintains detailed multi-season records of training sessions and race participation through the *TrainingPeaks* platform, providing a structured dataset suitable for quantitative analysis.

The resulting dataset therefore consists of longitudinal observations of professional athletes training and competing in real competitive conditions. From a methodological perspective, this represents a setting with high ecological validity: the data reflect actual training practices, race schedules, and workloads experienced by

professional riders during a competitive season. At the same time, focusing on a single team introduces a trade-off between environmental consistency and external generalization. While the shared coaching philosophy and training environment ensure internally coherent observations, the results may not directly generalize to teams operating under different training methodologies or competitive contexts.

Overall, Team Coop-Repsol offers an ideal real-world environment for analyzing how machine-learning methods can support performance analysis in endurance sports, combined the availability of past training data and race outcomes, providing an important starting point for the case study developed in the following sections.

3.2 Data Collection and Data Sources

The first step of the research is the data collection phase, so the collection of the input necessary for the analysis and model training. The data are taken from two sources: *TrainingPeaks*, which stores details of past training and race logs through devices, and PCS, which provides official race results of all the world. Together, these sources enable the integration of training history with performance outcomes, as illustrated in Figure 3.2, providing a high-level overview.

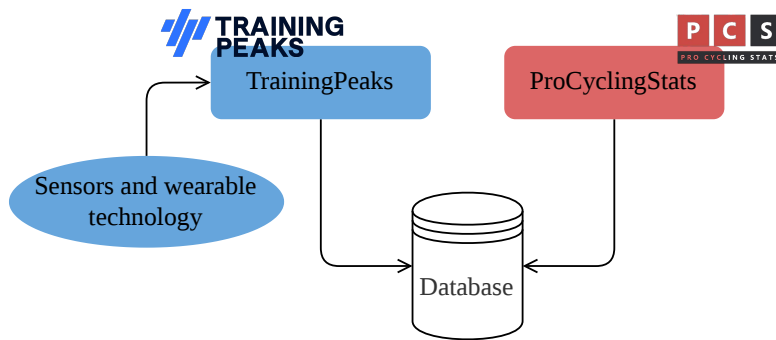


Figure 3.2: Data collection and data sources.

Sensors and Wearable Technology

The dataset analyzed in this study originates from multi-sensor recordings of professional cyclists collected through on-bike and wearable devices during both training and competition. These devices capture a range of physiological and biomechanical variables, including heart rate, power output, cadence, speed, elevation, and estimates of energy expenditure [23].

From a methodological perspective, these measurements can be grouped into internal load variables (physiological responses such as heart rate) and external load variables (mechanical outputs such as power, speed, and distance). Distinguishing between these types of variables is crucial for modeling performance, as they reflect different aspects of training adaptation, providing a comprehensive representation of athletes' physiological and mechanical workload and supporting data-driven performance analysis [2].

All sensor data are subject to potential measurement errors and noise, arising both technical limitations (e.g., signal dropouts, latency, or inaccuracies in wearable sensors) and operational conditions (e.g., environmental interference or variability in device placement) [24, 25]. These issues may lead to irregular sampling intervals, missing samples, duplicated timestamps or short dropouts in the recorded workout files. For these reasons, a rigorous and structured data pre-processing and cleaning phase are essential to ensure data reliability before any feature engineering or modeling step.

The raw measurements recorded by these devices are typically saved in Flexible and Interoperable Data Transfer (FIT) and synchronized to training platforms or health and fitness applications. These platforms often employ machine learning or AI tools to analyze the collected data and create insights for athletes and coaches. This reflects a trend: wearable technology is creating opportunities for AI research in Sport Sciences. The continuous collection of multi-sensor physiological and biomechanical data enables the development of algorithms for performance monitoring, injury risk assessment, fatigue modelling, and personalized training optimization [1].

In this work, raw time-series data are aggregated into training session summaries, reflecting physiologically informed feature extraction rather than just technical convenience. By summarizing sessions into representative metrics, noise is reduced and variables are contextualized in the framework of performance adaptation. This representation enables the construction of a structured dataset suitable for machine-learning models while preserving the interpretability of training-related features.

TrainingPeaks

TrainingPeaks is the platform used by the team for daily monitoring of training load and physiological metrics. Each activity includes time-series data (e.g., power, heart rate, cadence, altitude) together with summary statistics and derived metrics developed by the software itself. Since these data play a central role in the modelling phase, they are described in detail in Section 3.5. Furthermore, *TrainingPeaks* allows exporting different types of files. The workout files in *fit* format contain the time-series raw data recorded by the training device, described in detailed in the previous paragraph Sensors and Wearable Technology. *TrainingPeaks* generates one file per activity for each rider: the platform provides workout summary files in *csv* format. In this case, each rider has a single summary file, where each row corresponds to one training session and represents an aggregated description of the respective *fit* file: this summary includes overall statistics of the activity. Finally, the platform provides custom metrics files - also in *csv* - which record self-reported indicators such as sleep quality, fatigue, soreness, stress, appetite, and body weight. These optional subjective metrics are explored during the work but are not used due to their sparse and inconsistent availability across riders, as reported in Table 3.1.

Although the team has several years of training history for more than twenty riders, this study focuses only on five male riders from the 2023 and 2024 seasons, despite for Athlete D for which only 2023 season is available. This selection reflects the fact that only a subset of riders agreed to participate in the research and to share their complete training data. In summary, Table 3.1 provides an overview of the files available and used in the work for each rider.

Rider	Workout Files	Workout Summary	Custom Metrics
Athlete A	2023–24	2023–24	Not recorded
Athlete B	2023–24	2023–24	2023–24
Athlete C	2023–24	2023–24	2023–24
Athlete D	2023	2023	Unavailable
Athlete E	2023–24	2023–24	2023–24

Note: For privacy reasons, riders are anonymized throughout this thesis using identifiers Athlete A to Athlete E.

Table 3.1: Overview of available files per rider.

ProCyclingStats

All information related to race results is obtained from PCS (Table 3.2), an open-data platform that provides statistics on professional cycling.

Column	Description
<code>rider</code>	Rider name.
<code>result</code>	Final race position.
<code>race_name</code>	Name of the race.
<code>level_race</code>	UCI race level/category.
<code>distance</code>	Race distance [km].
<code>WorkoutDay</code>	Date of the event.
<code>Type</code>	Race.

Table 3.2: Variables extracted from ProCyclingStats.

Since the data are not available through an official Application Programming Interface (API), the extraction is performed via web scraping using Python libraries such as `requests`, `BeautifulSoup4`, and `pandas`. This procedure produces a single dataset containing the historical race calendar of each rider for the 2023 and 2024 seasons, comprising 372 rows. The extracted information are summarized in Table 3.2. Furthermore, Table 3.3 reports an excerpt of the race-results dataset; only a few representative rows are shown for clarity.

rider	result	race_name	level_race	stage_name	distance	WorkoutDay	Type
Athlete A	DNF	Lotto Famenne Ardenne...	1.1	one_day	188.1	2023-10-01	race
Athlete A	DNF	Grand Prix Cerami	1.2	one_day	152.6	2023-09-30	race
Athlete A	DNF	Memorial Fred De Bruyne	NAT	one_day	160.8	2023-09-27	race
Athlete A	43	Zwevezele Koers	NAT	one_day	153.5	2023-09-22	race
Athlete A	7	GP Dr. Eugeen Roggeman...	NAT	one_day	169.4	2023-09-21	race
...	...	...	...	...	...	...	...

Table 3.3: Excerpt of race results from PCS (5 sample rows shown).

To conclude, Table 3.4 summaries the set of files used in the next processing steps, which lead to the construction of the final dataset described in Section 3.6.

File Type	Description	Action
Workout Files (fit)	Raw sensor recordings, one file per activity per rider.	Extraction, standardization, and cleaning.
Workout Summary (csv)	Aggregated session-level statistics, one row per activity per rider.	Extraction, standardization, and cleaning.
Riders Calendar (csv)	Race history extracted from PCS.	Extraction and cleaning of the column <code>result</code> .

Table 3.4: Files used before the data-processing pipeline.

3.3 Data Pre-Processing

The second step of the workflow consists of the data pre-processing phase [26]. In this stage, the raw data are extracted and organized to obtain well-structured inputs. Several operations may be carried out during this phase; the following subsections describe what it is implemented in this case study. It is important to say that this stage deals only with existing variables, while the creation and the description of new derived metrics - aimed at improving predictive performance - is in the feature engineering phase in Section 3.5.

Another important aspect is that some of processing stages are carried out separately for **workout files** and the **workout summary** because these sources provide complementary information. Workout files contain high-resolution time-series measurements, useful for detailed inspection and potential derivation of new features. On the other hand, the workout summary already aggregates key session metrics and ensures a standardized structure across riders. By processing them separately, it is possible to validate, clean, and then align both sources into a unified classification file without losing information. Moreover, because the initial datasets include a large number of variables - which not all are relevant to the objectives of this study - only the most important columns are mentioned. A workflow of the data pre-processing is reported in the following Figure 3.3.

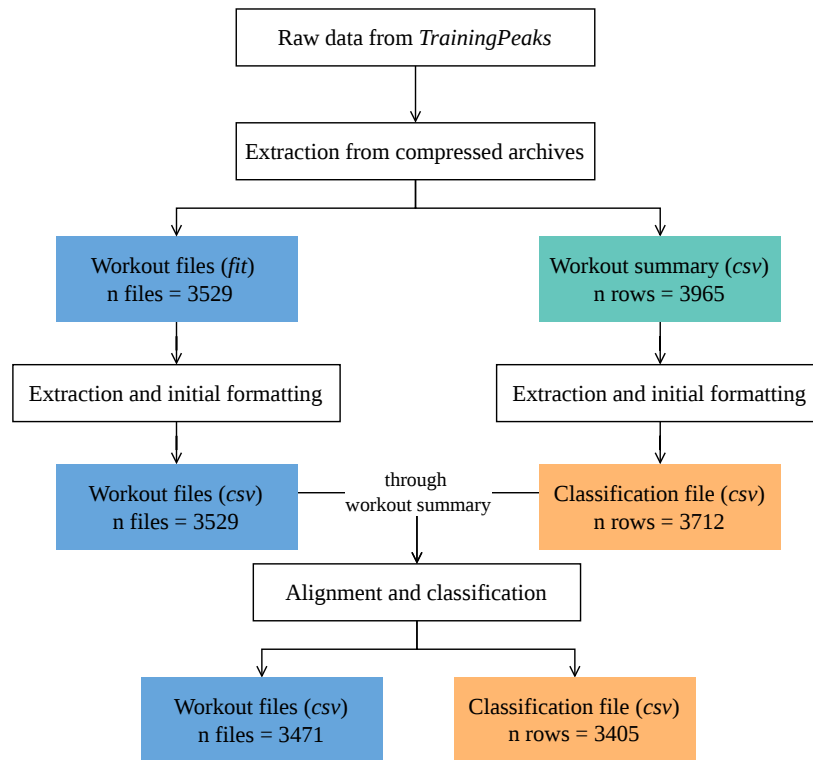


Figure 3.3: Data pre-processing.

The initial version of both files is illustrated through representative excerpts in the following tables. Table 3.5 reports a short excerpt of the workout files, where each row corresponds to one second of activity, while Table 3.6 displays the structure of the workout summary file, where each row corresponds to a single activity summarizing the various metrics.

altitude	distance	speed	heart_rate	power	cadence	...	rider	year	WorkoutDay	time
4.4	0.00	0.00	69	0		...	Athlete A	2023	2023-01-04	15:33:18
8.2	0.79	0.00	69	0		...	Athlete A	2023	2023-01-04	15:33:19
8.2	0.79	2.37	77		41	...	Athlete A	2023	2023-01-04	15:33:24
8.2	4.24	2.82	79	186	65	...	Athlete A	2023	2023-01-04	15:33:25
8.2	8.22	3.39	80	137	82	...	Athlete A	2023	2023-01-04	15:33:26
8.2	13.4	3.96	82	119.0		...	Athlete A	2023	2023-01-04	15:33:28
8.4	16.34	3.96	83	99.0		...	Athlete A	2023	2023-01-04	15:33:29
8.4	20.27	3.66	83	129.0		...	Athlete A	2023	2023-01-04	15:33:30
...	...	...	...	...	...	...	...	...	...	...

Table 3.5: Excerpt of the workout files (time-series); only 8 seconds and a subset of columns are shown. Empty cells indicate missing values.

Title	WorkoutType	WorkoutDay	PowerAvg	PowerMax	...	IF	TSS	rider
4t lett langtur	Bike	2023-01-02	194	676	...	0.5615	126.56	Athlete A
Lett sykling (1-2t)	Bike	2023-01-03	167	896	...	0.4611	21.33	Athlete A
Basisøvelser	X-Train	2023-01-03			...			Athlete A
1t trill for...	Bike	2023-01-04	185	605	...	0.6086	32.37	Athlete A
D1 - Spain...	Bike	2023-01-05	208	1419	...	0.6821	186.14	Athlete A
D2 - Spain...	Bike	2023-01-06	214	1548	...	0.7349	268.23	Athlete A
...	...	...	...	...	...	...	...	...

Table 3.6: Excerpt of the workout summary; only a subset of columns and 6 rows are shown. Empty cells indicate missing values.

3.3.1 Data Representation Hierarchy

Before describing the pre-processing procedures in detail, it is useful to clarify the hierarchical structure through which the raw data are transformed into modelling inputs. The dataset used in this study is organized across multiple levels of aggregation.

At the lowest level, the raw data consist of high-frequency sensor measurements recorded during each activity. These time-series include variables such as power output, heart rate, cadence, speed, and altitude, typically sampled every second. While these measurements capture the instantaneous physiological state of the rider, they are mainly used as the basis for computing higher-level training metrics. Instantaneous signals reflect momentary fluctuations in effort, which may vary in a single activity due to terrain, pacing strategy, or short physiological adjustments. Because of this, they do not directly represent the training stimulus that drives performance adaptation. For this reason, the raw signals are aggregated into session-level metrics that summarize the overall workload and intensity of the activity.

These measurements are aggregated at the session level, where each activity is summarized through workload indicators such as Training Stress Score (TSS), Intensity Factor (IF), average power, and duration. A session therefore represents a single training stimulus and constitutes the fundamental unit of the training history.

Sessions are then grouped into training blocks, representing short temporal windows preceding a race. In each block, multiple sessions contribute to the accumulated physiological load experienced by the athlete. These blocks capture the temporal structure of training and provide a representation of recent workload and recovery patterns.

Finally, modelling is performed at the race level. For each race, the corresponding training block preceding the event is used to predict. In this formulation, race performance is not used as the result of a single activity but as the outcome of the accumulated training adaptation generated over time.

This hierarchical representation, reported in Figure 3.4, clarifies the role of the pre-processing pipeline: raw sensor data are first organized into sessions, sessions are aligned and cleaned across riders, and training histories are structured so that meaningful race-level prediction instances can be constructed.

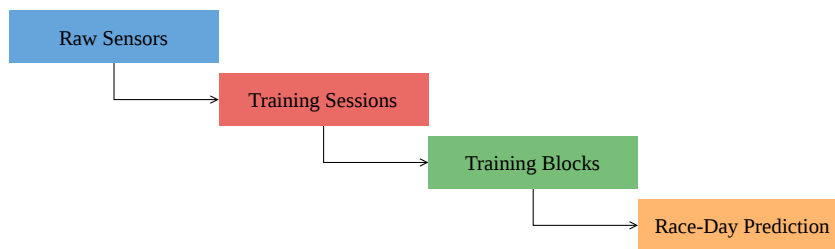


Figure 3.4: Data Representation Hierarchy.

This aggregation step is not only a technical simplification but also a domain-informed representation choice. In endurance sports, performance on race day is not determined by instantaneous effort but by the cumulative physiological adaptation generated by multiple training sessions over time. Representing the data at the session level allows individual activities to be combined into training blocks, which capture the temporal structure of workload accumulation preceding each race. These blocks then form the basis for constructing race-level prediction instances used in the modelling stage.

3.3.2 Extraction and Initial Formatting

Workout Files

The *TrainingPeaks* data are originally delivered as compressed archives, each containing one *fit* file per activity for each rider. These archives are extracted and converted into *csv* format to enable inspection and processing. An initial step is performed to standardize the structure of the workout files across riders. This included the identification of columns (such as *rider*, *year*, and *WorkoutDay*), removal of unused and inconsistent fields, rounding of numerical values, transformation of dates into a unified format, and elimination of rare columns appearing in fewer than 5% of all sessions. Although the extracted time-series contain detailed measurements, these data are not directly used in the modelling pipeline as inputs. Instead, the analysis is based on the session-level metadata provided by the platform, where key training-load metrics are already computed. As discussed later, this choice reflects a domain-informed feature representation rather than a purely technical

simplification. In endurance sports, performance is primarily influenced by the cumulative physiological stress generated by training sessions rather than by instantaneous fluctuations in a single activity. Aggregated session metrics provide a more meaningful representation of training stimuli, capturing quantities such as total workload, intensity, and duration that are directly related to adaptation processes.

Additionally, summarizing time-series signals into session-level descriptors reduces the impact of measurement noise and short signal irregularities that may occur in raw sensor recordings, as already described in Sensors and Wearable Technology. This representation enables the construction of a structured dataset suitable for machine-learning models while preserving the physiological interpretation of training load variables used in feature engineering.

Workout Summary

The workout summary files provided by the platform are also standardized. A rider identifier is added to each file, and only the records for five riders with complete time-series data are used. Columns with all NaN missing values, and workout types appearing in fewer than 5% of rider's sessions are removed; for instance, activities labeled as *custom*, *walk*, *xc-ski*, or *other*. In addition, all *MTB* sessions are categorized under the general *Bike* class.

To provide an overview of the resulting dataset after this first step, Table 3.7 summarizes the distribution of workout types across all riders, while Table 3.8 reports the same statistics by athlete. The new dataset, referred to in this work as the *classification file*, includes a total of 3,712 structured training sessions in the workout summary, dominated by cycling workouts, which account for more than 80% of all activities.

Workout Type	Count	Percentage (%)
Bike	3104	83.6
Run	243	6.5
Day Off	187	5.0
Strength	178	4.8
Total Workouts	3712	100

Table 3.7: Overview of the dataset: total sessions and distribution of workout types.

Rider	Bike	Day Off	Run	Strength
Athlete A	649	35	0	57
Athlete B	727	66	97	59
Athlete C	661	51	63	62
Athlete D	344	14	0	0
Athlete E	723	21	83	0

Note: as expected, since for Athlete D only 2023 season is available, the number of workouts is less than the other.

Table 3.8: Workout type distribution for each rider.

As expected for a professional cycling team, the majority of sessions correspond to bike workouts, while running and strength training appear mostly as complementary activities. Day-off events represent approximately 5% of the total training volume.

3.3.3 Alignment and Activity Classification

Once all the files are available in *csv* format, the next step consists in aligning the new source and the workout files through workout summary, that every time-series file corresponds to the correct row in the new metadata file.

Before describing the alignment procedure, it is important to clarify that the new *classification file* and the *workout summary* represent two different sources of information. In fact, the classification file contains one row for every activity for which a time-series workout file actually exists, together with the corresponding metadata from the workout summary. In practice, it can be defined as a cleaned and filtered version of the original workout summary. On the other hand, the *TrainingPeaks* workout summary metadata file contains one row for every

session recorded in the *TrainingPeaks* calendar, including cases where no time-series file is available, where duplicate entries exist, or where activities are logged on rest days.

Initial Matching

An iterative classification procedure is implemented to obtain a preliminary workout type for all available time-series files, starting from Table 3.8. Since the time-series files do not contain any information indicating the type of workout (e.g., Bike, Run, Strength), it is impossible to define the workout modality from the raw data alone.

For this reason, for each file, the *WorkoutDay* and the *rider* are extracted from the header of the corresponding *csv*. These identifiers are then matched:

- if exactly one row with the same *rider* and *WorkoutDay* is found, the workout gets the corresponding *WorkoutType*;

(a) Workout file time-series without *WorkoutType*.

altitude	...	rider	year	WorkoutDay	time	WorkoutType
4.4	...	Athlete A	2023	2023-01-04	15:33:18	
8.2	...	Athlete A	2023	2023-01-04	15:33:19	
8.2	...	Athlete A	2023	2023-01-04	15:33:24	
8.2	...	Athlete A	2023	2023-01-04	15:33:25	
...	...	...	...	...	...	...

(b) Workout summary with the corresponding matched row: both files have in common *rider* and *WorkoutDay*, so it is taken the value of the column *WorkoutType*.

Title	WorkoutType	WorkoutDay	PowerAvg	PowerMax	...	IF	TSS	rider
4t lett langtur	Bike	2023-01-02	194	676	...	0.5615	126.56	Athlete A
Lett sykling (1-2t)	Bike	2023-01-03	167	896	...	0.4611	21.33	Athlete A
Basisøvelser	X-Train	2023-01-03			...			Athlete A
1t trill for...	Bike	2023-01-04	185	605	...	0.6086	32.37	Athlete A
D1 - Spain...	Bike	2023-01-05	208	1419	...	0.6821	186.14	Athlete A
...	...	...	...	...	...	...	...	...

- if no match or multiple matches are found, the activity has temporarily saved as *Unknown*;
- files containing the string *zwift* in the filename - a famous virtual cycling and running app that allows you to train at home - are directly classified as *Bike*. For example, a filename as follows:

zwift-activity-1295728843650236464.csv

This step produces an initial mapping between time-series files and their workout types.

Resolution of Unknown Activities

After the initial matching stage, a subset of time-series files can not be assigned a workout type because the metadata contain no entry or more than one entry for the same rider and date. These activities are temporarily labeled as *Unknown*. To reduce their number, two more steps are then applied:

(c) Updated workout file time-series with assigned *WorkoutType*.

altitude	...	rider	year	WorkoutDay	time	WorkoutType
4.4	...	Athlete A	2023	2023-01-04	15:33:18	Bike
8.2	...	Athlete A	2023	2023-01-04	15:33:19	Bike
8.2	...	Athlete A	2023	2023-01-04	15:33:24	Bike
8.2	...	Athlete A	2023	2023-01-04	15:33:25	Bike
...	...	...	...	...	...	...

Table 3.9: Initial matching procedure between time-series workout files and workout summary. The matched summary row assigns the *WorkoutType* to the corresponding workout file.

1. Metadata-driven resolution: each *Unknown* activity is matched again against the cleaned workout summary file. For a given rider and date, if the summary contains exactly one valid (i.e., non-missing and unique) *WorkoutType*, this type is assigned to the corresponding time-series file.
2. Time-series-based resolution: for the remaining cases, the structure of the raw time-series file is explored. The following method uses:
 - files in which header includes pedal-related variables are reclassified as *Bike*;
 - sessions with zero distance and no pedal information in the columns are assigned to *Strength*.

After these steps, 3,529 time-series files (out of the 3,712 summary entries in Table 3.7) have a match to metadata. However, 104 files still remain *Unknown*. To resolve this issue, a final rider-specific step is implemented.

Rider-Specific Corrections

For the three athletes with *Unknown* sessions (Athlete B, Athlete C, and Athlete E), all remaining workouts are inspected by rider and date. For each rider, the dates of the *Unknown* activities are cross-checked with the workout summary, and when a given date appears exactly once among the unknown files and the corresponding metadata in the summary clearly indicated a *Run* session, the workout type is updated from *Unknown* to *Run*. This process reduces the number of ambiguous sessions and corrects a lot of running workouts that could not be resolved based on the time-series file alone. After this, only two *Unknown* workouts remained, both of Athlete B. Given their very small number, these sessions are manually inspected and assigned to the appropriate workout type. As a result, no time-series file have the *Unknown* label.

Non-Corresponding and Day-Off Sessions

After solving all *Unknown* activities, another inspection reveals a second class of problematic rows: sessions for which no match can be found between the time-series files and the workout summary. These include, for example, duplicated entries, activities recorded on rest days, or missing metadata. These cases are labeled as *NoCorrespondence*, and a total of 49 sessions fall into this category. All sessions flagged as *NoCorrespondence* are removed. The workout summary also contains a number of *Day Off* as workout, which represent non-training days. A small subset of these days wrongly contains an associated time-series file. Since these files do not correspond to an actual workout, they are removed.

After all these steps, the alignment of the workout type between the workout files and the classification file through the workout summary is done, creating a set of 3,471 valid sessions in the classification table for which:

- a time-series file exists;
- a consistent metadata row exists;
- a unique workout type is assigned (Bike, Run, Strength).

To clarify the structure of the data pipeline, three main data sources are now available:

- Workout files (time-series): raw recordings extracted from devices;
- Workout summary: session metadata computed by *TrainingPeaks*;
- Classification file: aligned dataset linking each workout file to a cleaned and validated metadata row.

3.3.4 Final Alignment and Classification Activity

After producing an aligned classification activity, a final step is the validation and correction of the TSS associated with each session, because TSS represents an important training-load metric (described in detailed in Section 3.5), and it is important to check the miss-match between the activity and the TSS value. This stage integrates information using the two sources in different useful way:

- the workout summary, which contains the official TSS computed by the platform;
- the time-series files, from which heart-rate information can be extracted to perform checks.

Starting from the 3,471 valid sessions, each row is matched with its corresponding row in the workout summary using the columns:

rider | WorkoutDay | WorkoutType

When a unique match is found and the metadata file provides a non-missing TSS, the value is copied into the classification file. This covers the majority of sessions. However, for a subset of activities, for instance, those with duplicated metadata entries or irregular device recording, the automatic procedure do not find a match. These sessions receive a temporary placeholder value ($TSS = 0$) and are processed separately. In fact, a manual check through a file-specific inspection is done with a dedicated mapping for finding the correct TSS value. This phase recovers the TSS for the majority of the placeholder rows. However, 66 sessions lacks a valid value for the metric and they are removed, reducing the dataset from 3,471 to 3,405 sessions.

To ensure the correctness of the training load, the consistency between the maximum heart rate recorded in the time-series and the value reported in the workout summary is evaluated. For each session:

1. the time-series file is checked and, when available, the heart rate column is extracted;
2. the maximum observed heart rate `HeartRateMaxFit` is taken;
3. this value is compared to the corresponding `HeartRateMax` from the official column in workout summary to ensure the alignment between the sources.

For all the sessions where both values of the columns `HeartRateMaxFit` and `HeartRateMax` are present but differ, a check is computed using a key to search a match based on:

rider | WorkoutDay | WorkoutType | HeartRateMaxFit

When a unique match is found, the TSS in the classification table is updated. As a final verification step, the resulting dataset classification file is checked. It contains 3,405 sessions and 22 variables and a missing-value analysis confirmed that no column contains NaN entries, indicating that:

- each session has a valid time-series file;
- all metadata variables are complete;
- no ambiguous, duplicated, or partially recorded sessions remain.

Furthermore, the Table 3.10 displays the structure of the classification file.

file	WorkoutDay	WorkoutType	rider	Distance	Type	TSS	IF	...
AthleteA2023\001.zwift	2023-01-02	Bike	Athlete A	106024.59	workout	126.56	0.562	...
AthleteA2023\002.zwift	2023-01-03	Bike	Athlete A	38704.14	workout	21.33	0.461	...
AthleteA2023\003.tp	2023-01-04	Bike	Athlete A	25200.75	workout	32.37	0.609	...
AthleteA2023\004.tp	2023-01-05	Bike	Athlete A	124538.00	workout	186.14	0.682	...
AthleteA2023\005.tp	2023-01-06	Bike	Athlete A	153585.88	workout	268.23	0.735	...
AthleteA2023\006.tp	2023-01-07	Bike	Athlete A	214452.56	workout	278.68	0.680	...
...	...	...	...	...	...	...	...	...

Table 3.10: Excerpt of the structured workout dataset. Only a subset of columns and rows is shown.

To conclude, Table 3.11 summaries the distribution of workout types across all riders, while Table 3.12 reports the same statistics by athlete.

WorkoutType	Count	Percentage (%)
Bike	3022	88.8
Run	233	6.8
Strength	150	4.4
Total Workouts	3405	100.0

Table 3.11: Overview of classification file: total sessions and distribution of workout types.

Rider	Bike	Run	Strength
Athlete A	640	0	41
Athlete B	703	96	55
Athlete C	641	61	54
Athlete D	335	0	0
Athlete E	703	76	0

Table 3.12: Workout type distribution for each rider after alignment.

After the data pre-processing phase, the columns in the classification file are reported in Table 3.13.

Column	Description
file	Path of the workout file, used as the key to link the corresponding time-series data.
WorkoutDay	Date of the session.
WorkoutType	Assigned activity type (Bike, Run, Strength).
rider	Athlete identifier.
Type	Session category (Workout, Race, Cool-down, Warm-up).
TSS	Training Stress Score of the session.
IF	Intensity Factor of the session.
DistanceInMeters	Total distance covered.
TotalTimeInHours	Duration of the workout.
Heart-rate zone durations	
HRZone1Minutes	Minutes spent in each HR zone: from 1 to 7.
HRZone <i>n</i> Minutes	
Power zone durations	
PWRZone1Minutes	Minutes spent in each power zone: from 1 to 6.
PWRZone <i>n</i> Minutes	

Table 3.13: Columns included in the classification file after the data pre-processing phase.

3.4 Data Cleaning and Data Reduction

The raw dataset contains thousands of workout files recorded with different devices, heterogeneous sampling frequencies, inconsistent metadata, and several missing or duplicated fields. Before any feature extraction or modelling can be performed, it is essential to build a clean, standardized, and robust data structure. This section describes the full cleaning pipeline applied to both the workout files and the classification metadata. The process includes filtering non-cycling activities, selecting and standardizing the relevant variables, handling missing or inconsistent values, harmonizing the sampling frequency across all files, and correcting implausible outliers. The following subsections detail each step of this workflow, as reported in the Figure 3.5.

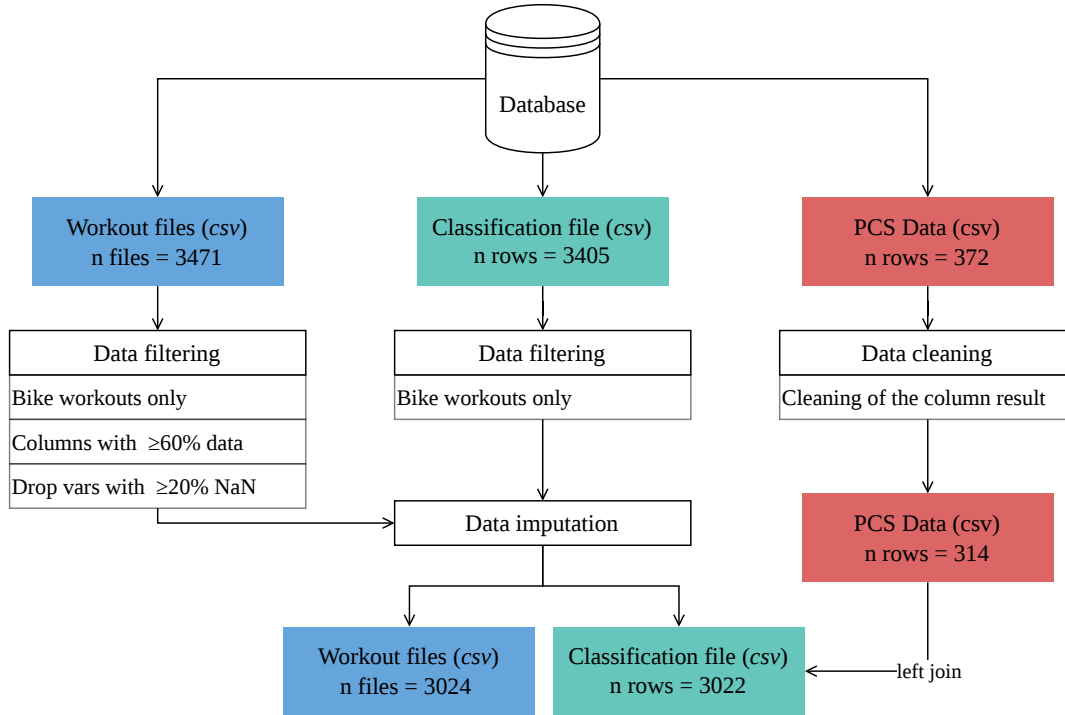


Figure 3.5: Data cleaning and data reduction.

3.4.1 Manage Workout Files Data Columns

First, since now there is also the `WorkoutType` column in the time-series files, only cycling workouts are kept, since all performance metrics and derived features used in this study are specific to cycling. In particular, key variables such as power output, which is fundamental for the computation of training-load metrics, are not meaningful in non-cycling activities such as running and strength training. For this reason, sessions where `WorkoutType` is not classified as `Bike` are excluded from the analysis.

Second, to ensure a consistent structure across riders and devices, the remaining workout files are then standardized. Starting from a global check of all the FIT-converted files, they contain over 60 different variables, but many of them appear sporadically. So, in a first step, to keep useful measurements, a column is kept only if it appears in at least 1,800 files ($\approx 60\%$ of the 3,024 total time-series sessions after filtering for `Bike` as `WorkoutType`, with 22 variables remain). In a second step, for each variable it is analyzed the number of missing values: for each column, the overall proportion of NaN entries across all files is computed, and variables with 20% or more are removed. This step removes four other columns over the total. Furthermore, also the columns `position.long`, `position.lat`, and `temperature` are removed because not strictly related to the scope of the work. At this moment, the number of workout files is 3,024 with 15 columns. In the Table 3.14 is reported the number of NaN for each column in the workout files (time-series):

Column	NaN (%)
left_right_balance	9.73
distance	7.49
heart_rate	2.86
fractional_cadence	1.70
speed	1.69
cadence	0.96
accumulated_power	0.87
enhanced_speed	0.63
power	0.63
altitude	0.37
enhanced_altitude	0.26
time	0.00
year	0.00
WorkoutDay	0.00
rider	0.00

Table 3.14: Final set of 15 variables retained in the time-series workout files.

So, it is necessary one final operation: the management of the remaining NaN values in the time-series workout files, and also the zero values in the classification file, since it has no column containing NaN entries.

3.4.2 Data Imputation of Missing and Inconsistent Values

An imputation procedure is applied to address missing or inconsistent values. Since the kept variables are essential for the computation of training-load and performance metrics, it is important to have a complete structure across all sessions. Then, the procedure is divided into three steps.

Step 1

The first step consists of having a uniform column structure across all workout files (time-series). A global list of variables is created, and any missing column in a file is added and filled with zeros. Likewise, columns that exist but contain only NaN values are replaced with zeros. These zeros are not used as meaningful measurements; they simply guarantee that every file follows the same schema.

Step 2

A second step focuses on specific variables, based on the type of source.

Workout Files

- **distance** contains NaN values which are 29 over more than 27 million samples (approximately 0.0001%). For these files, missing entries are first imputed by linear interpolation along the time axis. Any remaining NaN are then filled using forward and backward-filling.

The reduction in NaN proportions between Table 3.14 and Table 3.15 is due to the imputation strategy applied in Step 1 and Step 2 for the **distance** column. Structural missing values - derived from columns absent in some files or entirely composed of NaN - are replaced with zeros to ensure a consistent schema. As a result, only true sensor-level missing values remain in the final dataset. Table 3.15 reports the new version of the time series files showing that most fields have very low proportions of missing values (typically below 1%). The only exception is the **left_right_balance** metric, which, although available for all files, contains approximately 8% missing entries. This issue is addressed in Step 3 through a dedicated imputation strategy.

Column	NaN %
left_right_balance	8.19
cadence	0.95
fractional_cadence	0.81
power	0.62
enhanced_speed	0.61
heart_rate	0.60
speed	0.48
enhanced_altitude	0.25
altitude	0.24
accumulated_power	0.11
distance	0.00
time	0.00
year	0.00
WorkoutDay	0.00
rider	0.00

Table 3.15: The 15 variables retained in the time-series workout files.

Classification File

- **DistanceInMeters** is reconstructed where the classification file reports a zero value, which is impossible as a value for a distance in meters [m] for a bike session. In these cases, the final value from the corresponding time-series column **distance** is used to assign the total correct distance to the row metadata.
- **TotalTimeInHours** is recomputed directly from the time-series data by parsing timestamps in the **time** column, ensuring consistency between metadata and sensor data.

After that, the classification file is checked for both missing and zero values. Moreover, as for the workout files, sessions in classification file, where **WorkoutType** is not classified as **Bike**, are also excluded from the analysis.

Riders Calendar

The race calendar for each rider - scraped from PCS - also includes non-finish number position such as Did Not Finish (DNF), Did Not Start (DNS), and Over the Time Limit (OTL). Since these outcomes do not provide a meaningful finishing position, it is assigned them a placeholder value of 1000 and then all rows marked with the placeholder value are removed, ensuring that only valid finishing positions are kept. Furthermore, the PCS riders calendar is merged with the *classification file* using a left join on the common keys **rider** and **WorkoutDay**. This operation allows race-related sessions to be correctly classified as *race*, *warm-up*, *cool-down*, and *workout*, and, more importantly, enables the integration of race outcome information column **result** and **top20** of each race for each rider. This process results as a coherent dataset, where each activity is matched, classified, and temporally aligned with the rider's competition schedule. At this moment, the situation of both main files is reported in the following Table 3.16.

Dataset	Rows / Files	Columns	Next Actions
Workout-files (time-series)	3,024 files	15	Imputation of the NaN values in Table 3.15.
Classification file + PCS riders calendar	3,022 rows	24	No further actions are required.

Note: workout files and classification file do not need to contain the same number of rows/files; what matters is that each row in the classification file correctly links to its corresponding time-series file.

Table 3.16: Overview of the datasets during the data cleaning phase.

Step 3

The final step regards the imputation of the NaN values of the time-series files. A dedicated pipeline is applied to the workout time-series in order to (i) harmonize the sampling frequency across all files, (ii) impute missing values in a controlled way, and (iii) correct obviously implausible outliers.

(i) Harmonize the sampling frequency across all files. To ensure a consistent structure across riders, the workout files are resampled to a uniform frequency of 1Hz. In fact, the original FIT converted workout files are not guaranteed to be sampled at a regular frequency, due to sensor limitations, as discussed in Section 3.2: an initial diagnostic function scan all 3,024 sessions and flag files with missing timestamps, irregular time intervals or duplicated timestamps. In total, 2,599 files show at least one of these issues, confirming the need for resampling at 1 Hz. For each file, the `time` column is converted to a proper time index, duplicate timestamps are removed, and a new index with one-second resolution is introduced between the first and last recorded sample. The time-series variables are then re-indexed on this 1Hz grid. Only the newly created timestamps (i.e. those that did not exist in the original file) are imputed, while the original samples and their missing values are for the moment preserved. This resampling is applied column-wise according to a predefined grouping of variables and a group-specific imputation strategy. Both the column mapping and the corresponding strategies are reported in Appendix A.1 and A.2.

After this step, a second run of the function confirmed that all 3,024 files followed a regular 1Hz sampling with no duplicated timestamps.

(ii) Impute missing values in a controlled way. After forcing the 1Hz grid, some variables still contain NaN values. The same group strategies used in the sampling frequency is adopted. The columns are again processed by semantic groups, but some groups are explicitly excluded from imputation: rider name, temporal identifiers, and distance metrics, because as reported in Table 3.15, they do not have NaN values. On the other hand, for each remaining group (speed metrics, cardiac metrics, power metrics, cadence metrics, pedal metrics, and altitude metrics), the imputation followed a three-step strategy applied column-wise:

1. Linear interpolation: if at least two valid (non-missing) values were present in the column, missing entries are first filled by linear interpolation along the time axis, after sorting by `time`.
2. Group mean: if some NaN remain after interpolation, they are replaced by the mean value computed over the columns of the same group in that file.
3. Forward and backward filling: any remaining missing values are finally imputed using forward and backward-filling.

This procedure is applied to all 3,024 files. Summary statistics collected during the run show that, for instance, linear interpolation is applied thousands of times across the dataset for power, cadence, heart-rate and speed metrics, while no residual NaN remained in these groups after the full pipeline.

(iii) Correct obviously implausible outliers. As a final step, the global minimum and maximum values of the variables are inspected before and after imputation. Some extreme values, although not missing, are physiologically impossible or clearly inconsistent with cycling data, e.g., extremely high power values such as 7,406W or heart rates reaching 250bpm. So, a simple range-based correction is applied, defining plausible ranges based on domain knowledge; these thresholds are reported in Table 3.17 and are used to identify and correct outliers.

Variable	Min	Max	Unit
altitude	-100	3,500	m
enhanced_altitude	-100	3,500	m
cadence	30	150	rpm
heart_rate	60	220	bpm
power	0	1,500	W
speed	0	25	m/s

Table 3.17: Plausible value ranges defined for outlier correction.

An example workout file is reported for illustration in Table 3.18; all other workout files share the same structure, with uniformly sampled time-series data at 1Hz and no missing values.

time	WorkoutDay	altitude	cadence	...	heart_rate	rider	speed	year
11:43:20	2023-01-02	15.6	90	...	109	Athlete A	9.20	2023
11:43:21	2023-01-02	15.6	90	...	110	Athlete A	11.19	2023
11:43:22	2023-01-02	15.6	91	...	114	Athlete A	10.90	2023
11:43:23	2023-01-02	15.4	91	...	117	Athlete A	11.00	2023
11:43:24	2023-01-02	15.4	91	...	119	Athlete A	10.99	2023
11:43:25	2023-01-02	15.2	91	...	120	Athlete A	11.28	2023
11:43:26	2023-01-02	15.0	93	...	120	Athlete A	11.26	2023
11:43:27	2023-01-02	15.0	94	...	119	Athlete A	11.36	2023
11:43:28	2023-01-02	14.8	86	...	119	Athlete A	11.47	2023
11:43:29	2023-01-02	14.6	88	...	119	Athlete A	12.05	2023
...	...	...	...	...	...	...	...	...

Table 3.18: Excerpt of a workout time-series file after pre-processing and cleaning phase. Selected columns are shown.

Before introducing the feature-engineering phase, it is useful to summarize the structure of the datasets with only bike activity obtained after the full data pre-processing and cleaning steps. At this stage, two are the main data sources, reported in Table 3.19.

Dataset	Rows / Files	Columns	Description
Workout Files (time-series 1Hz)	3,024 files	15	Resampled and imputed second-by-second workout data.
Classification file + PCS riders calendar	3,022 rows	24	One row per workout; metadata, training load, summary metrics, and race results.

Table 3.19: Summary of the datasets data pre-processing and cleaning steps.

3.5 Feature Engineering

The feature engineering process is grounded in the fitness-fatigue framework commonly adopted in endurance sports science. According to this model, athletic performance at a given moment is determined by the interaction between two opposing physiological processes: long-term fitness adaptations accumulated through training, and short-term fatigue generated by recent workloads. From this perspective, different groups of features are designed to capture complementary aspects of the training process. In particular, chronic load indicators describe the long-term accumulation of training stimuli and are often interpreted as proxies for aerobic fitness. Acute load metrics instead reflect the short-term physiological stress induced by recent sessions and are associated with temporary fatigue. Additional indicators such as load variability and tapering dynamics aim to capture the temporal distribution of training stimuli, which has been shown to influence performance expression beyond total training volume alone.

Furthermore, training sessions are not treated as independent observations. Instead, they are organized into training blocks, reflecting the periodized structure typically adopted in professional cycling. This modelling choice assumes that physiological adaptation occurs over short training cycles rather than single sessions, and therefore several features are aggregated at the block level in order to represent the cumulative stimulus experienced by the athlete.

The following section is regarding the features generated during the feature engineering phase. There are three groups of variables: core features shared across all models, metrics-based features (used only in the metrics model family), and readiness-based features (specific to the readiness model), as Figure 3.6 displays.

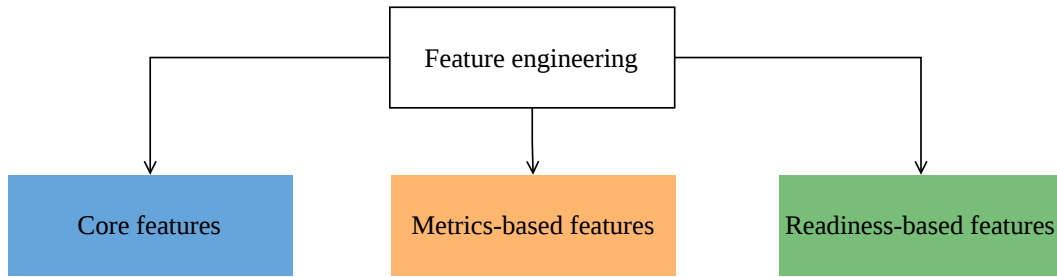


Figure 3.6: Three groups of features.

Furthermore, it is also reported the blocks construction logic, the final dataset, and the exploratory data analysis to emphasize the patterns and relationships between the created features.

3.5.1 Domain-Specific Training Metrics

TrainingPeaks uses several physiological metrics of a particular workout or portion of a workout such as Training Stress Score (TSS), IF, Chronic Training Load (CTL), Acute Training Load (ATL), and Training Stress Balance (TSB). Moreover, other important training load metrics used in endurance sports science are introduced. These include: Normalized Power (NP), TRIMPS Training Stress Score (tTSS), Lucia's Training Impulse (LuTRIMP), Acute:Chronic Workload Ratio (ACWR), kilojoules (kJ).

Normalized Power (NP) [3, 27]

Normalized Power provides an estimate of the power output that an athlete could have sustained for the same physiological cost if the effort had been perfectly constant. Unlike simple average power, NP tries to catch the nonlinear physiological response to changes in intensity and for the delayed time course with which the body reacts to rapid fluctuations in workload. NP is a core component of the TSS calculation, as already described before, and it is commonly used to quantify session intensity.

In this work, NP is computed from the power workout files (time-series) using a procedure consistent with the Training Peaks definition. The power values are first smoothed using a 30-second moving average, in order to approximate the delayed physiological response to rapid fluctuations in intensity. The smoothed power trace is then raised to the fourth power and averaged over the entire session. Finally, the fourth root of this average is taken to obtain NP. An example of this calculation applied to a real workout session is shown in Figure 3.7.

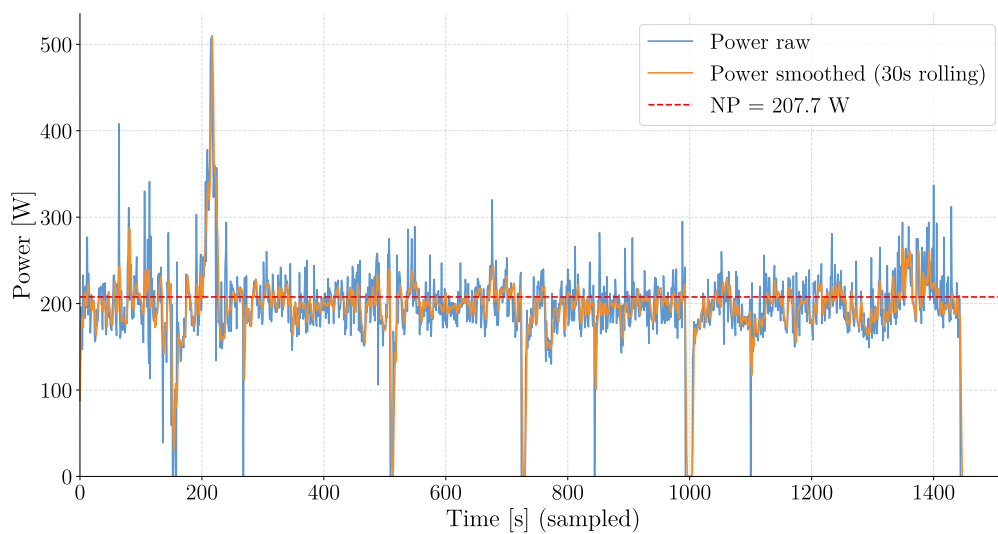


Figure 3.7: NP - A sampled 10 seconds workout session of Athlete A.

In practice, this procedure can be summarized as follows:

1. compute a 30-second rolling mean of the power signal;

2. raise all smoothed power values to the fourth power and take their arithmetic mean;
3. take the fourth root of this mean to obtain NP.

As observed in Figure 3.7, the NP value (dashed red line) remains consistently higher than the simple average of many segments, effectively accounting for the high-intensity peaks (e.g., around $t = 2000$ s) that contribute significantly to the athlete's physiological fatigue. This highlights how the fourth-power weighting emphasizes metabolic stress even when followed by periods of lower intensity or recovery.

This non-linear transformation assigns greater weight to high-intensity efforts, providing a more accurate approximation of the physiological cost compared to simple average power.

Intensity Factor (IF) [27, 28]

Intensity Factor expresses the relative intensity of a training session by comparing the NP of the workout to the athlete's Functional Threshold Power (FTP). It provides a fitness-adjusted measure of effort, allowing training sessions to be compared either within the same rider at different times of the season or between different athletes.

$$IF = \frac{NP}{FTP}$$

Since FTP represents the maximal power an athlete can sustain for approximately one hour, values of IF close to 1 indicate efforts performed near threshold intensity, while lower values correspond to sub-threshold or endurance paced workloads. The practical application of this metric is illustrated in Figure 3.8, which compares the IF distribution between structured workouts and races across all the athletes. As expected, races exhibit a significantly higher median IF, reflecting the higher relative intensity required during competition compared to general training sessions.

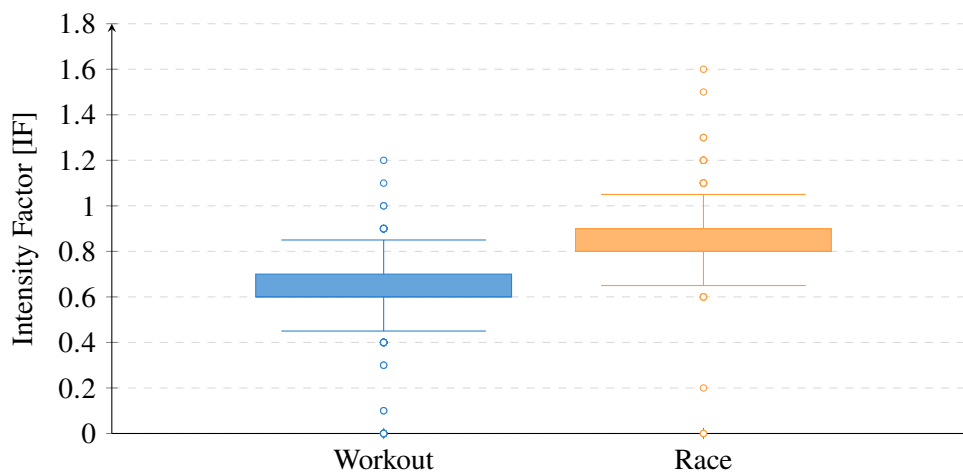


Figure 3.8: Distribution of IF by session type.

Because IF is normalized by FTP, it automatically accounts for changes in fitness over time: the same workout may tend to a different IF depending on whether the rider's threshold power has increased or decreased. IF is also used in the computation of TSS, where it provides the intensity component of the overall training load.

Training Stress Score (TSS) [4, 29, 27, 30]

Training Stress Score is a way of expressing the workload from a training session. While exercise intensity plays a key role in driving specific physiological adaptations, training frequency and duration together determine the total training volume. Importantly, intensity and volume do not act independently: their interaction shapes the overall training stimulus. At some point as intensity goes up volume must come down, and vice-versa, or else an you will become over-trained. To quantify the overall training load, *TrainingPeaks* uses the power data to calculate a TSS for every workout. Power-based TSS is the most accurate method of calculating a Training Stress Score, and is the default method for *Bike* workouts when there is enough data present to calculate.

Formally, TSS is defined as:

$$TSS = \frac{\text{sec} \times NP \times IF}{FTP \times 3600} \times 100 \quad (3.1)$$

with:

- sec - duration of the workout in seconds [s];
- NP - Normalized Power [W];
- IF - Intensity Factor given by the ratio between NP and the athlete's FTP;
- FTP - approximated as the maximal power the athlete can sustain for one hour;
- 3600 - number of seconds in one hour [s].

While individuals tend to differ in how much training they can tolerate, depending on their training background and natural abilities, the following scale can be used as an approximate guide:

- TSS less than 150 - low (recovery generally complete by following day).
- 150-300 - medium (some residual fatigue may be present the next day, but gone by 2nd day).
- 300-450 - high (some residual fatigue may be present even after 2 days).
- Greater than 450 - very high (residual fatigue lasting several days likely).

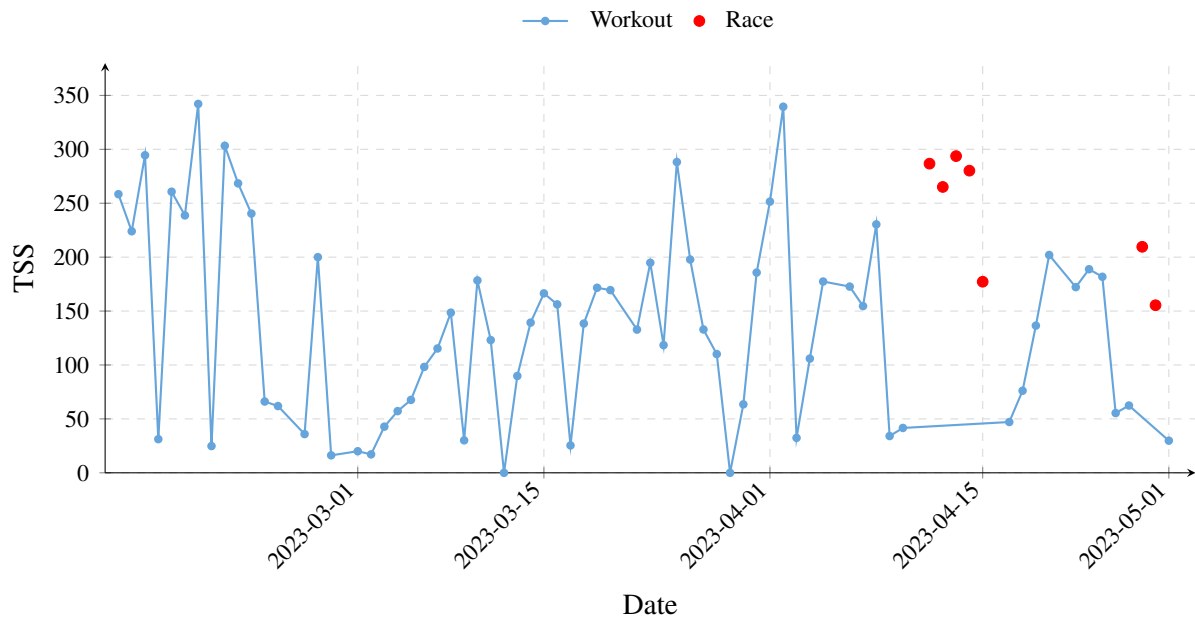


Figure 3.9: TSS of workouts and races of Athlete A for a certain period.

Figure 3.9 tracks the cumulative stress over time; it is possible to observe that race events consistently fall within the medium-to-high stress categories (150-350 TSS). This temporal distribution highlights the athlete's periodization, where high-load sessions are interspersed with recovery periods to manage residual fatigue, as defined by the *TrainingPeaks* guidelines. As well, the cumulative TSS per week or per month can be used help identify the maximum intensity and volume of training that still leads to improvements, rather than over-training.

Chronic Training Load (CTL) [31, 32]

Chronic Training Load is an estimate of the long-term training stress an athlete has accumulated and is commonly interpreted as a proxy for aerobic fitness. CTL reflects the average daily training load over a relatively long time horizon and increases when an athlete sustains higher workloads for consecutive weeks.

In endurance sports, CTL is typically computed using an Exponentially Weighted Moving Average (EWMA) of recent TSS values, with a time constant of 42 days. This formulation assigns more weight to the most recent sessions while gradually reducing the influence of older training stimuli. Formally, given a sequence of daily TSS values, CTL is defined as:

$$CTL_t = \alpha \cdot TSS_t + (1 - \alpha) \cdot CTL_{t-1}, \quad \text{with} \quad \alpha = \frac{2}{N + 1},$$

where $N = 42$ days is the chosen time constant. CTL captures the cumulative training load an athlete has been able to absorb over several weeks, and is widely used to monitor long-term fitness trends.

The relationship between daily training load and long-term fitness is visualized in Figure 3.10 which explains the same situation of the previous Figure 3.9 but with CTL.

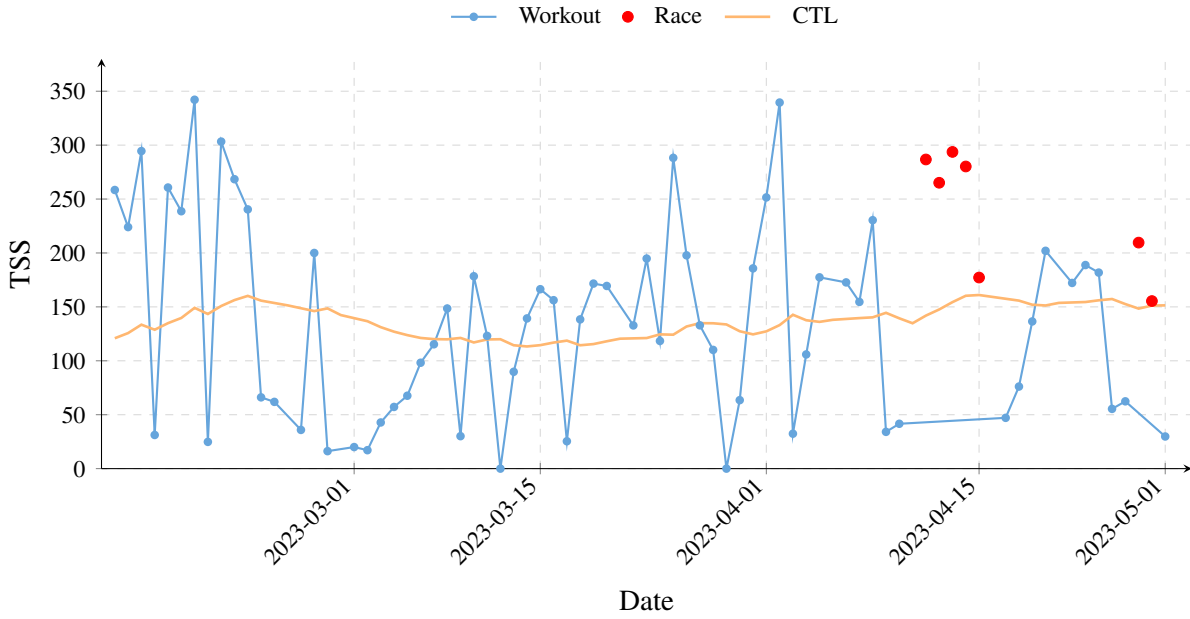


Figure 3.10: TSS and CTL of workouts and races of Athlete A for a certain period.

While the daily TSS (blue line) fluctuates significantly based on the specific session, the CTL (orange line) provides a more stable representation of the athlete's training state. As observed in the racing block during April 2023, the sustained high TSS values from consecutive competitions lead to a steady increase in CTL, reaching a peak that corresponds to the athlete's highest level of accumulated fitness for the period.

Acute Training Load (ATL) [32]

Acute Training Load represents the short-term training stress accumulated over the most recent days and is commonly interpreted as an indicator of an athlete's current level of fatigue. In endurance sports, ATL is typically computed using an EWMA of daily TSS values, with a time constant of approximately 7 days. This formulation places strong emphasis on recent sessions while rapidly diminishing the influence of older workouts. Given a sequence of daily TSS values, ATL is computed as:

$$ATL_t = \alpha \cdot TSS_t + (1 - \alpha) \cdot ATL_{t-1}, \quad \alpha = 1 - e^{-1/7},$$

where the time constant of 7 days reflects the relatively fast decay of fatigue compared with long-term fitness adaptation.

The dynamic nature of short-term fatigue is captured by the ATL trend shown in Figure 3.11.

Unlike the CTL, the ATL (teal line) responds rapidly to high-load sessions, such as the racing block in April 2023. During this period, the concentration of high-TSS events prevents the ATL from decaying, signaling a significant accumulation of acute fatigue that must be managed to avoid overreaching.

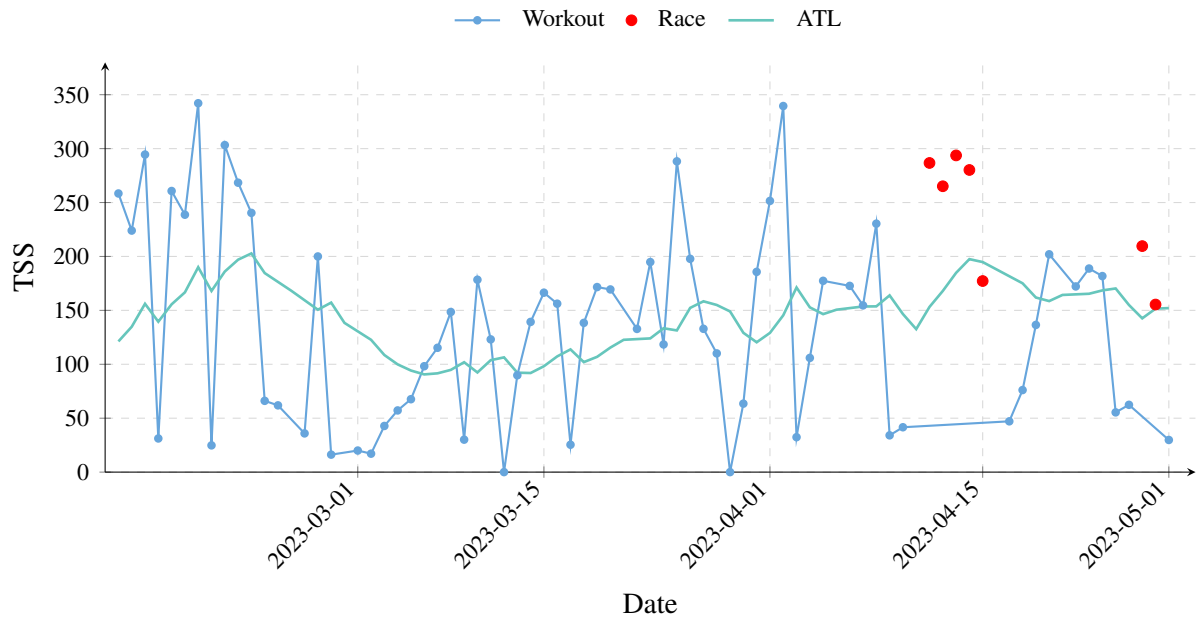


Figure 3.11: TSS and ATL of workouts and races of Athlete A for a certain period.

Training Stress Balance (TSB) [32, 33]

Training Stress Balance quantifies the relationship between an athlete's long-term training load (fitness) and short-term load (fatigue). It is defined as the difference between Chronic Training Load and Acute Training Load:

$$TSB_t = CTL_t - ATL_t$$

A positive TSB indicates that fatigue is relatively low compared with long-term fitness, a state typically associated with being rested or in a tapering phase. On the other hand, a negative TSB reflects high accumulated fatigue, which is expected during blocks of intensive training. TSB is used as a practical indicator of an athlete's readiness to perform.

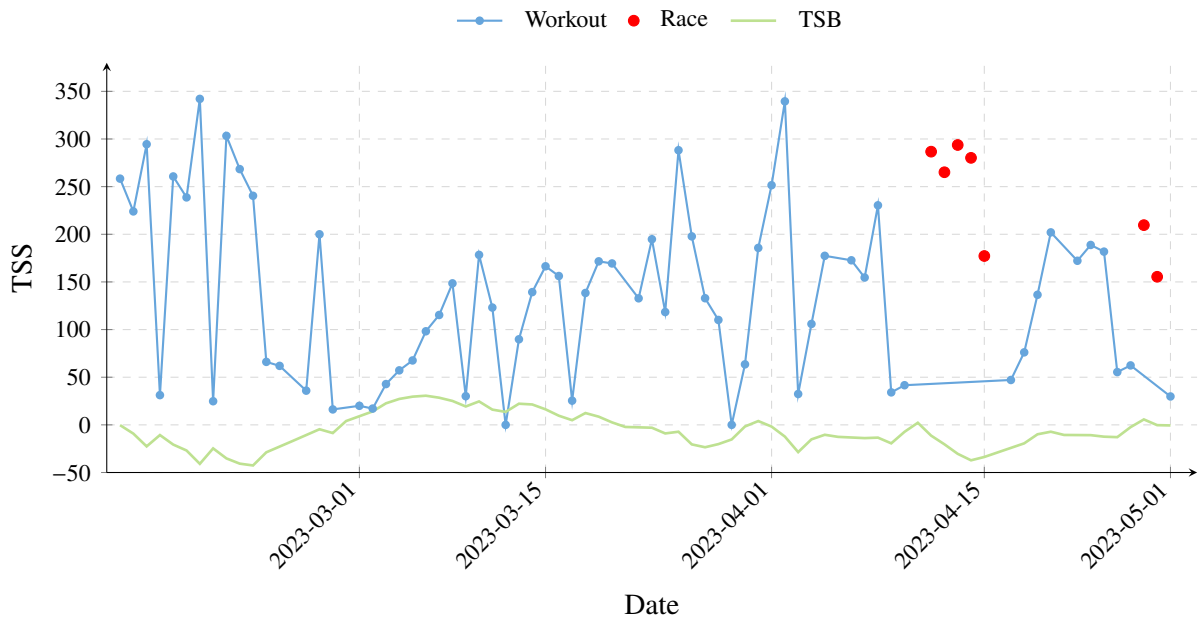


Figure 3.12: TSB of Athlete A for a certain period.

As illustrated in Figure 3.12, the TSB (green line) serves as a dynamic indicator of the athlete's balance between fitness and fatigue. During the intensive racing block in mid-April, the TSB drops significantly below

zero, reaching values near -40 , which indicates a high state of accumulated fatigue. Conversely, periods of reduced TSS allow the ATL to decay faster than the CTL, bringing the TSB back toward positive territory and signaling a state of readiness or recovery.

TRIMPS Training Stress Score (tTSS) [3, 34]

TRIMPS Training Stress Score estimates the internal training load of a session using heart-rate data. In fact, it combines the workout duration with the athlete's average heart rate and a nonlinear weighting factor that reflects the physiological strain associated with exercising at different heart-rate intensities. Unlike power-based TSS, tTSS uses only cardiovascular data and can therefore be used when power data are unavailable.

$$tTSS = \text{duration} \times HR \times f(HR) \quad (3.2)$$

where:

- duration - length of the workout in minutes,
- HR - average heart rate during the session,
- $f(HR)$ - a weighting function depending on individual heart-rate characteristics.

In this work, tTSS is computed from heart-rate data using a zone-based approach. The time spent in each heart-rate zone is multiplied by a predefined weight that reflects the increasing physiological cost of higher intensity levels. The resulting weighted sum provides the Training Impulse (TRIMP) value for the session. Specifically, the formula implemented is:

$$TRIMP = (1 \times Z1) + (2 \times Z2) + (4 \times Z3) + (6 \times Z4) + (10 \times Z5)$$

where Z1-Z5 represent the minutes spent in heart-rate Zones 1 to 5.

The TRIMP score is then rescaled to obtain tTSS:

$$tTSS = \frac{TRIMP}{250} \times 100$$

This formulation ensures that tTSS remains broadly comparable to power-based TSS values when power data are not available.

Lucia's Training Impulse (LuTRIMP) [35, 36, 37]

Lucia's Training Impulse is an internal training load metric designed to quantify the physiological strain of a workout based on the athlete's heart-rate response. Unlike the original Banister TRIMP [38], which applies an exponential weighting based on the percentage of maximum heart rate, LuTRIMP uses a simplified linear weighting system in which each heart-rate zone is assigned an integer coefficient.

$$\text{luTRIMP} = (\text{Duration}_{\text{zone 1}} \times 1) + (\text{Duration}_{\text{zone 2}} \times 2) + \dots + (\text{Duration}_{\text{zone 5}} \times 5)$$

where the five heart-rate zones are weighted with integer coefficients ranging from 1 (low intensity) to 5 (high intensity). Time spent in higher zones contributes proportionally more to the total internal load. As reported in Table 3.20, the Q1 (first quartile) for race sessions (230.00) is significantly higher than the median for workouts (156.00), confirming that even a *low-intensity* race typically exerts a higher internal load than an average training session.

Type	n	Mean	Std	Min	Q1	Median	Q3	Max
Workout	2153	167.85	108.87	0.00	72.00	156.00	248.00	543.00
Race	320	276.54	126.68	0.00	230.00	303.50	358.25	538.00

Table 3.20: Descriptive statistics of Lucia's TRIMP for workout and race sessions.

This approach makes the metric easier to compute and interpret while still capturing the progressive increase in cardiovascular stress as the intensity rises. As a result, LuTRIMP provides a practical and robust estimate of internal load, especially when heart-rate data are available but lactate-based intensity thresholds are not, as this case of study.

Acute:Chronic Workload Ratio (ACWR) [39]

The Acute:Chronic Workload Ratio expresses the relationship between short-term training load and longer-term load adaptation. It is defined as the ratio between Acute Training Load, which captures the training stress accumulated in the most recent days, and Chronic Training Load, which reflects the athlete's longer-term fitness state.

$$ACWR_t = \frac{ATL_t}{CTL_t}$$

Values greater than one indicate that short-term workload exceeds the current chronic capacity, potentially reflecting periods of intensified training. On the other hand, values below one suggest that recent training load is lower than the athlete's long-term training history, a pattern often seen during recovery phases or tapering. ACWR therefore provides a simple indicator of the balance between recent stress and long-term load tolerance.

As shown in Table 3.21, the majority of the athlete's sessions (67.2%) fall within the 1.0-1.3 range, which is often cited in sports science literature as the *sweet spot* for performance improvement without excessive injury risk. The relatively low percentage of sessions exceeding the 1.3 threshold (3.4%) suggests a well-controlled progression of training load, even during competitive phases.

ACWR Range	Race	Workout	Total	Percentage (%)
< 0.8	9	77	86	3.5
0.8 – 1.0	79	561	640	25.9
1.0 – 1.3	215	1447	1662	67.2
> 1.3	17	68	85	3.4
Total	320	2153	2473	100.0

Table 3.21: Distribution of sessions across different ACWR ranges across all the riders.

kilojoules (kJ) [37]

Kilojoules represent the total mechanical work performed by a cyclist during a training session and are commonly derived from power output data as the product of power and time. As an external load metric, kJ provide a measure of the overall energetic demands of a ride.

$$kJ = \frac{\text{Average Power (W)} \times \text{Duration (s)}}{1000}$$

Recent evidence from a study, highlights that the relationship between kJ and internal load metrics is strongly influenced by exercise intensity. In particular, the association between total kJ and physiological stress becomes weaker when a significant portion of the session is performed at high intensities, where metabolic and cardiovascular strain increases disproportionately relative to mechanical work. This suggests that kJ alone may underestimate the true internal load of intense sessions and should be interpreted alongside heart-rate or power-based intensity metrics when analyzing training load in professional cyclists.

In Figure 3.13, a subset of race observations exhibits unusually low total energy expenditure values. A closer inspection reveals that these cases are not random anomalies, but rather correspond to specific race contexts. Most low-kJ races are characterized by extremely short durations $\leq 0.3h$, consistent with prologues or neutralized stages, where limited mechanical work is required. Finally, low-kJ races often coincide with poor finishing positions, suggesting a limited active role of the athlete in the team strategy. For these reasons, low-kJ race outliers are retained in the dataset, as they reflect meaningful heterogeneity in race demands rather than data errors.

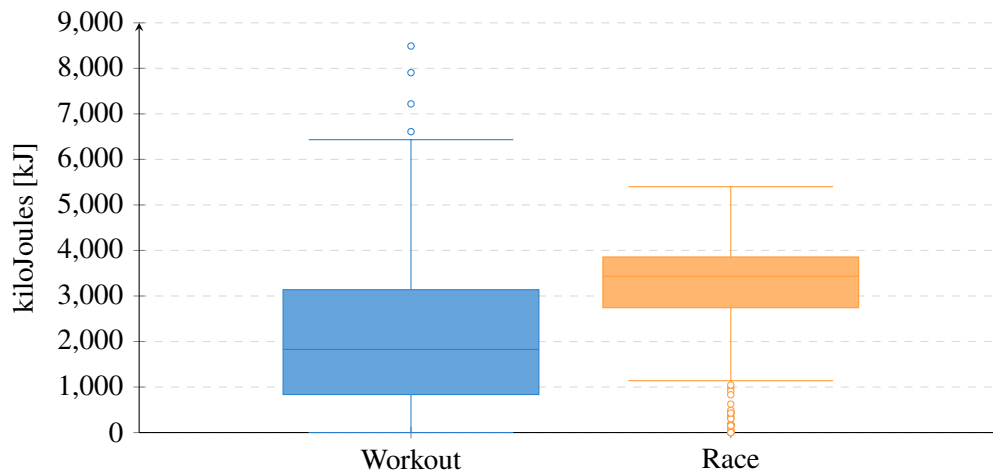


Figure 3.13: Distribution of kJ by session type.

Table 3.22 reports the main descriptive statistics for the core training load metrics included in this study, computed over all available sessions for all the athletes. The table shows the minimum, first quartile (Q1), median, third quartile (Q3), and maximum values, providing a quick overview of the scale and variability of each feature.

Feature	Min	Q1	Median	Q3	Max
TSS	0.0	35.1	105.0	192.3	444.1
IF	0.0	0.6	0.6	0.7	1.6
CTL	63.6	104.3	119.8	137.8	197.6
ATL	35.5	97.2	119.1	143.6	238.8
TSB	-62.6	-10.4	0.2	11.4	51.7
NP	0.0	207.6	240.6	281.6	1458.5
tTSS	0.0	27.6	80.8	162.8	557.6
LuTRIMP	0.0	60.0	146.0	256.0	543.0
ACWR	0.5	0.9	1.0	1.1	1.4
kJ	0.0	661.0	1766.0	3221.0	8490.0

Table 3.22: Descriptive statistics for the main training load metrics across all sessions.

As shown in Table 3.22, the TSS values exhibit a wide range, with a median around 105 points and maximums exceeding 400 points, reflecting the distinction between regular training sessions and high-intensity races. The IF distribution indicates that most sessions are performed at sub-threshold or endurance intensity, while occasional peaks correspond to short-duration maximal efforts.

CTL and ATL illustrate the expected temporal dynamics of long-term fitness and short-term fatigue: CTL varies more slowly and captures accumulated training load, whereas ATL responds quickly to recent sessions. Furthermore, TSB ranges from negative to positive values, highlighting periods of accumulated fatigue versus recovery or tapering.

NP and kJ confirm the large variability in absolute workload between sessions, with extreme values associated with sprint efforts, long endurance rides, or race stages. Internal load metrics such as tTSS and LuTRIMP provide complementary insight into the physiological cost of each session, especially when heart-rate data are available. Finally, ACWR remains mostly around 1, indicating that the short-term training load is generally well balanced relative to long-term capacity.

To conclude, Table 3.23 reports a sample of the dataset, illustrating the structure, scale, and typical values of the engineered features. All features are initially computed at the level of each available row in the classification file, corresponding to individual workout instances.

WorkoutDay	rider	...	NP	IF	TSS	CTL	ATL	TSB	tTSS	LuTRIMP	ACWR	kJ
2023-02-10	Athlete A	...	290.1	0.7	258.4	120.9	121.2	-0.3	223.2	343	1.0	3336
2023-02-11	Athlete A	...	237.0	0.6	224.0	125.7	134.8	-9.2	152.0	368	1.1	4447
2023-02-12	Athlete A	...	270.6	0.7	294.5	133.5	156.1	-22.6	210.4	401	1.2	4597
2023-02-13	Athlete A	...	201.1	0.5	31.2	128.8	139.5	-10.7	28.8	72	1.1	1212
2023-02-14	Athlete A	...	312.7	0.8	260.7	134.9	155.6	-20.7	171.6	295	1.2	3348
2023-02-15	Athlete A	...	247.7	0.6	238.7	139.7	166.7	-26.9	152.8	363	1.2	4524
2023-02-16	Athlete A	...	272.0	0.7	342.1	149.1	190.0	-40.9	240.0	462	1.3	5835
2023-02-17	Athlete A	...	200.6	0.5	24.8	143.4	168.0	-24.7	22.8	57	1.2	538
2023-02-18	Athlete A	...	300.9	0.8	303.3	150.8	186.0	-35.2	202.0	350	1.2	4198
2023-02-19	Athlete A	...	259.6	0.7	268.4	156.3	197.0	-40.7	153.6	371	1.3	5024
...	...	...	...	...	...	...	...	...	...	...	...	...

Table 3.23: Excerpt of the classification dataset; only a subset of sessions and computed features are shown.

3.5.2 Training Blocks Construction

Before introducing the entire feature engineering process, it is important to describe the logic of the data structure at this step. The time-series workout files are not used directly as inputs to the prediction models: in fact, they serve only to compute a number of training-related metrics, as described in the previous section. The core dataset used for modelling is instead the *classification file*: a metadata table in which each row corresponds to a single activity and includes summary statistics, training-load metrics, and all derived variables used in the prediction phase. An excerpt of the file is reported in Table 3.24.

WorkoutDay	WorkoutType	rider	TSS	Type	...	LuTRIMP	ACWR	kJ
2023-02-10	Bike	Athlete A	258.4	workout	...	343	1.0	3336
2023-02-11	Bike	Athlete A	224.0	workout	...	368	1.1	4447
2023-02-12	Bike	Athlete A	294.5	workout	...	401	1.2	4597
2023-02-13	Bike	Athlete A	31.2	workout	...	72	1.1	1212
2023-02-14	Bike	Athlete A	260.7	workout	...	295	1.2	3348
...	...	...	...	...	...	...	...	...

Table 3.24: Excerpt of the classification dataset showing session-level metadata.

In particular, each rider's activities are first sorted chronologically by date (*WorkoutDay*) and classified as either *race* or *non-race* sessions, as Table 3.25 shows.

WorkoutDay	WorkoutType	rider	TSS	Type	...	LuTRIMP	ACWR	kJ	is_race
2023-02-10	Bike	Athlete A	258.4	workout	...	343	1.0	3336	False
2023-02-11	Bike	Athlete A	224.0	workout	...	368	1.1	4447	False
2023-02-12	Bike	Athlete A	294.5	workout	...	401	1.2	4597	False
2023-02-13	Bike	Athlete A	31.2	workout	...	72	1.1	1212	False
2023-02-14	Bike	Athlete A	260.7	workout	...	295	1.2	3348	False
...	...	...	...	...	...	...	...	...	...
2023-04-12	Bike	Athlete A	286.7	race	...	370	1.1	3883	True
2023-04-13	Bike	Athlete A	265.1	race	...	340	1.1	3857	True
2023-04-14	Bike	Athlete A	293.7	race	...	294	1.2	4006	True
2023-04-15	Bike	Athlete A	280.2	race	...	291	1.2	4282	True
2023-04-16	Bike	Athlete A	177.2	race	...	176	1.2	2412	True

Table 3.25: Excerpt of the classification dataset showing the boolean variable *is_race* identifies race sessions.

Training sessions in endurance sports are rarely independent events. Instead, they are typically organized according to periodized training cycles, where groups of consecutive sessions are designed to produce a cumulative physiological stimulus followed by recovery phases. Because of this, the performance observed in a race is not determined by a single training session but by the combined effect of multiple recent sessions and their temporal organization.

To account for this dependency structure, training data are organized into training blocks. Each block represents a short training cycle preceding a race and aggregates the training load accumulated during that period. This modelling choice reflects the assumption that physiological adaptation emerges from the cumulative stimulus applied across multiple sessions rather than from isolated workouts.

Aggregating sessions at the block level also reduces the noise associated with day-to-day fluctuations in training execution and allows the model to capture higher-level training patterns, such as load accumulation, recovery phases, and tapering dynamics before competition.

The block structure therefore serves both as a physiological abstraction of periodized training and as a practical modelling strategy to represent the temporal dependence between training sessions.

In Figure 3.14, non-race sessions correspond to training activities and are represented in the diagram by blue squares, while race sessions correspond to competitive events and are represented by red circles. Then, a new block identifier (`block_id`) is created whenever at least **seven consecutive non-race days** occur between two race events. As illustrated in Figure 3.14, each block is defined by a `block_start` and a `block_end` date and contains at least one *race target*. The threshold of seven consecutive non-race days is adopted as a practical heuristic to identify coherent preparation phases preceding competition. This duration corresponds to a short micro-cycle in endurance training, where training load is typically organized on a weekly basis. Nevertheless, the exact threshold remains a modelling assumption (see Chapter 6).

So, for each (`rider`, `block_id`) combination, the following elements are defined:

- `block_id`: the identifier of the training block;
- `block_start`: the first day of the block;
- `block_end`: the date of the last race contained in that block;
- `race_target`: the target race associated with the block.

If two races occur within fewer than seven training days, they are considered part of the same block, based on the fact that no meaningful preparatory period could be done between them. So, a block is reset only when a streak of at least seven consecutive non-race days is observed.

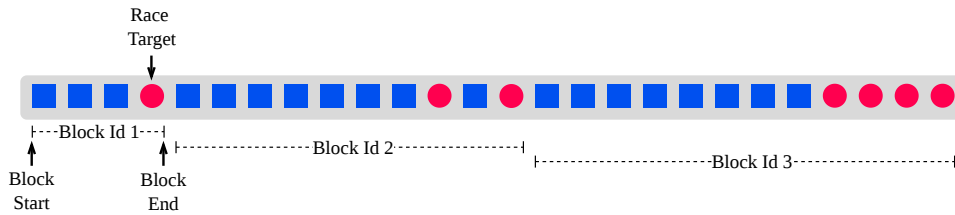


Figure 3.14: Blocks construction logic.

When multiple races are included in the same block, the targets are defined according to their chronological order: the first target corresponds to the earliest race in time, followed by subsequent races in sequence. After each race, the corresponding activity is also included in the subsequent training block and treated as part of the overall workload of that block. In this sense, the race can be interpreted as a specific form of training session in the same block, as illustrated in Figure 3.15 for block ID number 3.

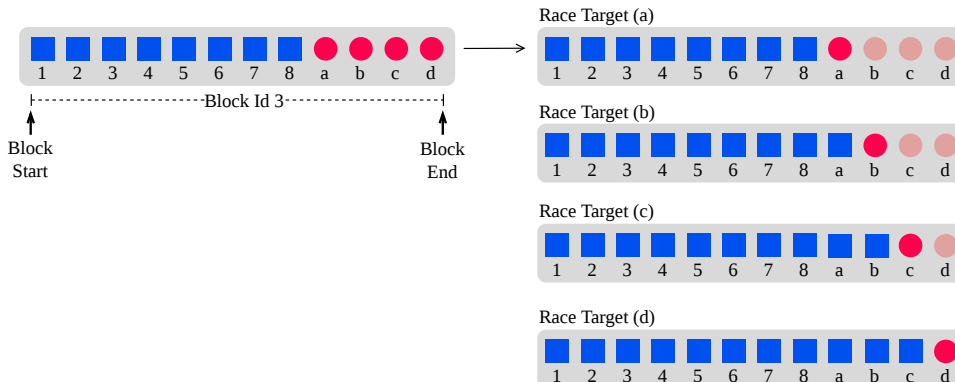


Figure 3.15: Block ID 3.

This modeling choice allows the race effort to directly contribute to the estimation of the rider’s physiological condition ahead of the next competition. For this reason, each block is defined to end on the date of the last race included in that block, ensuring that all relevant competitive efforts are incorporated into the condition assessment prior to the following event. Additionally, training sessions from the last part of the season without a set of races are removed, since they do not have any predictive purpose. Then, `block_start` and `block_end` are dropped to obtain a unique variable `block_days_length` which the number of days included in the block. This concept is applied to each row of the classification file to obtain for each row the dataset, as the Figure 3.26 shows.

rider	WorkoutDay	Type	block_id	block_start	block_end	race_day	result	is_race	...
Athlete A	2023-02-10	workout	1	2023-02-10	2023-04-16	2023-04-12	0	False	...
...	...	...	...	...	...	...	...	...	...
Athlete A	2023-04-11	workout	1	2023-02-10	2023-04-16	2023-04-12	0	False	...
Athlete A	2023-04-12	race	1	2023-02-10	2023-04-16	2023-04-13	77.0	True	...
Athlete A	2023-04-13	race	1	2023-02-10	2023-04-16	2023-04-14	112.0	True	...
Athlete A	2023-04-14	race	1	2023-02-10	2023-04-16	2023-04-15	105.0	True	...
Athlete A	2023-04-15	race	1	2023-02-10	2023-04-16	2023-04-16	37.0	True	...
Athlete A	2023-04-16	race	1	2023-02-10	2023-04-16	0	20.0	True	...
Athlete A	2023-04-19	workout	2	2023-04-17	2023-04-30	2023-04-29	0	False	...
...	...	...	...	...	...	...	...	...	...
Athlete A	2023-04-28	workout	2	2023-04-17	2023-04-30	2023-04-29	0	False	...
Athlete A	2023-04-29	race	2	2023-04-17	2023-04-30	2023-04-30	2	True	...
Athlete A	2023-04-30	race	2	2023-04-17	2023-04-30	0	2	True	...
Athlete A	2023-05-01	workout	3	2023-05-01	2023-05-14	2023-05-14	0	False	...
Athlete A	2023-05-02	workout	3	2023-05-01	2023-05-14	2023-05-14	0	False	...
Athlete A	2023-05-03	workout	3	2023-05-01	2023-05-14	2023-05-14	0	False	...
Athlete A	2023-05-04	workout	3	2023-05-01	2023-05-14	2023-05-14	0	False	...
Athlete A	2023-05-05	workout	3	2023-05-01	2023-05-14	2023-05-14	0	False	...
Athlete A	2023-05-06	workout	3	2023-05-01	2023-05-14	2023-05-14	0	False	...
Athlete A	2023-05-08	workout	3	2023-05-01	2023-05-14	2023-05-14	0	False	...
Athlete A	2023-05-10	workout	3	2023-05-01	2023-05-14	2023-05-14	0	False	...
Athlete A	2023-05-11	workout	3	2023-05-01	2023-05-14	2023-05-14	0	False	...
Athlete A	2023-05-12	workout	3	2023-05-01	2023-05-14	2023-05-14	0	False	...
Athlete A	2023-05-14	race	3	2023-05-01	2023-05-14	0	81	True	...
...	...	...	...	...	...	...	...	...	...

Table 3.26: Excerpt of workouts and races for Athlete A. Only selected columns and rows are shown.

Then, the outcome of this process is a structured dataset with blocks where each training period (each row) corresponds to a specific race day, containing the actual result and the Top-20 label for that day. All the features are then computed from the rider’s training data preceding the corresponding race day. An excerpt of the final dataset is reported in the Figure 3.27.

rider	block_id	race_day	result	top20	block_days_length	days_since_last_race	...
Athlete A	1	2023-04-12	77.0	0	56	0	...
Athlete A	1	2023-04-13	112.0	0	57	1	...
Athlete A	1	2023-04-14	105.0	0	58	1	...
Athlete A	1	2023-04-15	37.0	0	59	1	...
Athlete A	1	2023-04-16	20.0	1	60	1	...
Athlete A	2	2023-04-29	2.0	1	9	13	...
Athlete A	2	2023-04-30	2.0	1	10	1	...
Athlete A	3	2023-05-14	81.0	0	10	14	...
...	...	...	...	...	...	...	...

Table 3.27: Excerpt of race block summary for Athlete A. Only selected columns and rows are shown.

Note for the reader: although the models discussed in Subsections 4.3.1 and 4.3.2 in Chapter 4 share the same block-generation logic, they differ in the type of features used as predictors. A subset of variables is common to all models in Subsection 3.5.3, while the remaining features are specific to each modelling approach: the metrics-based model employs the load-related features described in Subsection 3.5.4, while the readiness-based model uses the readiness-derived indicators presented in Subsection 3.5.5.

3.5.3 Core Features

Both the metrics-based models and the readiness-based model have a common set of core features. These variables describe the training block, the accumulated workload and intensity before the race, and the athlete's historical performance. Each row in the dataset, as described before, corresponds to a race day, while all features are computed exclusively from the training sessions contained in the corresponding block.

Block structure. These features characterize the temporal structure of each preparation period:

- `block_days_length`: number of days included in the block;
- `days_since_last_race`: recency of the previous race.

Workload and volume. These variables quantify the external load accumulated during the block:

- `km_sum_block`: total distance covered;
- `kJ_sum_block`: mechanical work produced;
- `LuTRIMP_sum_block`, `TRIMP_sum_block`: internal load based on heart-rate-derived metrics;
- `altitude_gain_sum_block`: cumulative elevation gain;
- `load_monotony_time`: relationship between the mean and variability of daily training time.

Block-level intensity. These features summarize the physiological intensity of training:

- `CAD_mean_block`, `CAD_std_block`: average and variability of cadence;
- `HR_mean_block`, `NP_mean_block`, `PWR_mean_block`: average heart-rate, Normalized Power, and mean power over the block.

Historical intensity across blocks. These information capture longer-term conditioning trends by aggregating information from previous blocks:

- `HR_mean_prev_blocks`, `NP_mean_prev_blocks`, `PWR_mean_prev_blocks`: rolling averages of heart rate, Normalized Power, and power output computed across preceding blocks.

Performance history. These variables summarize the rider's competitive performance before the target race:

- `top20_count_cum.before`: cumulative number of Top-20;
- `top20_rate_cum`: proportion of career races finished in the Top-20;
- `mean_result_cum.before`: average finishing position up to that date;
- `success_rate_last3`: fraction of Top-20 results in the last three races;
- `days_since_last_top20`: days elapsed since the most recent Top-20 result.

Taper indicators. These features capture how the final days before the race differ from the overall block:

- `taper_distance_last7_over_block`: proportion of total block distance accumulated in the last seven days;
- `taper_HR_high_last7_over_block`: fraction of time spent in high heart-rate zones during the taper period.

Physiological and mechanical variables - cadence, heart rate, power output, and altitude - are extracted directly from the workout time-series and aggregated at block level for each row in the metadata file. Cadence summarizes neuromuscular demands and pacing strategy, while heart rate provides a classical measure of internal load and cardiovascular strain. Power metric quantifies external mechanical work, which is strongly linked to overall training intensity in cycling. Finally, total altitude gain captures the additional muscular and metabolic cost associated with climbing.

Furthermore, to account for short-term pre-race adjustments, taper-related features are included, such as the proportion of block distance accumulated in the last seven days and the fraction of high-intensity heart-rate time during taper. These indicators capture how the final days before competition differ from the overall preparation period, reflecting widely studied tapering strategies in endurance sports [40]. In addition, training monotony - defined as the ratio between mean daily load and its variability - is computed to describe how regular or variable the training stimuli are across the block. High monotony is associated with accumulated fatigue, while greater variability is typically considered more favorable [37]. Together, taper and monotony measures provide important contextual information about the structure and variability of the final preparation phase. More general considerations are reported in Subsection 3.5.6.

Together, these core features provide a general overview: they describe how the rider trains in the period before the event, how the block is demanding and intense for the athlete, and it is an attempt to combine the historicity of the season with how the rider had been performing recently. For this reason, they are features in common by both type of models.

From this, the two modelling approaches extend the feature space in different directions: the metrics-based models add training-load metrics such as TSS, CTL, ATL and TSB, while the readiness-based model incorporates all the indicators derived from the readiness framework.

3.5.4 Features Model with Metrics

The first modelling approach expands the core feature set by introducing a series of training-load metrics commonly used in endurance sports. These variables quantify both the short-term and long-term load dynamics and are derived from the *TrainingPeaks* features described in Subsection 3.5.1. The goal of these features is to represent the rider's physiological state using an overview of workload accumulation, fatigue, fitness, and the balance between chronic and acute load.

Block-level training load. These features summarize the internal and external load accumulated within the training block:

- `TSS_sum.block`: total Training Stress Score accumulated in the block;
- `tTSS_sum.block`: heart-rate-based TRIMP proxy when power-based TSS is unavailable;
- `IF_mean.block`: mean Intensity Factor over all sessions in the block.

Chronic and acute load dynamics. These variables capture the long-term and short-term training response, following an EWMA formulation:

- `CTL_mean.block`: Chronic Training Load (42-day EWMA);
- `ATL_mean.block`: Acute Training Load (7-day EWMA);
- `TSB_mean.block`: Training Stress Balance, defined as $CTL - ATL$;
- `TSB_last`: value of TSB on the race day;
- `ACWR_TSS_7_28`: ratio between acute (7-day) and chronic (28-day) workload.

Load stability and distribution. These features quantify the variability of training load within the block:

- `monotony_TSS`: ratio between mean daily TSS and its standard deviation, indicating load variability;
- `strain_TSS`: product of total block TSS and its monotony, representing cumulative stress.

Longer-term load windows. To capture extended trends in training load, additional rolling-window metrics are computed:

- `tsb_mean_30d`, `tsb_mean_45d`: mean TSB over 30 and 45 days;
- `atl_mean_30d`, `atl_mean_45d`: mean ATL over 30 and 45 days;
- `tss_sum_45d`: cumulative TSS accumulated over the previous 45 days.

3.5.5 Features Model with Readiness

The second modelling approach extends the core feature set by introducing a collection of readiness-based indicators. These features summarize not only the accumulated training load, but also how the rider has responded to that load over time. Unlike the metrics-based model, which uses classical load quantities, this model incorporates a data-driven readiness score describing short-term dynamics, recent form, and block-to-block trends. The goal is to approximate the rider's physiological *state* before each race, rather than modelling the training load itself. The concept and the idea behind the readiness framework and its motivation are discussed in detail in Section 4.4 in Chapter 4. The features introduced in this subsection represent its practical implementation.

Readiness level before the race. These variables quantify the overall readiness of the rider just before race day:

- `readiness_before`: readiness level on the day preceding the race;
- `readiness_mean_block`: mean readiness across the training block;
- `readiness_std_block`: variability of readiness within the block;
- `readiness_trend`: linear trend of readiness values across the block.

Short-term readiness dynamics. These features capture rapid fluctuations in the rider's physiological state:

- `readiness_mean_last7`: average readiness during the final seven days;
- `readiness_ema5_last`: short-term exponential moving average (half-life of five days);
- `mean_readiness_last28`: mean readiness over the last 28 days;
- `ACWR_readiness_7_28`: acute-to-chronic readiness ratio (7 vs. 28 days).

Taper-related readiness features. These variables describe how final pre-race freshness differs from the rest of the block:

- `readiness_last7_over_rest`: ratio between readiness in the last seven days and the rest of the block;
- `readiness_last7_minus_rest`: difference between recent and block-wide readiness levels;
- `readiness_taper_last7_over_block`: proportion of total block readiness accumulated during the taper period.

Stability and temporal structure. These indicators capture the temporal regularity and medium-term dynamics of readiness:

- `readiness_monotony`: ratio of mean readiness to its standard deviation, indicating the stability of the rider's physiological state;
- `readiness_ewm_hl14d`: exponential moving average (half-life of 14 days) representing medium-term readiness;
- `readiness_mean_prev_blocks`: average readiness across previous blocks;
- `mean_readiness_block_last3`: mean readiness across the three most recent blocks;
- `delta_readiness_vs_prev_block`: difference between current and previous block readiness.

Performance-aware temporal features. Finally, the readiness-based model also incorporates performance history metrics to relate physiological state to race outcomes:

- `ewm_top20_hl60d`: exponential moving average of Top-20 outcomes with a 60-day half-life;
- `success_rate_last3`: proportion of Top-20 finishes in the last three races.

By replacing traditional load metrics with readiness-derived indicators, this model aims to capture the rider's adaptive response to training rather than the training load itself. Readiness features integrate more complex dimensions of the athlete's condition-fatigue, freshness, load absorption, and recent performance into a unique variable. It tries to provide an alternative to the metrics-based model.

3.5.6 General Considerations

While classical metrics such as TSS, IF, CTL, ATL, and TSB capture fundamental aspects of the response to training, endurance sports science has increasingly emphasized the importance of trend- and variability-based indicators that describe how load evolves over time. Among these, tapering metrics quantify the progressive reduction in training volume prior to competition, a strategy shown to improve performance by optimizing recovery and restoring physiological freshness [40]. Monotony and strain further characterize the structure of the training cycle: monotony captures day-to-day variability in load, with higher values indicating repetitive, unvaried sessions that increase the risk of accumulated fatigue; strain combines monotony with total load, reflecting the overall stress imposed on the athlete [41]. In addition, EWMA provide a mathematically principled way to track short and long-term changes in training load by weighting recent sessions more heavily than older ones, offering a dynamic alternative to fixed-window metrics such as CTL or ATL [42].

Despite the widespread use of heart rate (HR) for quantifying internal intensity, its application in outdoor cycling presents several practical limitations. Although HR correlates strongly with exercise intensity in controlled laboratory settings, this relationship is significantly less in the real world. HR can drift upwards independently of actual load due to thermal stress, fatigue, dehydration, body position on the bike, and cumulative cardiovascular strain. As a consequence, HR may overestimate or underestimate effort during prolonged or variable sessions, especially in too hot or too cold environments [43, 44].

Power-based training addresses many of these issues, offering a direct measure of mechanical output that is unaffected by hydration, temperature or cardiovascular lag. However, power data in outdoor cycling tend to be highly variable on a second-by-second basis, due to continuous changes in gradient, wind, drafting conditions, and terrain. This makes it challenging for athletes to maintain precise power targets during short intervals, and may complicate their ability to remain in prescribed intensity zones [45]. For this reason, several coaching frameworks recommend prioritizing HR during low-to-moderate intensity training (zones 1–3, corresponding respectively to the moderate and heavy physiological domains), while relying more heavily on power data at higher intensities (zones 4–5, severe intensity domain), where the structured nature of intervals reduces variability and increases the reliability of power targets [46].

Together, these considerations highlight the need for a complementary approach that integrates both HR and power-derived metrics, combined with trend-based indicators such as monotony, strain, taper, and EWMA. Such a multidimensional perspective provides a more robust characterization of training load, allowing for both the detection of patterns (e.g., excessive monotony or rising strain) and the identification of readiness trends before a competition.

3.6 The Dataset

After the feature engineering phase, the resulting dataset used for modelling is structured at the race level. Each observation corresponds to a specific race day for a given rider and represents the prediction unit used in the modelling framework. All features associated with a race are computed only from the training sessions contained in the corresponding block, following the block-construction logic illustrated in Subsection 3.5.2. This restriction ensures that only information available before the race is used, preventing temporal data leakage in the prediction task.

The final modelling dataset is constructed at the race level. Each observation corresponds to a race instance associated with the training block that precedes it. The prediction represents a race-day performance outcome together with the set of training-derived features describing the preparation phase leading to that event. This

representation reflects the assumption that race performance emerges from the cumulative effects of recent training rather than from isolated training sessions. So, the model learns relationships between aggregated training patterns in a block and the performance observed in the corresponding race.

In particular, the initial classification file (one row per workout) is transformed now into a race-level dataset (one row per target race, one for each type model): two datasets containing the block-level features are available after completing all the processing steps described, as reported in Table 3.28.

Dataset	Rows	Columns
Data Metrics Blocks	314	44
Data Readiness Blocks	314	45

Table 3.28: Summary of the final datasets.

Furthermore, Table 3.29 shows an example of five race-day rows, illustrating the race-level structure from the data readiness blocks file. The completed set of features for both data metrics blocks and data readiness blocks files is reported in Appendix A.4.

rider	block_id	race_day	result	top20	block_days_length	days_since_last_race	...
...	...	...	...	...	...	...	...
Athlete A	1	2023-04-16	20.0	1	60	1	...
Athlete A	2	2023-04-29	2.0	1	9	13	...
Athlete A	2	2023-04-30	2.0	1	10	1	...
Athlete A	3	2023-05-14	81.0	0	10	14	...
Athlete A	4	2023-06-04	111	0	15	21	...
...	...	...	...	...	...	...	...

Table 3.29: Excerpt of the final race-level readiness dataset.

Both modelling datasets contain:

- 314 race-day observations across the five Team Coop-Repsol riders;
- one row per (rider, race_day) pair;
- the race `result` (finishing position) and binary `top20` outcome;
- the core features described in Subsection 3.5.3;
- the metrics-based features (Subsection 3.5.4) or the readiness-based features (Subsection 3.5.5) depending on the model.

3.6.1 Target variable

For each race instance, the target variable represents where the rider achieved a strong performance in that event. This binary outcome is derived from race results using the criterion described in Subsection 5.6.2, which converts raw rankings into a classification label indicating successful performance. The modelling task can then be formulated as a binary classification problem in which the objective is to estimate the probability that a rider will achieve a strong performance in an upcoming race given the characteristics of the preceding training block.

To ensure the validity of the predictive framework, the feature set is restricted to information available before the race takes place. All variables used by the model as inputs are computed only from training sessions that occur in the training block preceding the target race.

Race-day information and any data recorded after the start of the competition are excluded from the feature construction process. This restriction prevents information leakage and ensures that the model simulates a realistic prediction scenario in which race readiness must be inferred from past training data only.

3.6.2 Threats to validity

Temporal leakage A first potential threat concerns temporal leakage, which could occur if features incorporate information from future sessions relative to the target race. This risk is mitigated by the block construction procedure, which ensures that all features are computed strictly from sessions preceding the race: no post-race information is included, preserving the temporal ordering and the causal interpretation of the features.

Repeated riders across splits A second limitation relates to the presence of repeated riders across the dataset. Since multiple races are associated with the same athlete, observations are not fully independent. Individual training habits, physiological characteristics, and performance profiles may introduce dependencies between samples, potentially increase model performance estimates when the same athlete appears in both training and test sets. While this is inherent to datasets of this type, it should be considered when interpreting generalization results to unseen athletes.

Single-team bias Finally, the dataset is derived from a single professional cycling team. Although this ensures a consistent training environment and enhances ecological validity, it may limit the generalization of results to teams with different training philosophies, competitive levels, or athlete characteristics. Therefore, caution is advised when extrapolating findings beyond the studied team.

3.6.3 Feature comparison between datasets

A comparison of the two datasets highlights:

- 29 common features, corresponding to the core features;
- 15 metrics-specific features, including TSS-based, CTL, ATL, TSB windows, and load monotony descriptors;
- 16 readiness-specific features, including readiness dynamics, taper indicators, and medium-term trends.

A full overview of all features and their classification into core, metrics-specific, and readiness-specific groups is provided in Appendix A.3.

3.6.4 Data quality checks

Both datasets are also validated through an integrity check:

- all numeric features fall within plausible physiological ranges with no missing values;
- the keys (`rider`, `race_day`) match exactly between the two datasets, confirming perfect alignment.

3.7 Exploratory Data Analysis

The goal of this exploratory analysis is to characterize the structure and the distributional properties of the dataset before modelling. In particular, this section examines the availability of race-day observations per rider, the class imbalance of the prediction target, the temporal organization of training blocks, and the relationships between key workload variables. Understanding these aspects provides essential context for the modelling choices introduced in Chapter 4, such as the use of block-level features and personalized thresholds. The following subsections summarize the main patterns observed in the data and motivate the methodological decisions adopted later in the thesis.

3.7.1 Dataset Overview

As shown in Figure 3.16, the number of available race-day observations varies across riders. As already mentioned, only the 2023 season is available for Athlete D, which explains the lower number of race-day entries.

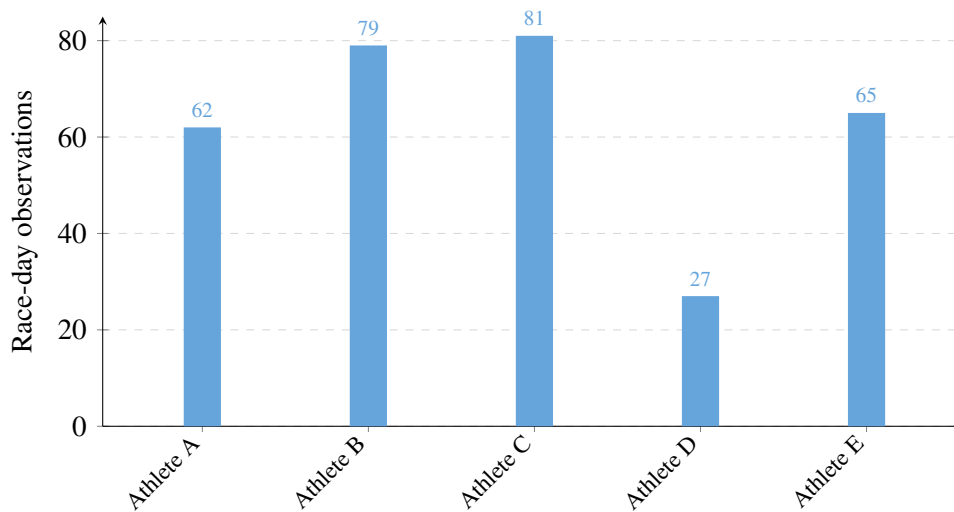
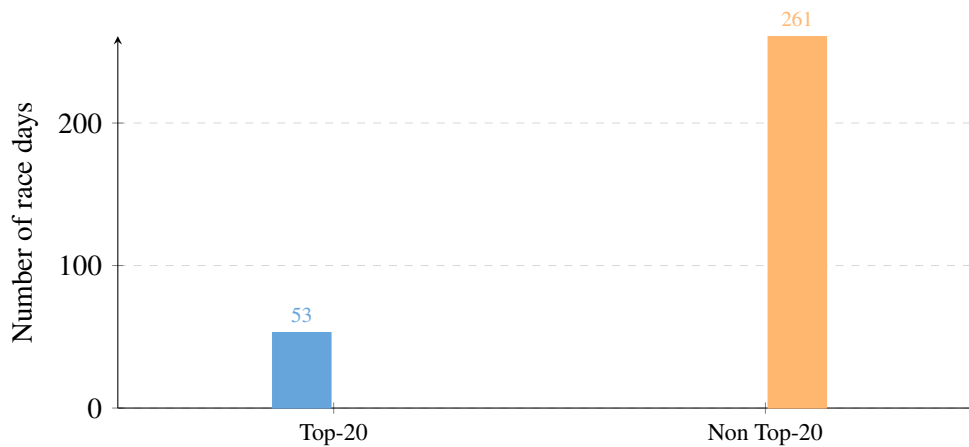


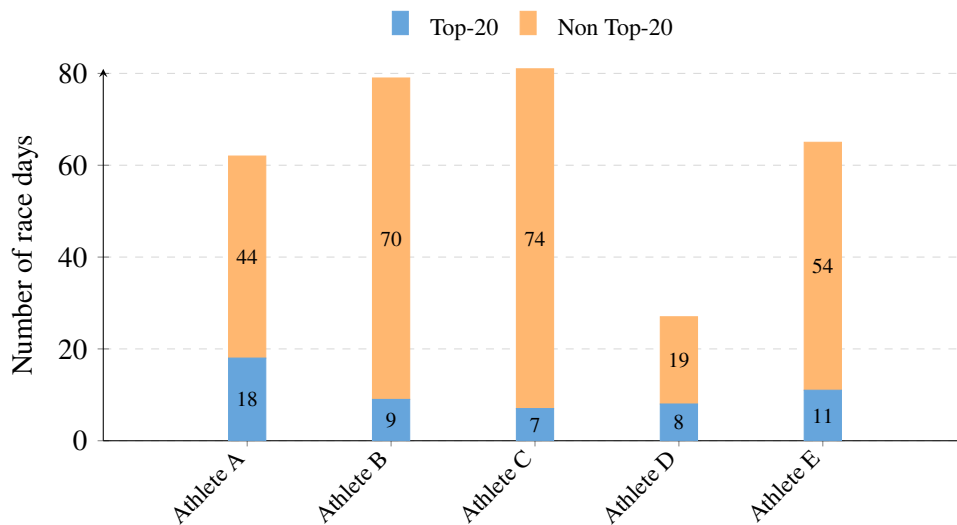
Figure 3.16: Race-day observations per rider.

Class distribution

The prediction target is finishing in the Top-20, a proxy for strong performance. The class is imbalanced, with fewer Top-20 outcomes than Non-Top-20, as shown overall and per rider in Figure 3.17.



(a) Overall class distribution.



(b) Class distribution per rider.

Figure 3.17: Overall and per-rider class distribution of Top-20 and Non Top-20 race days.

Panel (a) shows the global ratio between Top-20 and Non-Top-20 race days, while panel (b) highlights how this imbalance is present for all riders. This class imbalance problem motivates the use of techniques such as SMOTE and personalized thresholding (Youden index) to calibrate better the model, which are discussed in Chapter 4.

3.7.2 Temporal Structure of Training Blocks

The distribution of block lengths, reported in Figure 3.18, is predominantly concentrated in the 2-4 week range, reflecting the typical training structure of the riders.

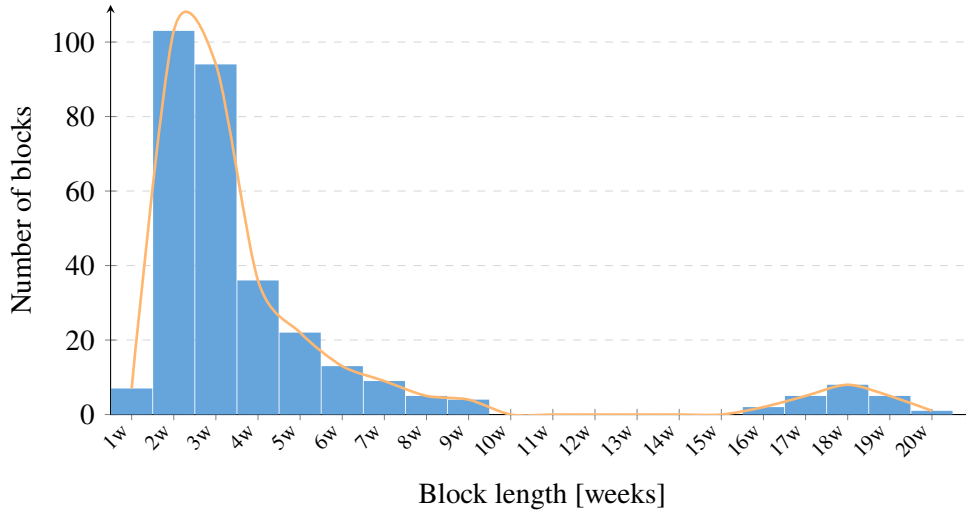


Figure 3.18: Distribution of training block lengths grouped into weekly classes.

Interestingly, there is a complete absence of blocks lasting 10-15 weeks. This gap is consistent with the competitive calendar: during the racing season, riders almost never spend more than 5-7 weeks without competing, while blocks exceeding 11 weeks occur only during the winter off-season (November-January). This explains both the cluster of very long blocks around 16-20 weeks and the zeros observed between 10 and 15 weeks.

3.7.3 Relationships Between Key Variables

Block TSS and Mechanical Work (kJ). Block-level TSS and total mechanical work (kJ) have a positive relationship, as reported in Figure 3.19. Higher external work corresponds to higher accumulated TSS across riders, although the relative slope differs slightly depending on individual training habits. This confirms that block TSS can be interpreted as a good integrated measure of the overall external load completed in a training block.

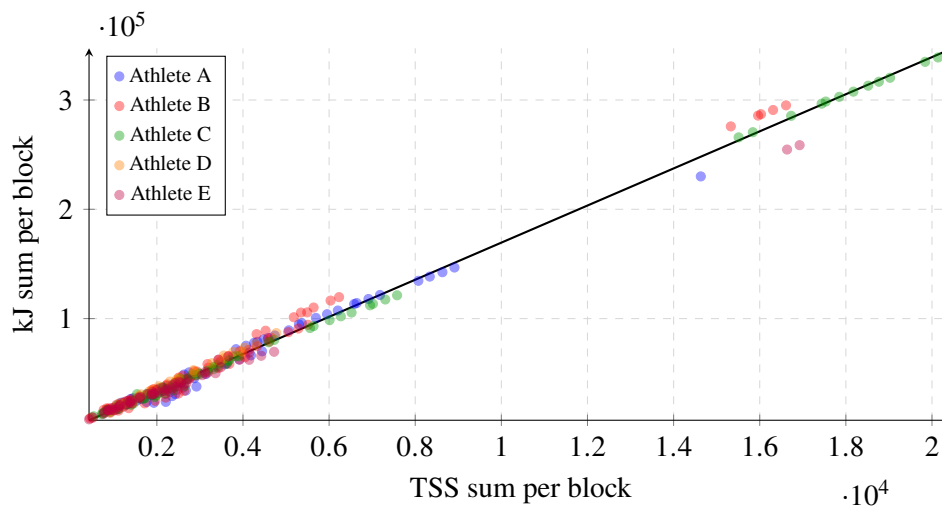


Figure 3.19: Relationship between block TSS and mechanical work (kJ), coloured by rider.

Mechanical Work (kJ) and Internal Load (TRIMP). Mechanical work and TRIMP show an overall positive trend, but with some variability, as reported in Figure 3.20. The scatter reflects how internal load is not determined only by mechanical output: two blocks with similar kJ can have different TRIMP depending on intensity distribution, terrain, type of intervals, and athlete-specific physiological response. This supports the idea that internal and external load capture complementary aspects of workload.

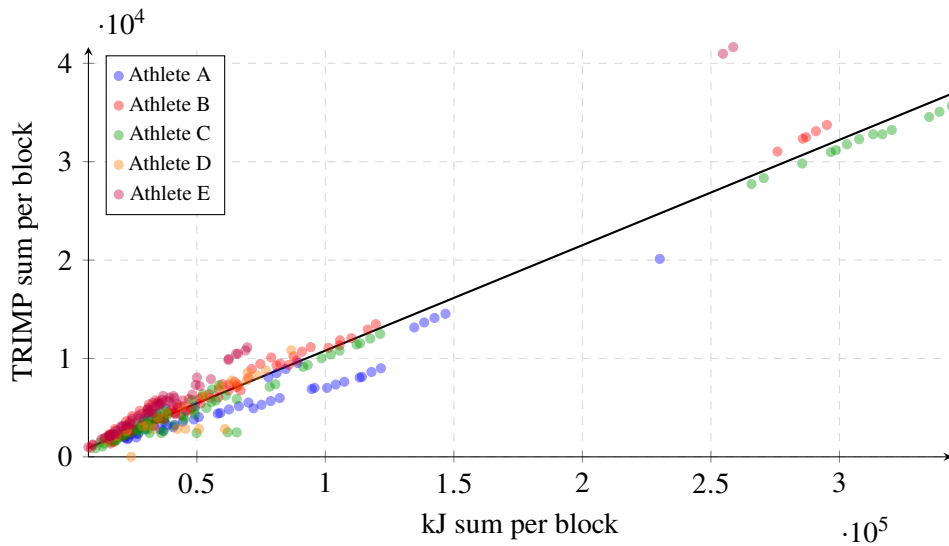


Figure 3.20: Relationship between mechanical work (kJ) and internal load (TRIMP) accumulated within the training block.

Training Monotony and Training Strain. Since strain is defined as the product between total block TSS and training monotony, blocks with higher monotony naturally exhibit greater strain values. However, monotony alone does not determine the magnitude of strain: as reported in Figure 3.21, the large vertical spread of points at each monotony level indicates that the total accumulated TSS strongly modulates the final strain. Blocks that combine high monotony with high workload produce extremely elevated strain values, while blocks with similar monotony but lower TSS remain comparatively less stressful. This illustrates how monotony amplifies the physiological cost of training, particularly when sustained over high-load periods.

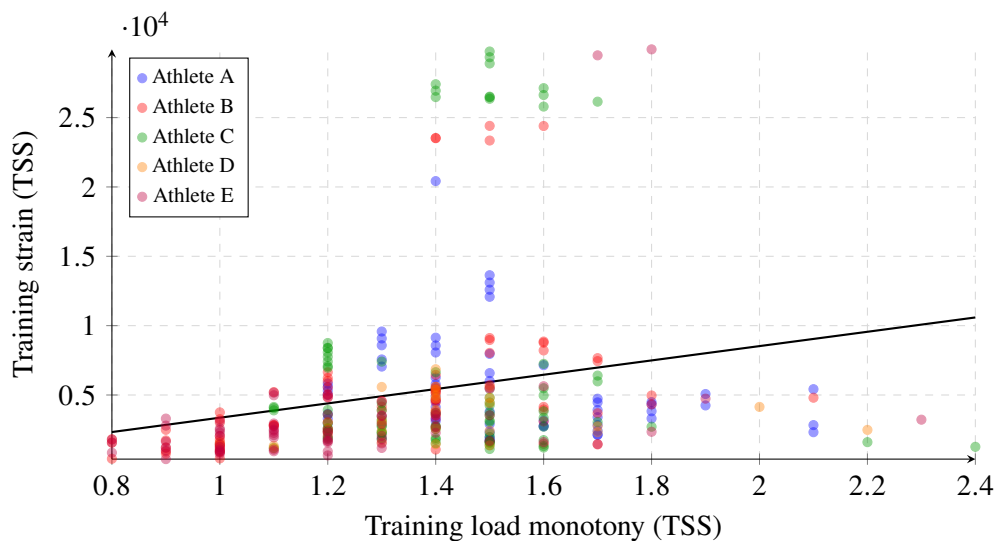


Figure 3.21: Relationship between training load monotony and strain at block level.

Readiness-Based ACWR (7-28 days) and Race Outcome. The readiness-based ACWR (7-28 days) displays (Figure 3.22) slightly higher median values on Top-20 race days compared to Non-Top-20 days. This suggests

that good race performances often occur during periods in which short-term readiness is elevated relative to the preceding month. Nevertheless, the considerable overlap between the two distributions highlights that ACWR alone cannot distinguish performance outcomes, and should instead be interpreted in combination with other readiness indicators.

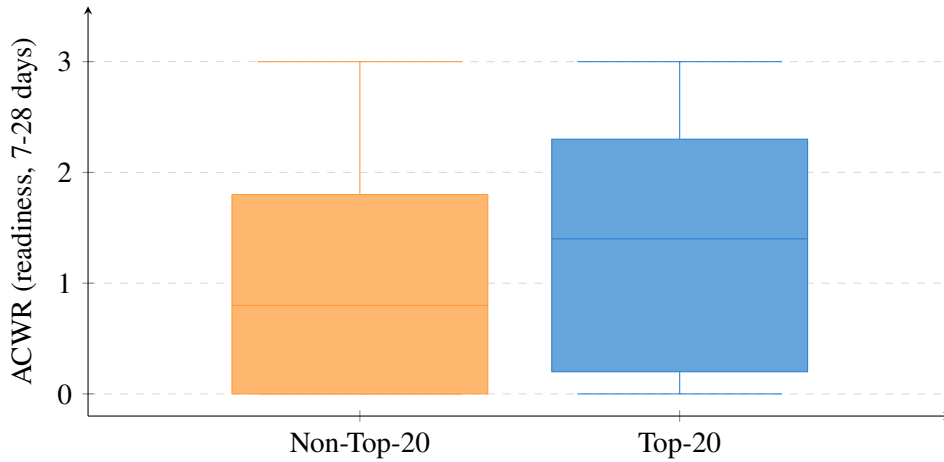


Figure 3.22: ACWR readiness (7-28 days) by race outcome.

3.7.4 Correlation Matrix

The preliminary analysis of linear relationships between the independent variables and the target was performed through the computation of Pearson correlation coefficients, summarized in the correlation matrix (Figure 3.23). Examination of the dataset reveals high multicollinearity among the external workload descriptors; in particular, an almost perfect correlation ($\rho \geq 0.98$) is observed between `block_days_length`, `km_sum_block`, `kJ_sum_block`, and the various TRIMP indices, suggesting informational redundancy that could compromise coefficient stability in classical linear models. Regarding internal load and fatigue management, a strong association is found between ATL and CTL ($\rho = 0.93$). Concerning the target variable `result`, linear correlations are weak with the most notable values associated with `monotony_TSS` ($\rho = -0.19$) and `tsb_mean_45d` ($\rho = 0.11$). This absence of strong linear relationships motivates and justifies the use of a non-linear learning algorithm, such as XGBoost, capable of capturing complex interactions and not evident patterns.

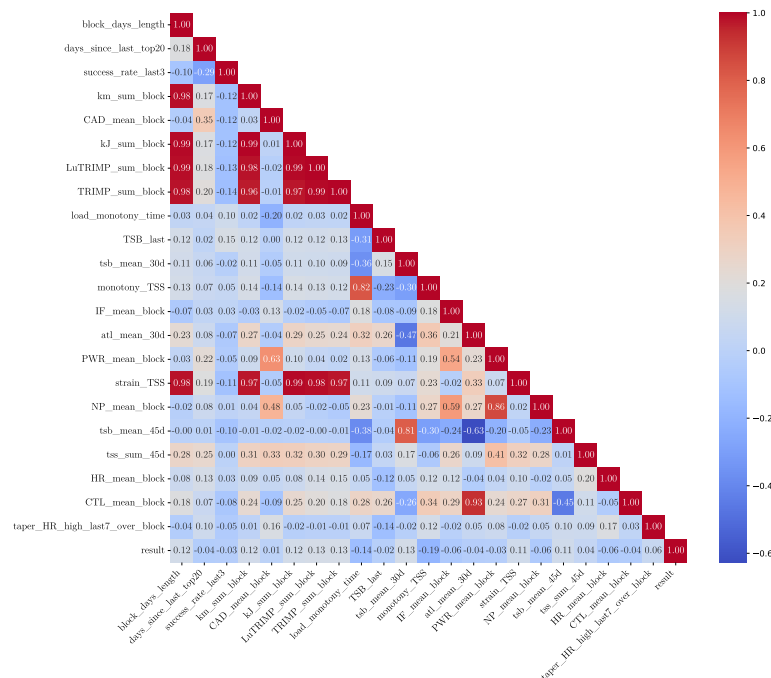


Figure 3.23: Correlation matrix metrics-based.

The second correlation matrix (Figure 3.24) integrates external load parameters with Readiness indicators, aimed at monitoring the athlete's recovery state and physiological response. It is immediately apparent that the readiness metrics exhibit a significant internal autocorrelation structure: for instance, `mean_readiness_last28` and `readiness_ewm_hl14d` show a coefficient of 0.95, indicating a strong temporal stability in perceived or measured fitness state. In particular, there is also the emergence of moderate correlations between readiness variability (`readiness_std_block`) and historical performance (`mean_result_cum_before`, $\rho = 0.50$), suggesting that greater fluctuations in readiness parameters may be associated with more consistent performance patterns. Redundancy between training volume and energy load remains evident, with `kJ_sum_block` and `TRIMP_sum_block` maintaining an almost perfect correlation ($\rho = 0.97$). However, with respect to the target variable `result`, the inclusion of readiness metrics provides more encouraging signals compared to the previous matrix: `readiness_std_block` shows a positive correlation of 0.20, the highest observed value with the target. This finding highlights how adaptability and variation in freshness state can be more effective predictors of final performance than simple linear training volume.

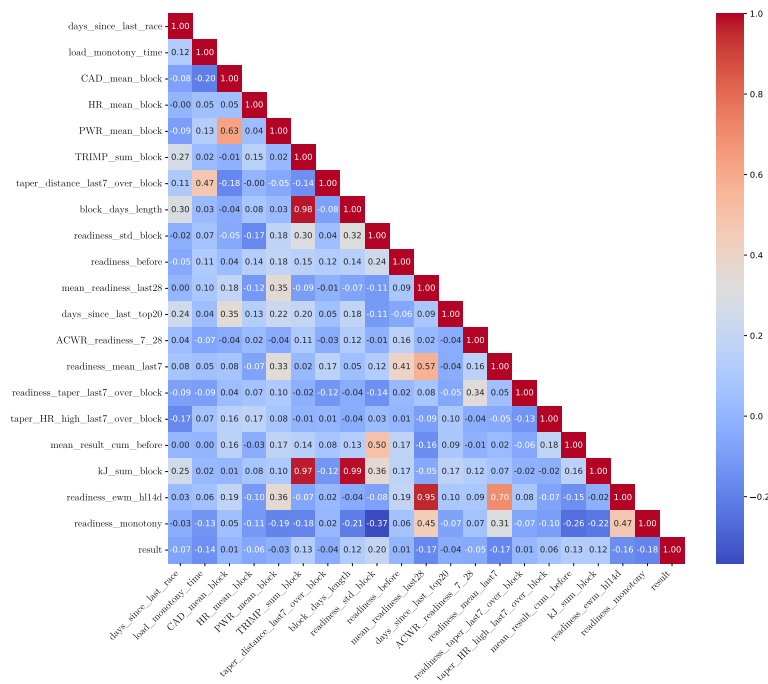


Figure 3.24: Correlation matrix readiness-based.

Chapter 4

Ideas, Research Iterations, and Final Approach

4.1 First Attempts and Early Failures (Exploratory Modelling)

Before describing the final modelling idea, a wide range of preliminary approaches were explored. These attempts were essential to understand the structure of the data, the limitations of raw time-series modelling, and the role of physiological training metrics. Although none of these early models produced credible predictions, each contributed to clarifying *what does not work* and guided the development of the final methodology presented in Section 4.2.

The following subsection summarize the most relevant exploratory directions, grouped conceptually rather than chronologically, to illustrate the progressive of the modelling strategy.

4.1.1 Naive Time-Series Modelling Without Feature Engineering

The first exploratory phase focused on simple time-series models based purely on raw metadata and workout files, considering all the workout types (*Bike, Strength, Run*). A basic chronological split (70-30) combined with a standard XGBoost regressor yielded extremely high coefficients of determination ($R^2 \approx 0.99$), which initially suggested a strong fit but soon revealed an high overfitting. The model relied on noisy, high-frequency variables and lacked any representation of training structure, tapering, or accumulated load. Errors were particularly large on extreme race outcomes (e.g., DNFs or peak performances), demonstrating that raw daily metrics were insufficient to capture meaningful readiness information.

This attempt showed that meaningful prediction required structured aggregation, not raw series.

4.1.2 Expanding the Feature Set Without Structural Context

A second exploratory direction involved the creation of new features set to include all available workout types, full time-series statistics, and a more complete metadata file. Despite the increased dimensionality, the core issues did not change: the models still overfitted, failed to generalize, and lacked any physiological representation of training dynamics.

These experiments confirmed that only adding features does not improve predictive power when the underlying temporal and physiological structure is absent.

4.1.3 Testing Classical Training Metrics in Isolation (ATL, CTL, TSB)

A third set of attempts focused on predicting training metrics such as ATL, CTL, and TSB. Multiple configurations were tested - using these metrics alone, excluding them, or combining them. Although certain models achieved high performance on the training split, results were unstable: removing a single metric often caused the test accuracy to collapse.

This instability made it clear that these metrics, when used directly, were acting as proxies for temporal trends rather than capturing true physiological effects. Traditional load metrics proved insufficient as standalone predictors; they require historical context, rolling windows, or block-level interpretation.

4.1.4 Preliminary Attempts at Training Blocks

A simplistic version of training blocks described in Subsection 3.5.2 in Chapter 3 was also explored. However, the resulting blocks produced inconsistent train-test splits and high variability in the number of activities per block. Model performance on the test set got worse, showing that blocks did not achieve the expected performance. This result demonstrated that block definition must follow a consistent and meaningful criterion, although the concept itself can be valuable when applied correctly.

4.1.5 Synthesis of Early Failures

Across all exploratory models, the same limitations emerged: extreme overfitting, temporal leakage, and an inability to represent the physiological structure of training load. Time-series models were overly sensitive to noise, traditional load metrics behaved unpredictably when used in isolation, and early block-based attempts lacked consistency.

These failures highlighted the need for:

- rider-specific models rather than pooled approaches;
- robust feature engineering and load aggregation;
- physiologically meaningful training blocks and rolling windows.

All these elements created the basis for the final modelling strategy presented in Section 4.2.

4.2 Final Approach

After the exploratory phase described in Section 4.1, the modelling strategy shifted from raw time-series to a more robust formulation based exclusively on the metadata classification file. In fact, time-series data are used only to compute training metrics, while the modelling phase operates on aggregated, structured features associated with each (`rider`, `race_day`). Since the block-generation logic and the feature-engineering process have already been introduced, this section focuses only on the modelling phase.

4.2.1 Problem Reformulation and Target Definition

While the initial attempts explored the prediction of specific physiological metrics, the final approach redefines the task as a *performance prediction problem*. For each race day, the target variable is defined as a binary indicator

$$\text{Top-20} \in \{0, 1\},$$

which specifies whether the rider finished inside the first twenty positions.

The Top-20 outcome is treated as an observable proxy to infer an underlying, latent state of performance readiness and of strong race performance. It is important to clarify that finishing in the Top-20 of a race and the concept of performance readiness are not interchangeable. Performance readiness represents a latent and qualitative state, reflecting an athlete's physiological condition, preparation level, and capacity to perform well on a given day. In contrast, race outcome is an observable and discrete result, influenced not only by the athlete's condition but also by external factors such as competition level, team tactics, and race dynamics.

However, in professional cycling, a Top-20 finish can reasonably be interpreted as an indicator of a good and strong performance, particularly in the context of a Continental-level team. Achieving such a result typically requires an adequate level of preparation and physical readiness, even though it does not capture all dimensions of performance quality. Since the notion of *good* and *strong performance* is inherently qualitative and not directly measurable, the presence in the Top-20 is adopted in this work as a pragmatic and operational proxy.

4.3 Two Modelling Families

The final pipeline evaluates two parallel families of models, each reflecting a different interpretation of the relationship between training and performance - both models share core features described in Subsection 3.5.3 in Chapter 3:

1. **Metrics-based models:** these models use the core features and the full set of metrics-based features described in Subsection 3.5.4 in Chapter 3 to predict the Top-20 outcome. The *metrics-based* family includes three configurations:
 - (a) Logistic Regression: baseline without feature selection, threshold = 0.5
 - (b) XGBoost: after feature selection, threshold = 0.5
 - (c) XGBoost: after feature selection, with personalized thresholds (Youden index per rider).

2. **Readiness-based model:** this model use a synthetic indicator of preparation (readiness), which aggregates the readiness-based features described in Subsection 3.5.5 in Chapter 3. The readiness concept and its construction are detailed in Section 4.4, while the description of the model is reported in Subsection 4.3.2. The *readiness-based* family relies exclusively on the XGBoost model with personalized thresholds, again trained with the same hyperparameter tuning procedure, as initial experiments indicated that this configuration provides the most stable and informative probabilistic estimates for race performance.

Both families share the same data structure, block definition, and chronological organization of training history.

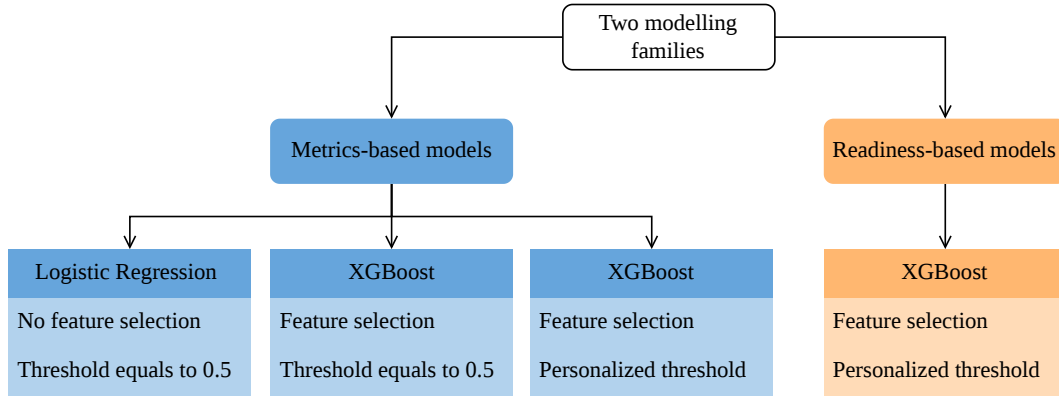


Figure 4.1: Two modelling families.

Training and Test For each rider, activities are sorted chronologically by `race_day`, and a 60-40 temporal split is applied: the first 60% of races form the training set, while the remaining 40% are reserved for testing [47, 48]. A 60-40 split is preferred over the conventional 70-30 because the low prevalence of **Top-20** outcomes would otherwise produce an insufficient number of positive cases in the test set, resulting in unstable evaluation metrics.

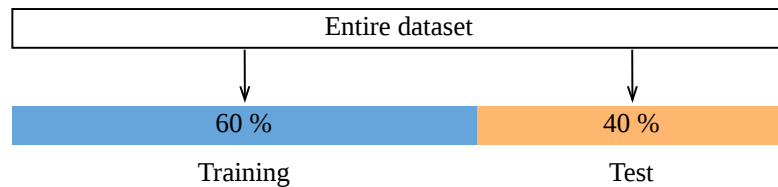


Figure 4.2: Training and test.

Furthermore, if the initial split yields a training set with a single target class, the split index is shifted by one race to ensure that both classes are represented. This procedure guarantees a realistic and leakage-free evaluation while maintaining a sufficient number of positive (**Top-20**) cases in the test set, as reported in the following Table 4.1.

Before introducing the models, it is important to address class imbalance. Although the proportion of **Top-20** outcomes is similar in the training and test sets, the training split contains more than fifty positive cases in total, which makes oversampling feasible. For this reason, the training data are balanced using an oversampling strategy: for each rider, if the minority class contains at least two observations, SMOTE is applied with a number of neighbors adapted to the available positive cases; otherwise, simple random oversampling is used [49]. Oversampling is applied *only* to the training set, so that the test set represents perfectly a real world problem and preserves the original distribution of race outcomes.

Rider	N Races	n_train	Top-20 train (%)	n_test	Top-20 test (%)
Athlete A	62	37	29.7%	25	28.0%
Athlete B	79	47	10.6%	32	12.5%
Athlete C	81	48	4.2%	33	15.2%
Athlete D	27	16	31.2%	11	27.3%
Athlete E	65	39	20.5%	26	11.5%
Total	314	187	16.6%	127	17.3%

Table 4.1: Train-test partition and class distribution (Top-20 prevalence).

4.3.1 Metrics-based Models

In the metrics-based family, three configurations are evaluated in order to increase the model performance. All models operate on the same race-level dataset (*data metrics blocks file*), where each row corresponds to a (rider, race_day) and all predictors are computed using only information available before the target race.

Logistic Regression (baseline). The first configuration is a linear baseline based on Logistic Regression [50]. For each rider, all selected features are used as predictors without additional feature selection. The Logistic Regression model is fitted on the balanced training data and produces, for each test race, a predicted probability of Top-20 - the training and test split used is described in Table 4.1. A fixed decision threshold of 0.5 is then used to obtain binary predictions, from which confusion matrices and performance metrics (accuracy, precision and recall) are computed at both rider level and aggregated level, as reported in the Section 5.2 in the Chapter 5.

XGBoost with fixed threshold. The second configuration replaces the linear classifier with a gradient-boosted decision tree model (XGBoost), in order to capture non-linear interactions between training-load features [51]. In particular, the algorithm used is a boosted decision tree model. Decision trees are composed of nodes, which represent choices or conditions, and branches, which represent the resulting decision paths. Each tree consists of internal nodes and terminal leaves: the nodes encode the input features, while the leaves represent the predicted output value. The term *boosted* refers to a family of algorithms - known as boosting - that transform weak learners into a stronger ensemble model. Boosting is an ensemble-learning technique designed to reduce both bias and variance [52]. In practice, the model works by combining the predictions of many decision trees. Each tree is trained on a subset of the data, and their individual predictions are aggregated to produce the final output [53]. Starting from an initial model, prediction errors (residuals) are computed and used to build the next model in the sequence. These error-focused models are then added to the ensemble. To make a final prediction, the contributions of all previously learned trees are summed.

Overall, the model construction process combines multiple weak learners to obtain a more powerful predictor. During training, XGBoost optimizes an objective function that measures the discrepancy between predictions and the true target values. Its goal is to minimize this discrepancy through iterative, sequential additions of new weak models. Each added tree is trained specifically to reduce the remaining residuals, thereby improving the overall accuracy. Thanks to this ensemble approach, the final predictions are generally more reliable than those produced by a single decision tree.

About the split, the same 60-40 chronological split by rider is used, as reported in Table 4.1. Furthermore, for each rider, the training data are further divided into an internal training and validation portion: the last 20% of the rider's training races are reserved for validation.

A lightweight random search is then performed over 24 randomly sampled hyperparameter configurations, including:

$$n_estimators, max_depth, learning_rate, subsample, colsample_bytree, min_child_weight, \gamma, \lambda, \alpha$$

During model selection, the XGBoost classifier is tuned using a rider-specific randomized hyperparameter search. In each iteration, the model is trained on the internal training portion and evaluated on the internal validation set, as Figure 4.3 displays.

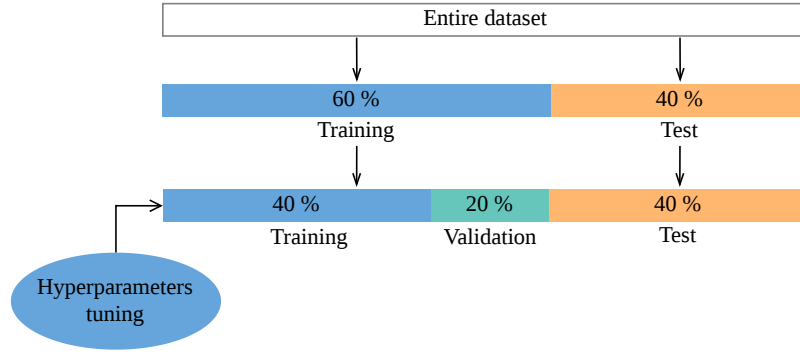


Figure 4.3: Training, validation, and test.

The complete hyperparameter search space and the best parameter set selected for each rider are reported in Appendix B.1 and Appendix B.2, respectively. It is important to note that both the fixed-threshold and the personalized-threshold configurations use the same underlying XGBoost model for each rider. Hyperparameters are selected once through the rider-specific randomized search described above, and the final model is retrained using the full training set. The thresholding strategy (fixed at 0.5 or adapted via the Youden index) affects only the conversion from predicted probabilities to binary outputs and does not alter model training or the learned hyperparameters.

The optimization metric is the Area Under the Precision-Recall Curve (AUC-PR), which is particularly suitable for strongly imbalanced classification problems such as the prediction of Top-20 outcomes. Unlike Receiver Operating Characteristic Area Under the Curve (ROC-AUC), which may remain artificially high when the negative class dominates, the Precision-Recall (PR) curve focuses on the performance of the positive class. PR curves are more informative than Receiver Operating Characteristic (ROC) curves when positive samples are rare, because precision explicitly accounts for false positives and recall quantifies the ability to detect the positive class [54]. For this reason, AUC-PR is widely recommended in imbalanced settings and provides a more discriminative signal during hyperparameter selection.

The ROC curve represents the trade-off between the true positive rate and the false positive rate for all possible decision thresholds. The ROC-AUC is a widely used metric that quantifies the model's ranking ability: it can be interpreted as the probability that a randomly chosen positive example is assigned a higher score than a randomly chosen negative example [55].

Although ROC-AUC is useful in balanced problems, its interpretability decreases when one class is much more frequent than the other, because the false positive rate may understate the effect of misclassifications in highly skewed datasets.

The, the best-performing hyperparameter configuration for each rider is then refitted on the full training set (after oversampling), and test predictions are generated as probabilities of Top-20. In this configuration, a fixed decision threshold of 0.5 is again used to obtain binary labels, allowing a direct comparison with the logistic regression baseline.

XGBoost with rider-specific thresholds (Youden index). The third configuration builds on the previous XGBoost setup but replaces the fixed decision threshold with a personalized threshold for each rider. The model, feature set, train-test split of Table 4.1, and hyperparameter tuning procedure are identical to the fixed-threshold configuration; only the probability-to-label conversion changes. While the fixed threshold of 0.5 provides a natural baseline, it is not optimal in imbalanced classification settings. In fact, for every rider, the final decision threshold is therefore calibrated individually for each rider using the Youden index [56]. For a given a threshold τ , the Youden index is defined as:

$$J(\tau) = \text{Sensitivity}(\tau) + \text{Specificity}(\tau) - 1,$$

The optimal threshold τ^* maximizes $J(\tau)$ over all candidate values on the ROC curve. Geometrically, in Figure 4.4, τ^* corresponds to the point on the ROC curve with the maximum vertical distance from the diagonal of random guessing, thus providing the best trade-off between sensitivity and specificity.

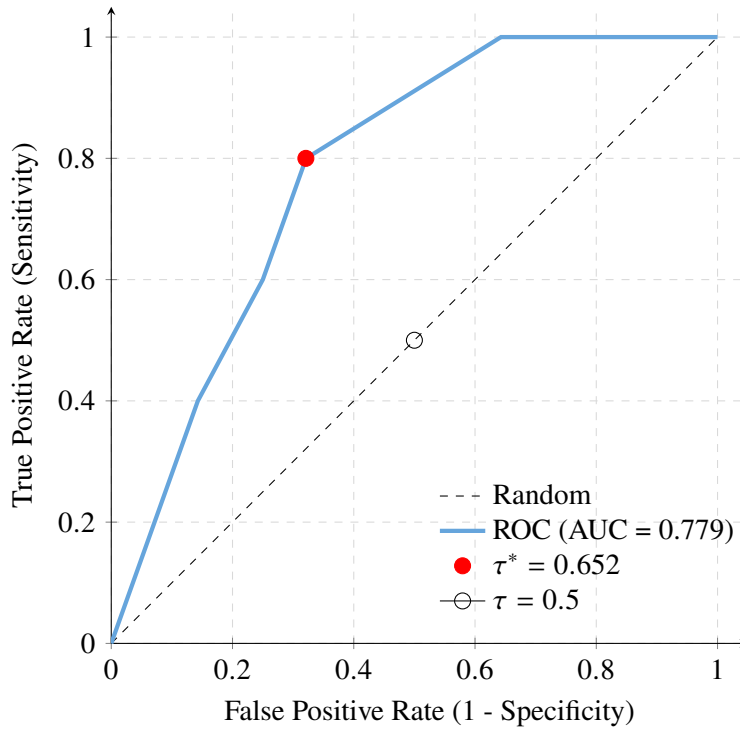


Figure 4.4: ROC curve for Athlete C. The optimal threshold τ^* is selected via the Youden index as the point that maximizes the vertical distance from the random-guessing diagonal.

Note: Among the five athletes, Athlete C displayed the highest ROC-AUC (0.78) and also had the largest number of race observations (33), resulting in a smoother and more interpretable ROC curve. The optimal Youden threshold for this rider ($\tau^ = 0.652$) yields a clear trade-off between sensitivity (True Positive (TP) = 0.80) and specificity (False Positive (FP)). For these reasons, Athlete C is selected as the representative example used to illustrate the ROC curve and the Youden thresholding procedure.*

The resulting rider-specific thresholds are then used to convert predicted probabilities into binary decisions. This configuration is specifically designed to improve the prediction of rare Top-20 events while respecting the individual distribution of results for each athlete. The final decision thresholds obtained via the Youden index for each rider are reported in the following Table 4.2:

Rider	Best Threshold (Youden)
Athlete A	0.844
Athlete B	0.033
Athlete C	0.652
Athlete D	0.269
Athlete E	0.054

Table 4.2: Rider-specific thresholds based on the Youden index for the XGBoost metrics-based model.

4.3.2 Readiness-based Model

XGBoost with rider-specific thresholds (Youden index). The readiness-based family model consists only of an XGBoost classifier combined with rider-specific decision thresholds. This model uses the same modelling pipeline of the metrics-based approach: the chronological 60-40 split by rider (Table 4.1), oversampling of the minority class with SMOTE, internal validation for hyperparameter optimization, and the same randomized search used in the previous configuration. The full XGBoost search space is reported in Appendix B.1, while the rider-specific best hyperparameter settings are provided in Appendix B.3. Naturally, this version is trained on the dataset enriched with both core and readiness-based features (*data readiness blocks file*). All technical aspects of model training and threshold calibration with the Youden index follow exactly the procedures described in

the previous subsections.

To avoid redundancy with the previous subsections, only the conceptual differences are emphasized here. Unlike the metrics-based family, no logistic-regression baseline is introduced: the goal of this section is not to benchmark different algorithms, but rather to focus if the readiness construct itself provides information that goes beyond traditional workload metrics. Early experimentation shows that the XGBoost configuration - combined with personalized thresholds - yielded the most informative probabilistic estimates of performance, and is therefore selected as the sole modelling framework in the readiness-based family.

In this setup, the XGBoost model predicts, for each race, the probability that the athlete finishes in Top-20. These probabilities are then converted into binary outcomes using rider-specific decision thresholds, derived through the Youden index of the readiness-based model reported in Table 4.3.

Rider	Best Threshold (Youden)
Athlete A	0.280
Athlete B	0.056
Athlete C	0.541
Athlete D	0.600
Athlete E	0.267

Table 4.3: Rider-specific thresholds based on the Youden index for the XGBoost readiness-based model.

Finally, the predicted probabilities are transformed into a continuous Readiness Index (RI), as formally described in Section 5.6, allowing a more interpretable representation of the athlete's predicted preparedness level. To understand better the real concept of readiness and how it is created, the next Section 4.4 reported the logic behind its idea.

4.4 Readiness Concept

While the metrics-based family of models directly exploits workload descriptors and training-load metrics, the second modelling approach aims to represent *physiological preparedness* through a more compact, synthetic variable: *readiness*. The intuition is grounded in endurance sports science, where performance capacity emerges not from individual metrics in isolation (e.g., TSS, IF, CTL), but from the combined interaction of accumulated load, fatigue, recovery, and short-term sharpness. Readiness is therefore designed as a rider-specific score summarizing this multivariate interaction.

Different athletes may respond differently to the same training stimulus, due to variations in physiology, fitness, and recovery kinetics. To address this, the readiness score is constructed so that each rider receives a personalized weighting of training-related variables, optimized to predict performance on race days.

Only race sessions are used to estimate the readiness coefficients. This is because the goal is not to model training behavior itself, but rather the *relationship* between pre-race physiological state and competitive performance (Top-20). For every rider, race days are sorted chronologically, and a consistent temporal split is applied:

60% training portion, 40% testing portion.

This preserves the forward flow of time and avoids any form of data leakage, to maintain totally unseen the test set created and reported in Table 4.1.

For each rider, a separate Logistic Regression is fitted on the training portion using the workload-related variables:

{IF, NP, CTL, ATL, TSB, TSS, tTSS, ACWR}

as predictors, and the binary indicator Top-20 as the target. The choice of a linear model is deliberate: Logistic Regression produces an interpretable set of coefficients β_i that quantify the direction of each feature's contribution to race performance. Although the regression is trained exclusively on race days, the learned coefficients encode athlete-specific sensitivities to training-related metrics. These coefficients are therefore

applied to the *full* training history of each rider, producing a daily readiness timeline:

$$R(t) = \beta_1 \tilde{x}_1(t) + \dots + \beta_k \tilde{x}_k(t).$$

Because $R(t)$ depends on the athlete's coefficient scale, readiness values are min-max normalized for each rider:

$$\text{ReadinessScore}_{\text{rider}}(t) = 100 \times \frac{R(t) - R_{\min}}{R_{\max} - R_{\min}},$$

yielding a score in the interval $[0, 100]$. This score forms the basis of all readiness-based predictions of the model described in Subsection 4.3.2.

To understand if readiness is meaningful and discriminates performance, a statistical analysis is conducted on race days only, as reported in Table 4.4. Here, *Mean Top-20* denotes the average readiness on races where the rider finished within the first twenty positions, while *Mean No-Top-20* represents the average readiness on all other race days. An higher Mean Top-20 compared to Mean No-Top-20 indicates that the rider tended to achieve better results on days with higher readiness, providing a first descriptive signal of the relationship between readiness and performance.

Rider	N Races	Mean Top-20	Mean No-Top-20	t-stat	p-value	AUC
Athlete A	62	42.65	37.63	1.19	0.243	0.721
Athlete B	79	81.96	73.54	5.82	< 0.001	0.865
Athlete C	81	88.50	78.19	2.81	0.025	0.782
Athlete D	33	81.55	73.15	1.64	0.123	0.691
Athlete E	65	61.11	51.43	2.96	0.008	0.719
Global AUC across all athletes: 0.529						

Table 4.4: Readiness score analysis on race days: group comparison and discriminative power per rider.

Additionally, first, the mean readiness of Top-20 and non-Top-20 races is compared using Welch's t-test [57]. The t-test evaluates when the mean readiness differs between Top-20 and non-Top-20 races. The test computes a t-statistic, which quantifies the magnitude of the difference relative to the within-group variability. The associated p-value represents the probability of observing a difference at least this large under the null hypothesis of no true effect.

Second, the p-value quantifies the probability of observing a difference in mean readiness between the Top-20 and non-Top-20 groups that is at least as large as the one measured in the data, under the null hypothesis that no difference exists [58]. In other words, it measures the strength of evidence against the hypothesis that readiness has no relationship with performance. A small p-value indicates that the observed separation between readiness distributions is unlikely to be explained by random variation alone. Following common practice in sports science and applied physiology [59], a threshold of $\alpha = 0.05$ is used to determine statistical significance. Athletes B, C, and E show statistically significant differences between Top-20 and non-Top-20 races ($p < 0.05$), with Area Under the Curve (AUC) values between 0.72 and 0.87, indicating a strong individual association between readiness and performance. For Athletes A and D, the separation is weaker and not statistically significant. It is important to say that statistical significance does not imply practical relevance. To complement the hypothesis test, the discriminative ability of readiness is evaluated using the AUC, which measures the probability that a randomly chosen Top-20 race has an higher readiness score than a non-Top-20 race [55]. In imbalanced datasets, AUC provides an additional perspective on the predictive value of readiness beyond the p-value alone.

To conclude, of course, the global AUC (0.529) is low, as expected, because readiness scores are computed independently for each rider and are not meant to be comparable across athletes. The readiness concept is therefore effective at a personalized level, but not as a universal predictor across the entire peloton.

Chapter 5

Results

This chapter presents the experimental results derived from the two modelling families introduced in Chapter 4. The evaluation begins with a detailed description of the used metrics and validation strategy (Section 5.1). Then, the performance of the Logistic Regression baseline is analyzed (Section 5.2), followed by the results obtained from the metrics-based XGBoost model (Section 5.3) and the readiness-based XGBoost configuration (Section 5.4). Finally, a comprehensive comparison across models and a practical dashboard are provided (Sections 5.5 and 5.6).

5.1 Evaluation Setup and Metrics

Multiple complementary metrics are used for evaluating the models prediction, since relying on a single metric may be misleading in imbalanced classification settings [60]. For clarity and to ease the reading of the following sections, the four components of the confusion matrix are summarized in the following Table 5.1.

Symbol	Meaning
TP	True Positives (Top-20 correctly predicted)
FP	False Positives (Non-Top-20 predicted as Top-20)
TN	True Negatives (Non-Top-20 correctly predicted)
FN	False Negatives (Top-20 not predicted as such)

Table 5.1: Definition of the confusion-matrix components used in the evaluation metrics.

Accuracy. Accuracy measures the overall proportion of correctly classified race outcomes:

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN}$$

Although commonly reported, accuracy is not fully informative in imbalanced settings, as a model predicting almost always the majority class may still achieve high accuracy [60].

Precision. Precision quantifies how reliable positive predictions are:

$$\text{Precision} = \frac{TP}{TP + FP}$$

In this context, precision measures the proportion of predicted Top-20 races that are actually Top-20. A low precision indicates many *false alarms* (Non-Top-20 predicted as Top-20).

Recall (Sensitivity). Recall, or sensitivity, measures the model’s ability to detect true Top-20 performances:

$$\text{Recall} = \frac{TP}{TP + FN}$$

This metric indicates how many real Top-20 outcomes are successfully identified. Low recall corresponds to missing relevant performances.

Specificity. Specificity measures the model’s ability to correctly recognize Non-Top-20 outcomes:

$$\text{Specificity} = \frac{TN}{TN + FP}$$

Specificity complements sensitivity, and both are used in the Youden index for threshold optimization.

Youden index. The Youden index [56], as already described, combines sensitivity and specificity into a single quantity:

$$J = \text{Sensitivity} + \text{Specificity} - 1$$

For each rider, the threshold τ^* is selected as the value maximizing J , leading to a personalized classification threshold adapted to the individual distribution of results.

ROC-AUC. The Receiver Operating Characteristic Area Under the Curve (ROC-AUC) quantifies the overall separability between Top-20 and Non-Top-20 outcomes, independently of the decision threshold [61]. An AUC of 0.5 corresponds to random guessing, while higher values indicate better discrimination between the classes.

Together, these metrics provide a comprehensive view of the model’s performance, balancing the evaluation between all the possible cases. Given the small sample size and the imbalanced nature of the data, no single metric is sufficient on its own; instead, all metrics are together interpreted to provide a robust assessment of model performance [60, 61].

5.2 Baseline Modelling: Logistic Regression

The first set of experiments evaluates the Logistic Regression baseline introduced in 4.3.1. All available features, common and metrics-specific features - reported in Table A.1 in Appendix A.3 - are used as predictors, without any feature selection, and a fixed decision threshold of 0.5 is applied to the predicted probability of **top20**. Figure 5.1 shows the workflow behind the model.

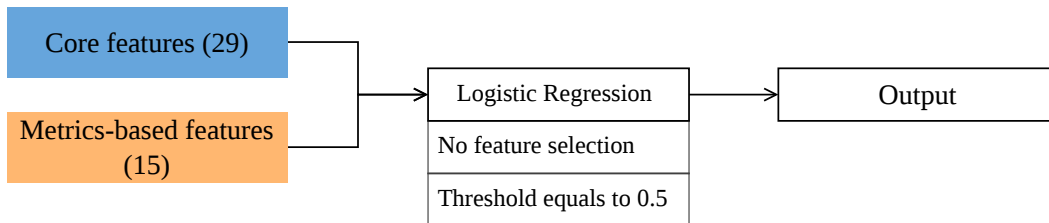


Figure 5.1: Linear Regression workflow model.

Table 5.2 reports the confusion-matrix summary on the test set, both per rider and aggregated across all athletes.

Rider	Total races	Top-20 correct	Top-20 missed	Non-Top-20 as Top-20	Non-Top-20 correct	Precision	Recall	Accuracy
Athlete A	25	5/7	2/7	10/18	8/18	0.333	0.714	0.520
Athlete B	32	0/4	4/4	0/28	28/28	0.000	0.000	0.875
Athlete C	33	0/5	5/5	6/28	22/28	0.000	0.000	0.667
Athlete D	11	0/3	3/3	0/8	8/8	0.000	0.000	0.727
Athlete E	26	0/3	3/3	1/23	22/23	0.000	0.000	0.846
Total	127	5/22	17/22	17/105	88/105	0.227	0.227	0.732

Table 5.2: Confusion-matrix summary for the Logistic Regression baseline (fixed threshold = 0.5).

The Logistic Regression baseline achieves an overall accuracy of 0.732 on the test set. However, this value is largely driven by class imbalance: 105 out of 127 races are non-Top-20 outcomes. The model predicts the negative class, correctly identifying 88 non-Top-20 races while detecting only 5 out of 22 Top-20 performances (recall = 0.227). In practice, this means that the classifier is strongly conservative: it rarely predicts a Top-20 finish and consequently misses the majority of actual high-level performances.

The baseline reveals the structural limitations of modelling race performance as a linear and additive combination of traditional training-load metrics. When the main training metrics load and related variables are treated as independent predictors within a linear decision boundary, the model collapses into a conservative strategy that minimizes overall error by favoring the majority class.

The low recall on strong performances highlights a fundamental difficulty: peak race outcomes are rare events emerging from nonlinear interactions between accumulated adaptation, short-term load modulation, and individual readiness. A linear classifier with a fixed global threshold is poorly suited to represent such dynamics.

This behavior is also underlined at rider level where the heterogeneity of results reinforces this interpretation. Athlete A exhibits moderate recall (0.714), but at the cost of substantial false positives, indicating instability in the linear boundary. For all other riders, the model effectively converges to a "never Top-20" rule. This suggests that a single linear formulation fails to capture athlete-specific performance expression patterns.

Therefore, this baseline provides empirical evidence that traditional load metrics, when modelled without interaction structures or individualized decision rules, are insufficient to detect strong race performances. This finding directly motivates the transition toward nonlinear modelling approaches and athlete-specific thresholding strategies. The inability of the linear baseline to detect rare high-level performances suggests that strong performance can not be approximated as a simple linear accumulation of workload indicators, but emerges from higher-order interactions that require a different, flexible, and more complex modelling structures.

5.3 Model XGBoost with Metrics

The second set of experiments focuses on the XGBoost metrics-based models. In this family, the model operates on a reduced set of features obtained through a gain-based feature selection procedure: feature importance scores are averaged across riders, and features are sorted by mean gain. The first 22 features, reported in Table 5.3, account for 81.8% of the total gain, and are retained as predictors in the final configuration, as described in 5.3.3.

Category	Features	
Core Features (7)	block_days_length	days_since_last_top20
	success_rate_last3	km_sum_block
	CAD_mean_block	kJ_sum_block
	LuTRIMP_sum_block	
Metrics-only (15)	TRIMP_sum_block	load_monotony_time
	TSB_last	tsb_mean_30d
	monotony_TSS	IF_mean_block
	atl_mean_30d	PWR_mean_block
	strain_TSS	NP_mean_block
	tsb_mean_45d	tss_sum_45d
	HR_mean_block	CTL_mean_block
	taper_HR_high_last7_over_block	

Table 5.3: Subset of features selected for the XGBoost metrics-based model, divided into Core Features and Metrics-only features.

5.3.1 Fixed Threshold (0.5)

Figure 5.2 shows the workflow behind the model, while Table 5.4 reports the confusion-matrix summary for the XGBoost metrics-based model with feature selection and a fixed threshold of 0.5 on the predicted probability of top20.

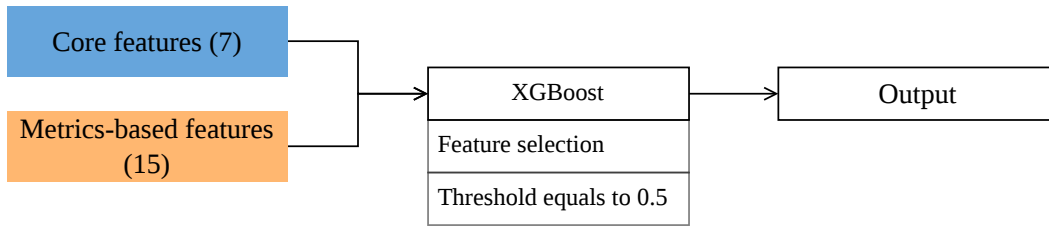


Figure 5.2: XGBoost with Metrics fixed threshold workflow model.

Rider	Total races	Top-20 correct	Top-20 missed	Non-Top-20 as Top-20	Non-Top-20 correct	Precision	Recall	Accuracy
Athlete A	25	4/7	3/7	12/18	6/18	0.250	0.571	0.400
Athlete B	32	1/4	3/4	4/28	24/28	0.200	0.250	0.781
Athlete C	33	5/5	0/5	18/28	10/28	0.217	1.000	0.455
Athlete D	11	1/3	2/3	2/8	6/8	0.333	0.333	0.636
Athlete E	26	0/3	3/3	1/23	22/23	0.000	0.000	0.846
Total	127	11/22	11/22	37/105	68/105	0.229	0.500	0.622

Table 5.4: Confusion-matrix summary for the XGBoost metrics-based model with feature selection and fixed threshold (0.5).

Compared to the linear baseline (Table 5.2), the XGBoost model exhibits a fundamentally different learning behavior. Rather than collapsing into a conservative majority-class strategy, it identifies a substantially larger portion of Top-20 performances (recall = 0.500), indicating sensitivity to interaction effects that are not captured by the linear formulation. Precision on the positive class remains low (0.229), reflecting the intrinsic difficulty of discriminating rare peak performances within a highly imbalanced competitive environment.

This behavioral shift is not only a performance improvement: it suggests that race performance can not be approximated as a linear and additive function of workload descriptors. Instead, strong performances appear to emerge from higher-order interactions between accumulated adaptation, short-term load modulation, and recent performance context. Tree-based models, by construction, are able to partition the feature space into nonlinear regions, implicitly modelling such interaction structures.

At rider level, the heterogeneity of responses further reinforces the individualized nature of adaptation. Athlete C shows perfect recall (1.000) but at the cost of numerous false positives, indicating a broad probability region associated with potential peak expression. In contrast, Athlete E remains difficult to model, suggesting that for some athletes the relationship between training descriptors and race outcomes may be weaker, noisier, or influenced by unobserved factors.

Overall, this configuration behaves as a more sensitive detector of potential strong performances, but the fixed threshold of 0.5 imposes a global decision boundary that does not fully align with athlete-specific probability distributions. This limitation motivates the introduction of individualized calibration strategies to reconcile probabilistic model outputs with the individual and nonlinear expression of performance readiness.

5.3.2 Rider-Specific Thresholds (Youden Index)

To better adapt the model to each athlete, a second configuration - shown in Figure 5.3 - replaces the fixed threshold with rider-specific thresholds derived from the Youden index, as reported in Table 4.2 in Chapter 4.

With rider-specific thresholds, the model correctly identifies 14 out of 22 Top-20 races (recall = 0.636), improving over both the Logistic Regression baseline (0.227) and the fixed-threshold XGBoost configuration (0.500). Precision on the positive class remains low (0.250), reflecting the presence of 42 false positives.

However, the relevance of this configuration is not only quantitative. The introduction of rider-specific thresholds changes the interpretation of model outputs: rather than imposing a global decision boundary, it allows each athlete's probability distribution to define its own operational definition of "readiness".

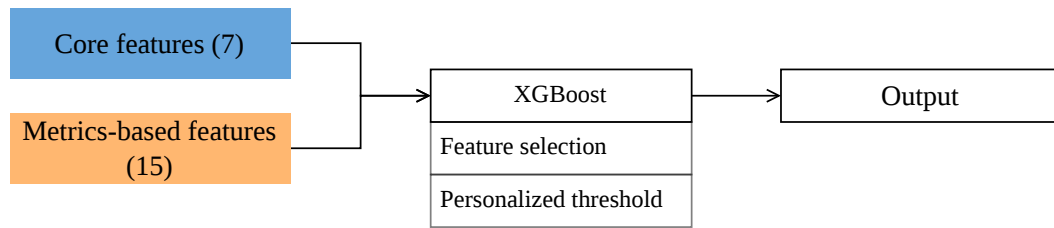


Figure 5.3: XGBoost with Metrics personalized threshold workflow model.

The resulting thresholds reveal inter-individual heterogeneity. For instance, Athlete A requires a very high probability (0.844) to classify a race as strong performance, suggesting a probability distribution concentrated at lower values and a more conservative readiness profile. In contrast, Athlete B operates with a much lower threshold (0.033), indicating that moderate predicted probabilities already correspond to a meaningful likelihood of strong performance within that athlete’s historical distribution. Athlete E represents again an extreme case: the model identifies all three Top-20 races in the test set, but misclassifies 17 non-Top-20 races as Top-20, resulting in low overall accuracy (0.346).

These differences should not be interpreted as model instability, but as empirical evidence that the mapping between workload-derived features and competitive outcomes is athlete-specific. The same predicted probability does not carry the same performance implication across riders.

Table 5.5 shows the corresponding confusion-matrix summary on the test set.

Rider	Total races	Top-20 correct	Top-20 missed	Non-Top-20 as Top-20	Non-Top-20 correct	Precision	Recall	Accuracy
Athlete A	25	1/7	6/7	0/18	18/18	1.000	0.143	0.760
Athlete B	32	3/4	1/4	11/28	17/28	0.214	0.750	0.625
Athlete C	33	4/5	1/5	9/28	19/28	0.308	0.800	0.697
Athlete D	11	3/3	0/3	5/8	3/8	0.375	1.000	0.545
Athlete E	26	3/3	0/3	17/23	6/23	0.150	1.000	0.346
TOTAL	127	14/22	8/22	42/105	63/105	0.250	0.636	0.606

Table 5.5: Confusion-matrix summary for the XGBoost metrics-based model with rider-specific thresholds (Youden index).

From a methodological perspective, rider-specific thresholds put into practice the individualized nature of endurance performance. They transform a statistical classifier into an athlete-aligned decision system, where the boundary between “ready” and “not ready” is derived from personal performance distributions rather than imposed globally, as the previous models.

This shift is conceptually relevant: it acknowledges that performance is not defined by an absolute probability level, but by its relative position within an athlete’s historical adaptation dynamics. In this sense, individualized calibration is not only a technical adjustment, but a structural alignment between machine learning methodology and the physiology of endurance training.

5.3.3 Feature Importance

Beyond predictive accuracy, the feature importance profile provides indirect evidence about the structure of performance encoded by the model. Table 5.6 reports the mean gain-based feature importance, averaged across riders, for the 22 features retained after feature selection.

The importance of block-level internal load (TRIMP_sum_block) and monotony indicators (load_monotony_time, monotony_TSS) suggests that adaptation is not primarily volume-driven, but structure-driven. In other words, the magnitude of the stimulus matters, but its temporal distribution appears equally relevant. High monotony values being informative implies that excessive load regularity may limit performance expression, supporting the hypothesis that variability within training blocks contributes to strong performance.

Feature	Mean gain importance
TRIMP_sum_block	0.904
load_monotony_time	0.846
TSB_last	0.813
tsb_mean_30d	0.786
monotony_TSS	0.777
block_days_length	0.756
IF_mean_block	0.733
atl_mean_30d	0.692
CAD_mean_block	0.623
days_since_last_top20	0.617
success_rate_last3	0.611
PWR_mean_block	0.552
strain_TSS	0.521
NP_mean_block	0.509
tsb_mean_45d	0.494
tss_sum_45d	0.465
HR_mean_block	0.464
kJ_sum_block	0.438
km_sum_block	0.397
CTL_mean_block	0.300
LuTRIMP_sum_block	0.284
taper_HR_high_last7_over_block	0.243

Table 5.6: Mean gain-based feature importance for the metrics-based XGBoost model.

Freshness-related metrics (TSB_last, tsb_mean_30d) indicate that race performance is strongly modulated by short- and medium-term recovery states. This supports the hypothesis that peak performance emerges not simply from accumulated fitness, but from the interaction between chronic adaptation and acute load reduction (tapering).

The simultaneous importance of intensity descriptors (IF_mean_block, NP_mean_block) and cumulative load measures (tss_sum_45d, CTL_mean_block) suggests that both stimulus quality and long-term load exposure contribute to readiness. The model therefore encodes a multi-timescale adaptation structure rather than giving more importance to a single load metric.

The relevance of recent performance history variables (days_since_last_top20, success_rate_last3) indicates that physiological load metrics alone do not fully explain strong outcomes. Their importance may reflect latent components such as tactical role, psychological confidence, race selection strategy, or contextual dynamics, suggesting that readiness is partially path-dependent. Past competitive success appears to shape the probability landscape of future peak performances.

The most influential predictors reflect how training load is accumulated and how the athlete approaches the race in terms of freshness and monotony. In particular, the model relies on three main groups of variables:

- Block-level internal load and monotony: TRIMP_sum_block, load_monotony_time, monotony_TSS.
- Short- and medium-term freshness indicators: TSB_last, tsb_mean_30d, block_days_length.
- Recent performance history: days_since_last_top20, success_rate_last3.
- Block-averaged intensity and power: IF_mean_block, PWR_mean_block, NP_mean_block.
- Accumulated mechanical and physiological load: kJ_sum_block, tss_sum_45d, CTL_mean_block, LuTRIMP_sum_block.

Overall, the feature importance structure suggests that race performance is an emergent property of interaction between stimulus magnitude, temporal distribution, recovery state, and recent competitive trajectory. This interpretation aligns with a dynamical view of endurance performance, where peak outcomes represent states emerging from the alignment of adaptation, recovery, and contextual readiness.

5.4 Model XGBoost with Readiness

This section presents the final model (Figure 5.4) adopted in this thesis and summarizes the reasoning that led to its selection. The XGBoost framework was chosen for its ability to capture non-linear interactions between training load, readiness, and race outcomes, reflecting the complex adaptation dynamics in professional cycling.

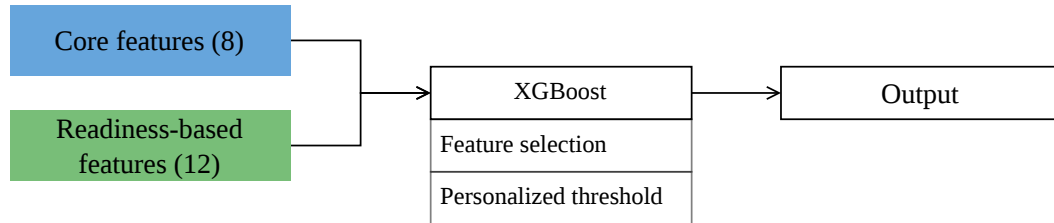


Figure 5.4: XGBoost with Readiness personalized threshold workflow model.

Starting from a linear Logistic Regression baseline and a metrics-based XGBoost configuration, progressively more flexible models were evaluated to address class imbalance, non-linear feature interactions, and inter-rider heterogeneity. Beyond purely statistical considerations, this modeling process explicitly aimed to represent physiological phenomena: how accumulated workload, recent recovery, and tapering influence the probability of achieving a Top-20 performance. Feature selection was performed using gain-based importance scores, retaining only the most informative predictors (more than 80%), reported in Table 5.7.

Category	Features	
Core Features (8)	days_since_last_race	load_monotony_time
	CAD_mean_block	HR_mean_block
	PWR_mean_block	TRIMP_sum_block
	taper_distance_last7_over_block	block_days_length
Readiness-only (12)	readiness_std_block	readiness_before
	mean_readiness_last28	days_since_last_top20
	ACWR_readiness_7_28	readiness_mean_last7
	readiness_taper_last7_over_block	taper_HR_high_last7_over_block
	mean_result_cum_before	kJ_sum_block
	readiness_ewm_hl14d	readiness_monotony

Table 5.7: Selected features for the XGBoost model, divided into Core Features and Readiness-based features.

In parallel, rider-specific decision thresholds were introduced to better balance sensitivity and specificity. The readiness-based XGBoost model represents the final outcome of this process, combining a compact and interpretable set of readiness-related features with personalized threshold calibration, providing a framework that reflects both performance potential and individual physiological response patterns.

An additional motivation behind the proposed modeling strategy is the transition from a generalized prediction approach to an individualized one. Initial baseline models aim to learn population-level relationships between training load and race outcomes, implicitly assuming that similar training stimuli produce comparable effects across athletes. However, in professional cycling this assumption is often unrealistic, as riders differ in physiology, training history, and response to workload. The readiness-based modeling approach explicitly addresses this limitation by shifting the focus toward individualized performance estimation, as reported in the Figure 5.5.

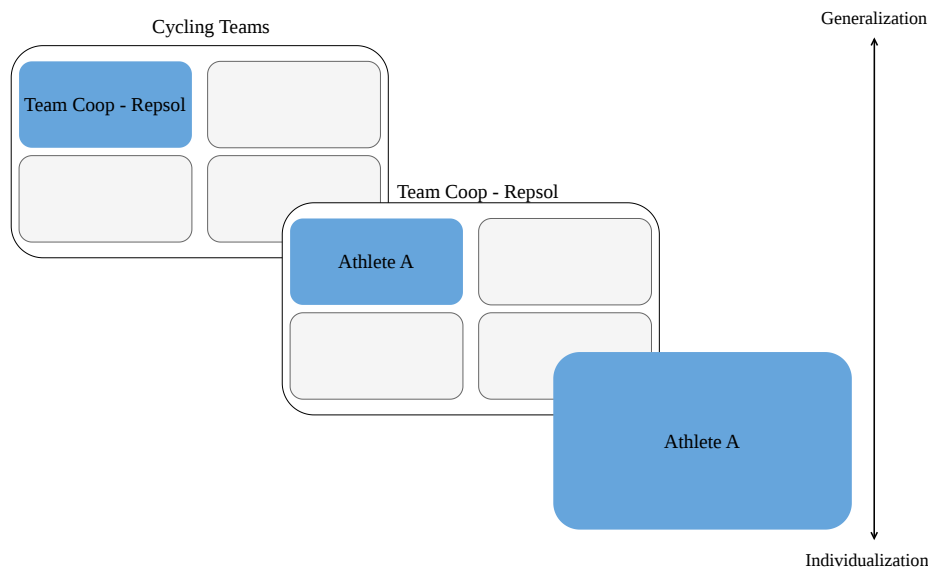


Figure 5.5: Conceptual shift from generalization to individualization.

Instead of relying on a single global decision criterion, the model learns personalized patterns of workload and performance and applies rider-specific thresholds. This transition allows the Readiness Index (RI) to better reflect individual readiness dynamics, supporting athlete-centered performance assessment rather than team-level generalization.

Overall Performance

Based on these considerations, the readiness-based configuration is selected as the final model, described in Subsection 4.3.2. Unlike the metrics-based configuration, which directly uses training-load descriptors, the readiness-based model builds on a compact set of readiness-related features and on the synthetic readiness construct introduced in Section 4.4. As the XGBoost metrics-based model with Youden index, rider-specific decision thresholds are calibrated (Table 4.3 in Chapter 4).

After computing the importance scores across riders, the top twenty features are retained for the final model; these twenty predictors account for 80.9% of the total gain, indicating that most of the discriminative capacity is concentrated in a relatively small subset of readiness descriptors, as reported in Subsection 5.4.1.

Table 5.8 reports the confusion-matrix summary for the readiness-based model on the test set.

Rider	Total races	Top-20 correct	Top-20 missed	Non-Top-20 as Top-20	Non-Top-20 correct	Precision	Recall	Accuracy
Athlete A	25	4/7	3/7	4/18	14/18	0.500	0.571	0.720
Athlete B	32	2/4	2/4	6/28	22/28	0.250	0.500	0.750
Athlete C	33	4/5	1/5	9/28	19/28	0.308	0.800	0.697
Athlete D	11	3/3	0/3	3/8	5/8	0.500	1.000	0.727
Athlete E	26	3/3	0/3	12/23	11/23	0.200	1.000	0.538
TOTAL	127	16/22	6/22	34/105	71/105	0.320	0.727	0.685

Table 5.8: Confusion Matrix per rider for the readiness-based XGBoost model with personalized thresholds.

At aggregated level, the readiness-based model correctly identifies 16 of the 22 Top-20 races in the test set (recall = 0.727), representing the highest sensitivity among all the models. False positives decrease compared to the metrics-based configuration with Youden thresholds (34 vs. 42), and overall accuracy increases from 0.606 to 0.685. Positive precision remains modest (0.320), as expected in a highly imbalanced setting, but the readiness-based configuration achieves the best trade-off between detecting Top-20 performances and limiting unnecessary alerts.

At rider level, several patterns emerge:

- **Athlete A:** the model balances sensitivity and specificity (4/7 Top-20 detected, only 4 false positives), achieving the highest accuracy among riders (0.720).
- **Athlete B:** performance improves compared to both Logistic Regression and the metrics-based model; the model captures half of the Top-20 races while maintaining a low number of false positives.
- **Athlete C:** the model preserves high recall (0.800) but reduces false positives compared to the metrics-based family (from 18 to 9), indicating better probability calibration.
- **Athlete D:** all three Top-20 races are correctly detected (recall = 1.000), with a moderate number of false positives (3/8).
- **Athlete E:** the most challenging athlete; although the model identifies all Top-20 races (3/3), the number of false positives remains high (12/23), suggesting the limited number of races for this rider: only the 2023 season is included, which is not sufficient to learn stable patterns and to produce good predictions.

5.4.1 Feature Importance

Table 5.9 reports the mean gain-based importance scores for the twenty features retained in the readiness-based XGBoost model. Unlike the metrics-based configuration, which emphasizes raw training load descriptors, the readiness-based model highlights a distinct hierarchy of predictors, reflecting physiological signals associated with performance readiness, recovery, and tapering.

Feature	Mean gain importance
days_since_last_race	1.455
load_monotony_time	1.156
CAD_mean_block	1.138
readiness_std_block	0.987
readiness_before	0.907
HR_mean_block	0.888
PWR_mean_block	0.789
TRIMP_sum_block	0.785
taper_distance_last7_over_block	0.618
mean_readiness_last28	0.573
days_since_last_top20	0.569
ACWR_readiness_7_28	0.568
readiness_mean_last7	0.560
readiness_taper_last7_over_block	0.502
taper_HR_high_last7_over_block	0.501
block_days_length	0.488
mean_result_cum_before	0.473
kJ_sum_block	0.348
readiness_ewm_hl14d	0.337
readiness_monotony	0.206

Table 5.9: Mean gain-based feature importance for the readiness-based XGBoost model.

The readiness-based model highlights a different structure of influential predictors compared to the metrics-based family, with a stronger emphasis on:

- Temporal freshness and acute recovery: `days_since_last_race` ranks highest, indicating that the recency of competition strongly affects readiness. Athletes require sufficient recovery to express maximal performance, and this feature captures the balance between fatigue and freshness.

- Monotony and block-level distribution: `load_monotony_time` and `block_days_length` emphasize that the structure of training blocks - not only total volume - affects performance adaptation. High monotony may explicit adaptation, while varied load distribution can potentiate physiological responses.
- Readiness variability and pre-race state: `readiness_std_block` and `readiness_before` reflect both short-term fluctuations and immediate pre-competition state. Their importance suggests that consistency and acute preparedness are critical in predicting Top-20 outcomes.
- Short and medium-term readiness trends: `readiness_mean.last7`, `mean_readiness.last28`, and `readiness_ewm.hl14d` capture temporal dynamics of physiological readiness. These features provide a moving window into how adaptation unfolds over days to weeks, revealing the influence of accumulated load and recovery on potential performance.
- Tapering effects: `taper_distance.last7_over_block`, `readiness_taper.last7_over_block`, and `taper_HR_high.last7_over_block` highlight how strategic reduction in load before races modulates readiness. The model identifies these tapering patterns as key determinants of peak performance windows.
- Traditional training load descriptors: features such as `CAD_mean_block`, `HR_mean_block`, `PWR_mean_block`, and `TRIMP_sum_block` retain moderate importance, confirming that accumulated workload still contributes to performance, but its predictive value is enhanced when contextualized by readiness and tapering dynamics.

Taken together, these results illustrate that performance readiness is not a simple function of total training load. Instead, it emerges from the interaction between accumulated workload, recovery, tapering, and individual variability, supporting the thesis that personalized modeling and feature selection grounded in physiological interpretation are essential to understand and predict competitive outcomes.

Overall, the readiness-based model provides the highest recall, indicating that incorporating readiness measures and individual thresholds captures key physiological dynamics. In particular, the model identifies athletes' peak preparedness windows and their individual responses to workload accumulation and tapering, confirming that performance is not only determined by total training volume but also by recovery patterns and inter-individual variability.

5.4.2 Readiness Trajectory and Race Outcomes for Athlete A: An Example

Figure 5.6 illustrates the readiness trajectory of Athlete A over the test races, together with the observed race outcomes.

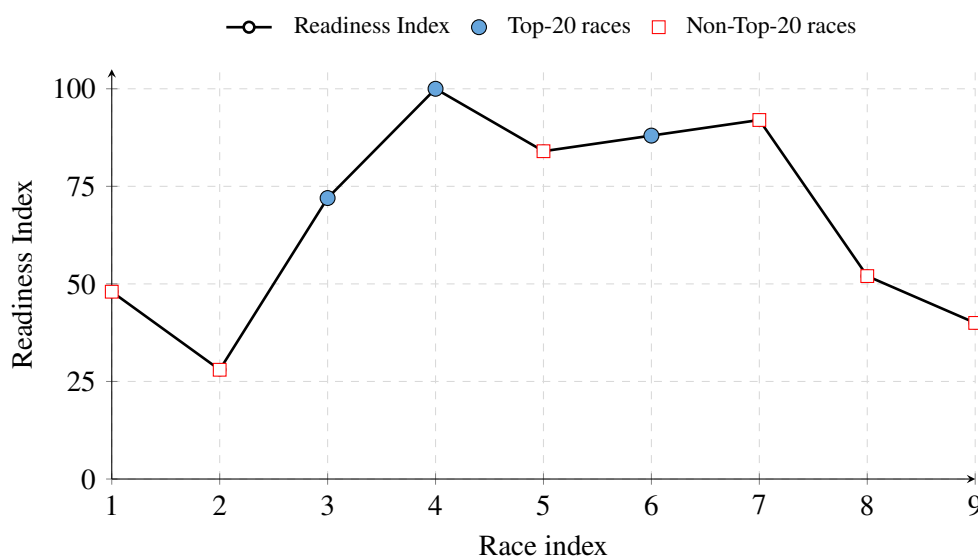


Figure 5.6: Readiness trajectory and race outcomes over time for Athlete A in the test set.

The plot provides an intuitive visual inspection of how readiness values fluctuate over time and how these values relate to actual performance (Top-20 vs. non-Top-20). Top-20 races are highlighted with blue filled

circles, while non-Top-20 races are marked with black squares. This is a first representation which helps assess whether higher readiness levels tend to coincide with successful performances, and whether the readiness pattern captures meaningful pre-race conditions.

A more illustrative example and the logic behind the creation of RI is reported in Section 5.6. However, it is also possible to observe races in which the athlete exhibits a high readiness value (e.g., Race 5 and Race 6) while the actual result falls outside the Top-20. These cases reflect the influence of contextual, tactical factors-such as race dynamics and team strategy, and model limitations.

5.5 Model Comparison

To summarize the behavior of the three models - the Logistic Regression baseline, the XGBoost model using metrics-based features, and the XGBoost model using readiness-based features - a comparative analysis is presented.

5.5.1 Global Model Comparison

Aggregated Comparison

Figure 5.7 compares the three models across the main evaluation metrics (precision, recall, and accuracy). The readiness-based model shows the highest recall (0.727), indicating a superior ability to identify true Top-20 performances, while maintaining competitive accuracy (0.685) and an improved precision compared to the other configurations. While, Logistic Regression achieves high accuracy only because it almost always predicts the majority class, resulting in minimal recall.

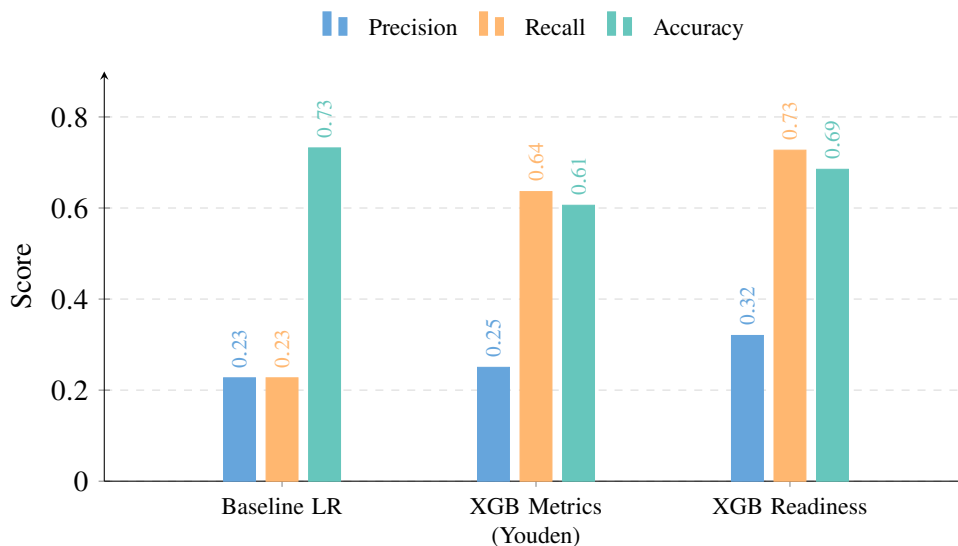


Figure 5.7: Comparison of precision, recall, and accuracy across the three main models.

Per-Rider Recall Comparison

Figure 5.8 presents the recall obtained by each modelling family across all riders. Logistic Regression baseline fails to detect any Top-20 race for four out of five athletes, while the metrics-based XGBoost model increases recall but introduces instability, with large fluctuations across riders. The readiness-based model provides the most consistent performance, matching or improving the recall of the metrics-based configuration for every athlete, and achieving perfect recall for Athletes D and E.

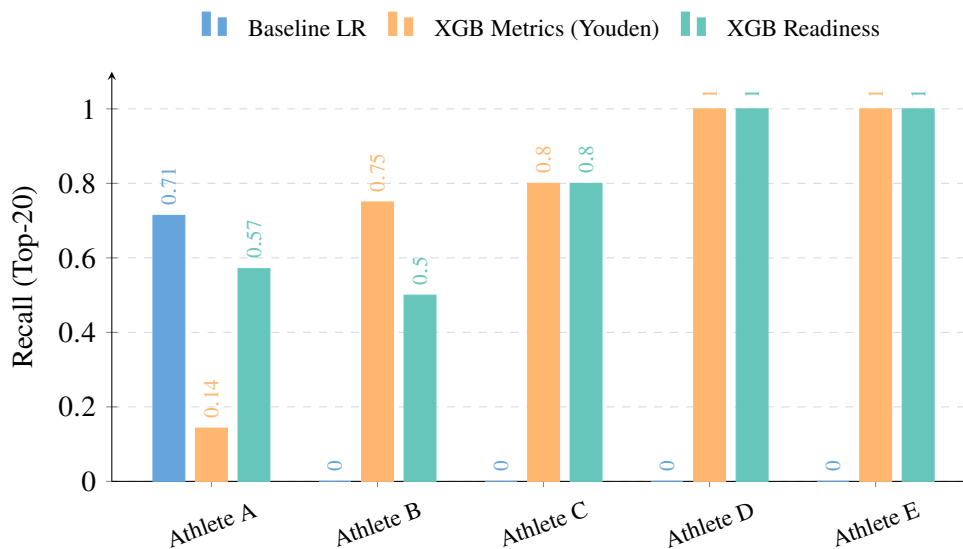


Figure 5.8: Per-rider recall on Top-20 races for the three main models.

5.5.2 Sensitivity to the Decision Threshold

Figure 5.9 illustrates how the precision and recall of the readiness-based model vary as the decision threshold is increased from 0 to 1. As expected in an imbalanced classification problem, recall starts very high at low thresholds - where almost all races are predicted as Top-20 - and progressively decreases as the threshold becomes more selective. On the other hand, precision shows the opposite trend: it increases only when the threshold becomes sufficiently strict, indicating that high-confidence predictions are rare. The crossover between the two curves occurs around thresholds 0.45-0.55, highlighting the trade-off between identifying as many Top-20 results as possible and limiting false positives. This behavior underlines the need for rider-specific threshold calibration (Table 4.2 and Table 4.3 in Chapter 4), since a single global threshold would underperform across athletes with different distributions of outcomes.

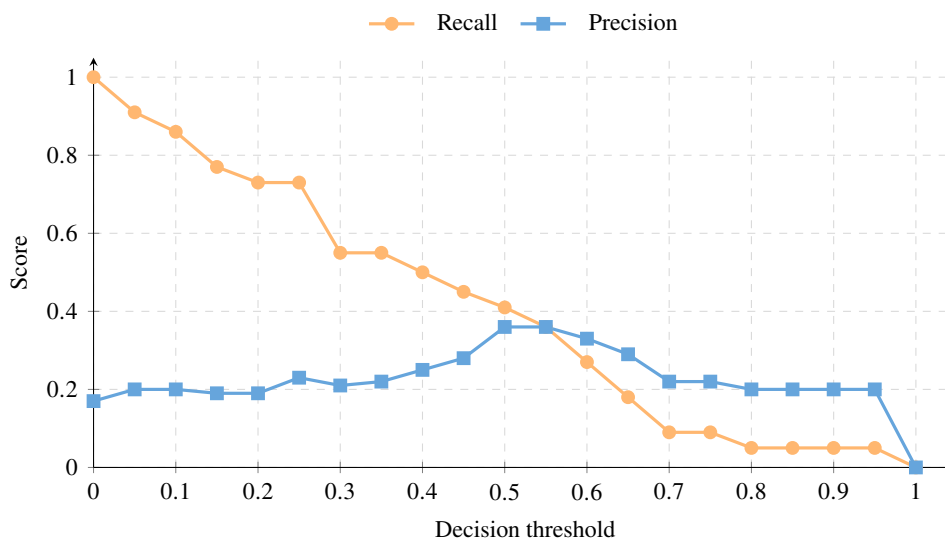


Figure 5.9: Effect of the decision threshold on precision and recall for the readiness-based model.

When analyzing the effect of the decision threshold, the curve exhibits a consistent downward trend as the threshold increases. This behavior is expected in highly imbalanced settings such as Top-20 prediction. At low thresholds, the model classifies a large proportion of races as positive, yielding high recall but low precision. As the threshold becomes more selective, the number of predicted positives decreases rapidly; in several cases, no races exceed the threshold at all. When this occurs, the model produces no true positives, causing both precision and recall to collapse toward zero. This downward trend does not indicate model degradation, but reflects the statistical behavior of precision and recall under strong class imbalance and limited per-rider sample sizes.

5.5.3 Ranking Ability and Cumulative Gains

Figure 5.10 illustrates the cumulative gains achieved by the readiness-based model when races are ranked from the highest to the lowest predicted probability of achieving a Top-20 result. The diagonal line represents a random selection strategy: inspecting $x\%$ of the races would retrieve, on average, $x\%$ of the Top-20 results. The readiness-based curve stays consistently above this baseline, showing that the model is able to rank races effectively.

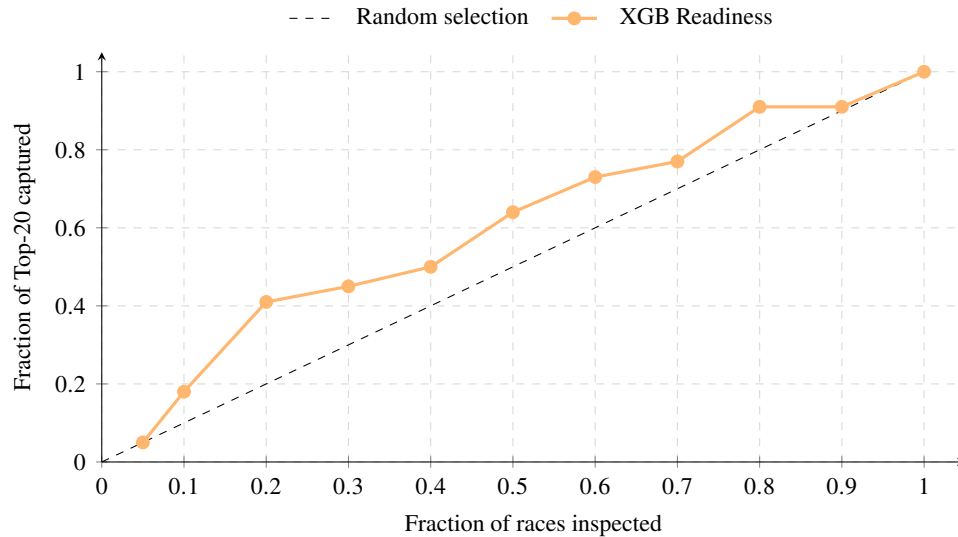


Figure 5.10: Cumulative gains chart for the readiness-based model.

For example, the top 20% of races (axis-x) already contain almost 40% of all Top-20 results (axis-y), and by inspecting half of the races the model retrieves more than 60% of the successful performances. Only in the very first segment does the curve briefly align with the random baseline, which is expected given the limited sample size. Overall, the gains chart highlights that the readiness model provides meaningful ranking ability, effectively prioritizing the races with the highest likelihood of a Top-20 finish.

5.5.4 Baseline vs Readiness: Detection Performance

Figure 5.11 provides a direct comparison between the Logistic Regression baseline and the readiness-based XGBoost model in terms of Top-20 detections. The difference is important: the baseline correctly identifies only 5 of the 22 Top-20 races, missing the remaining 17. In contrast, the readiness model detects 16 Top-20 outcomes and reduces missed events to just 6.

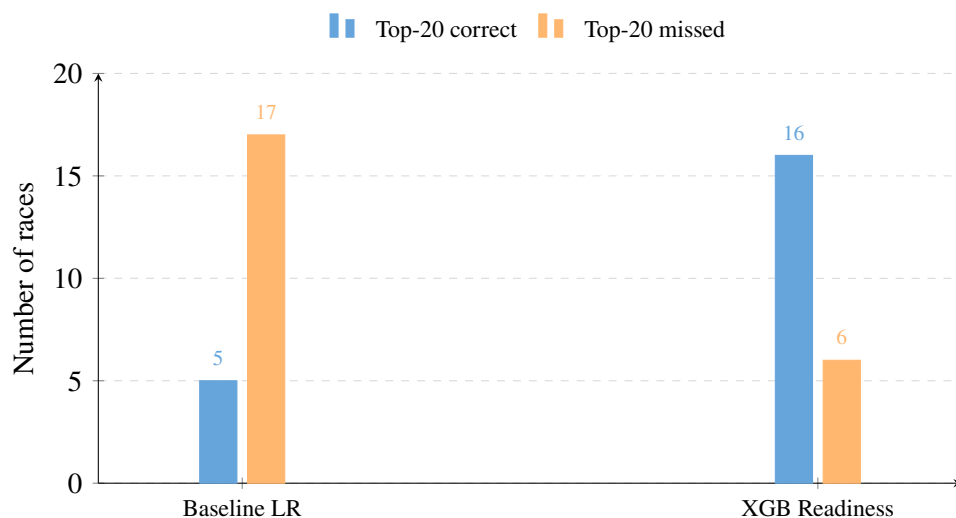


Figure 5.11: Comparison of Top-20 detections: Logistic Regression baseline vs. XGB Readiness.

This illustrates a clear improvement in sensitivity to high-performance races, confirming that readiness-related information captures relevant pre-race conditions that the simple linear baseline fails to recognize. Such improvement is consistent with the methodological refinements introduced in the readiness-based approach: a more targeted feature engineering process, the selection of performance-relevant training-load indicators, and the use of a non-linear model better suited to capturing complex interactions between load structure and race outcomes.

5.6 Dashboard Implementation and Practical Use

The objective of the machine learning framework developed in this thesis is to estimate, in a *pre-race setting*, the probability of a rider achieving a strong performance in an upcoming competition. Predictions are generated using training and physiological indicators collected during workout sessions preceding each race, as described in Chapter 4.

Among the evaluated models, the readiness-based XGBoost model presented in Subsection 4.3.2 demonstrated the best overall performance and was therefore selected as the reference model for downstream analysis and application. While classical performance metrics (e.g., accuracy, AUC, precision, recall) are useful for model comparison, they are not directly interpretable by coaches or performance analysts in an operational context. For this reason, the model output is transformed into representations that are both quantitatively sound and practically interpretable, forming the basis for the decision-support framework developed in this chapter.

5.6.1 Model Output and Readiness Index (RI)

For each race, the readiness-based XGBoost model outputs a continuous probability, $y_pred \in [0, 1]$, representing the estimated likelihood that a rider will achieve a Top-20 performance. While these probabilities reflect the model’s statistical confidence, they are not immediately actionable in a coaching context, since these values lack direct operational meaning. To bridge this gap, a Readiness Index (RI) is introduced as a standardized, rider-specific representation of expected performance.

Computation and rationale of the Readiness Index (RI) The Readiness Index (RI) is computed as the within-rider percentile rank of predicted probabilities. Conceptually, it transforms raw model outputs into a relative measure of readiness for each athlete, contextualized by their personal history:

$$\text{ReadinessIndex}_{r,i} = 100 \times P_{r,i}$$

where $P_{r,i}$ represents the percentile of race i ’s y_pred within all predicted probabilities for rider r . The resulting value, ranging from 0 to 1, is then scaled to a 0-100 range to obtain the final RI. This mapping achieves three objectives:

- Normalization across athletes: riders differ in likelihoods due to physiological heterogeneity, racing experience, and role in the team. Percentile ranking allows comparison of readiness relative to each athlete’s historical profile.
- Interpretability for coaching decisions: a 0-100 scale provides an intuitive visualization of readiness, facilitating identification of high- or low-readiness periods and supporting training or tactical adjustments.
- Linking machine learning outputs to latent performance states: rather than simply reporting probabilities, the RI uses the concept of “latent preparedness,” reflecting the integrated effects of training load, tapering, and recovery on expected performance.

Interpretive value Beyond classification accuracy, the RI serves as a quantitative tool for longitudinal monitoring, allowing coaches to:

- Identify periods of consistently high or low readiness within the season.
- Track the effect of tapering strategies, training load distribution, and recovery interventions on individual performance potential.
- Compare readiness trends across riders while respecting inter-individual variability.

In essence, the Readiness Index (RI) carry out machine learning outputs into an athlete-centered metric, translating probabilistic predictions into actionable insights that are meaningful for training prescription, race planning, and tactical decision-making. Its development transforms model outputs from abstract probabilities into a common language of performance readiness, bridging the gap between predictive modeling and coaching practice. The transformations described above enable a direct and intuitive visualization of model predictions, and constitute the conceptual foundation of the interactive dashboard presented in the Subsection 5.6.2.

Illustrative cases Table 5.10 summarizes four scenarios illustrating the interaction between y_{pred} , personalized thresholds, and the resulting RI. Each case highlights different aspects of model interpretation: during prediction, for each race, the model outputs a numerical value between 0 and 1, referred to as y_{pred} . It represents the *estimated probability that the athlete will achieve a strong performance*.

For instance:

rider	race_day	readiness_index	readiness_band	y_pred	y_true
...	...	...	...	...	...
Athlete A	2024-07-03	88.0	Excellent	0.3706	1
Athlete A	2024-07-04	92.0	Excellent	0.3885	0
Athlete A	2024-07-05	52.0	Neutral	0.2320	0
Athlete A	2024-08-04	24.0	Low	0.1253	1
...	...	...	...	...	...

Table 5.10: Excerpt of the prediction dataset with four different situations.

The different scenarios shown in the Table 5.10 are now analyzed, with particular emphasis on Case 1. Then, all the other cases are presented, but they share the analogous logic.

Case 1 - TP. The first case corresponds to a TP, where the model correctly predicts a Top-20 result that actually occurred.

rider	race_day	readiness_index	readiness_band	y_pred	y_true
...	...	...	...	...	...
Athlete A	2024-07-03	88.0	Excellent	0.3706	1
Athlete A	2024-07-04	92.0	Excellent	0.3885	0
Athlete A	2024-07-05	52.0	Neutral	0.2320	0
Athlete A	2024-08-04	24.0	Low	0.1253	1
...	...	...	...	...	...

Table 5.11: Excerpt of the prediction dataset in Case 1 - True Positive.

As reported in Table 5.11, the predicted probability $\hat{p}$ obtained from the XGBoost model is:

$$y_{\text{pred}} = 0.3706$$

which represents the model's estimated probability that the rider finishes in the Top-20 for that race (approximately 37%).

The personalized decision threshold for this rider, as reported in Table 4.3 in Chapter 4, computed using the Youden Index, is:

$$\text{best_threshold} = 0.280$$

This threshold determines the binary classification rule used to predict when the rider is expected to finish in the Top-20:

$$\hat{y} = \begin{cases} 1, & \text{if } y_{\text{pred}} \geq 0.280 \\ 0, & \text{if } y_{\text{pred}} < 0.280 \end{cases}$$

In this example:

$$0.3706 \geq 0.280 \Rightarrow \hat{y} = 1$$

Therefore, the model correctly predicts that the rider would achieve a strong performance, since the actual outcome is $y_{\text{true}} = 1$.

In the dashboard, the predicted probability can be displayed as:

Estimated probability of strong performance: 37%

Since y_{pred} already represents a probability (a continuous value in the $[0, 1]$ range), the percentage value is simply obtained as:

$$P(\text{Top-20}) = y_{\text{pred}} \times 100$$

For this race:

$$0.3706 \times 100 = 37.06\%$$

So, the model assigns to the rider a 37% probability of achieving a strong performance in that race.

Then, the RI is computed as the within-rider percentile rank of y_{pred} . For each rider, all predicted probabilities are collected and the position of a given y_{pred} is expressed as its percentile within that personal distribution. A general view of the logic of the development of this index for Case 1 is reported in the following Table 5.12:

rider	race_day	y_pred	percentile_rank	readiness_index
Athlete A	2024-07-03	0.3706	0.88	88

Table 5.12: Example of the computation of the readiness index for a given rider. The index represents the percentile rank of the predicted probability (y_{pred}) within the rider's historical distribution.

For this example, as reported in a draft of the dashboard Figure 5.12a in Subsection 5.6.2, the rider has a readiness index value of 88, categorized as *Excellent*. This transformation maps the predicted probability to a 0-100 scale, interpreted as a *readiness percentage*. In particular:

- low readiness (0-40): unfavorable physiological or training condition;
- neutral readiness (40-60): moderate condition;
- good readiness (60-80): above-average probability of good performance;
- excellent readiness (80-100): high probability of a strong performance.

This direct mapping ensures consistency between the model prediction and its visual representation in the dashboard.

An analogous logic holds for all other possible scenarios presented in Table 5.10, and they are discussed briefly in the following paragraphs.

Case 2 - False Positive FP. The second case corresponds to a FP, where the model predicts a Top-20 outcome that does not actually occur, as reported in Table 5.13. This happens when the predicted probability exceeds the personalized threshold, but the real result is outside the Top-20.

rider	race_day	readiness_index	readiness_band	y_pred	y_true
...	...	...	...	...	...
Athlete A	2024-07-03	88.0	Excellent	0.3706	1
Athlete A	2024-07-04	92.0	Excellent	0.3885	0
Athlete A	2024-07-05	52.0	Neutral	0.2320	0
Athlete A	2024-08-04	24.0	Low	0.1253	1
...	...	...	...	...	...

Table 5.13: Excerpt of the prediction dataset in Case 2 - False Positive.

In this situation:

$$y_{\text{pred}} = 0.3885 \geq \text{best_threshold} = 0.280 \Rightarrow \hat{y} = 1$$

Although the model predicts a strong performance, the real outcome is $y_{\text{true}} = 0$. This scenario reflects an overestimation of readiness or race-day performance, despite the extremely high readiness index (92, classified as *Excellent*).

Case 3 - True Negative (TN). The third case represents a TN, where the predicted probability is below the rider-specific threshold, and the real outcome confirms that no Top-20 finish occurred, as in Table 5.14.

rider	race_day	readiness_index	readiness_band	y_pred	y_true
...	...	...	...	...	...
Athlete A	2024-07-03	88.0	Excellent	0.3706	1
Athlete A	2024-07-04	92.0	Excellent	0.3885	0
Athlete A	2024-07-05	52.0	Neutral	0.2320	0
Athlete A	2024-08-04	24.0	Low	0.1253	1
...	...	...	...	...	...

Table 5.14: Excerpt of the prediction dataset in Case 3 - True Negative.

In this case:

$$y_{\text{pred}} = 0.2320 < \text{best_threshold} = 0.280 \Rightarrow \hat{y} = 0$$

The model correctly predicts that the athlete would not achieve a strong performance, consistent with the observed outcome. This reflects a typical mid-readiness situation (readiness index = 52), aligned with more uncertain or average performance expectations.

Case 4 - False Negative (FN). The fourth, and last, case corresponds to a FN, reported in table 5.15, where the model underestimates performance: despite predicting that the athlete will not achieve a strong performance, the rider actually achieves a Top-20 result.

rider	race_day	readiness_index	readiness_band	y_pred	y_true
...	...	...	...	...	...
Athlete A	2024-07-03	88.0	Excellent	0.3706	1
Athlete A	2024-07-04	92.0	Excellent	0.3885	0
Athlete A	2024-07-05	52.0	Neutral	0.2320	0
Athlete A	2024-08-04	24.0	Low	0.1253	1
...	...	...	...	...	...

Table 5.15: Excerpt of the prediction dataset in Case 4 - False Negative.

In this scenario:

$$y_{\text{pred}} = 0.1253 < \text{best_threshold} = 0.280 \Rightarrow \hat{y} = 0$$

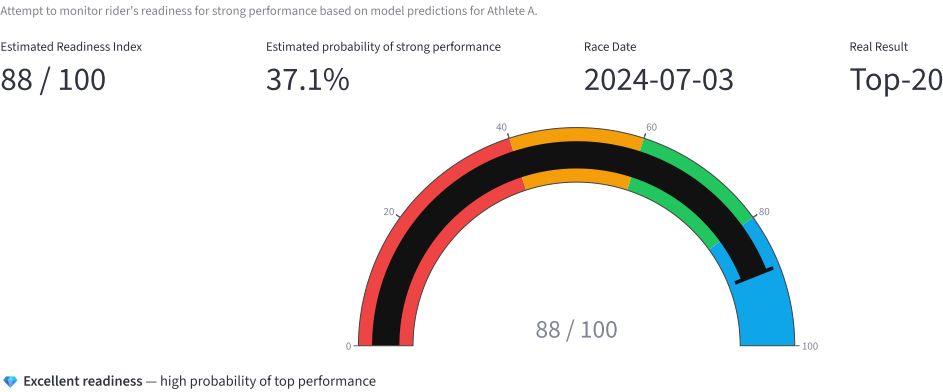
However, the true result is $y_{\text{true}} = 1$. For this example, as reported in a draft of the dashboard Figure 5.12b in 5.6.2, the rider has a readiness index value of 24, categorized as *Low*. However, the athlete achieves a strong race outcome despite a historically low readiness level - often attributable to tactical factors, race dynamics, favorable course characteristics, or simply limitations of the developed model, as described in Chapter 6.

5.6.2 Interactive Dashboard Prototype

To visualize and explore the model outputs described above, an interactive dashboard prototype was developed. The dashboard serves as a visual analytics layer on top of the predictive model, aiming to support interpretation rather than replace the modelling process itself. The dashboard integrates the outputs of the readiness-based XGBoost model with race-level information, providing rider-specific views of predicted readiness, estimated performance probability, and observed outcomes. The prototype was implemented using Streamlit for the application framework and Plotly for interactive visualizations.

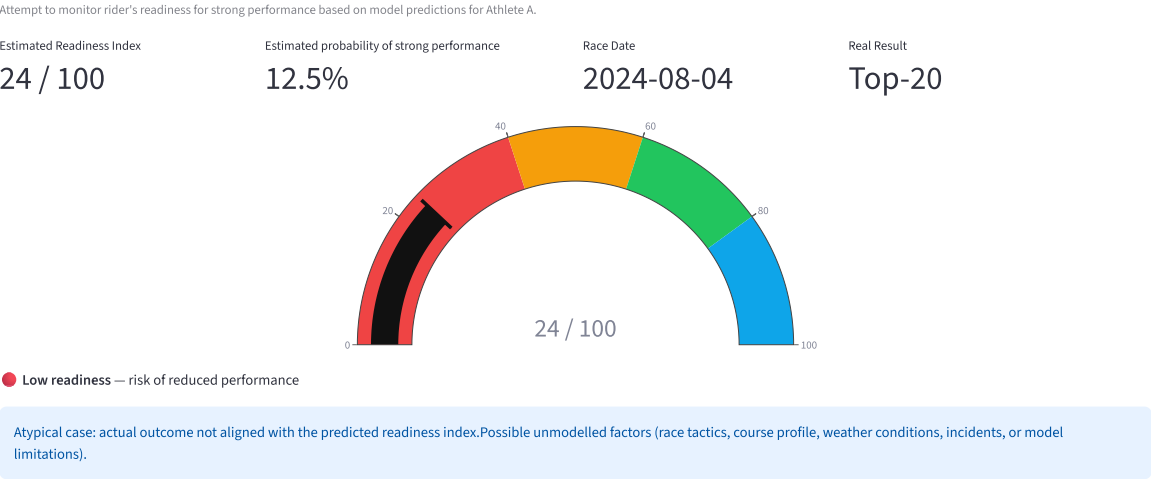
Key Performance Indicator (KPI) & Readiness Gauge The upper section of the dashboard displays key indicators for the selected race, including the readiness index, the estimated probability of a strong performance, and the actual race outcome. The gauge visualization provides an immediate qualitative interpretation of readiness (Figure 5.12a), while mismatches between predicted readiness and actual results highlight atypical cases potentially driven by unmodeled factors such as race dynamics or external conditions (Figure 5.12b).

Readiness & Performance Dashboard - Athlete A



(a) Example of predicted excellent RI and correct prediction.

Readiness & Performance Dashboard - Athlete A



(b) Example of predicted low RI but incorrect prediction.

Figure 5.12: Key sections of the Readiness Index & Performance Dashboard.

Timeline and Confusion Matrix The timeline in Figure 5.13 view allows the user to analyze readiness evolution across the season on a race-by-race basis.

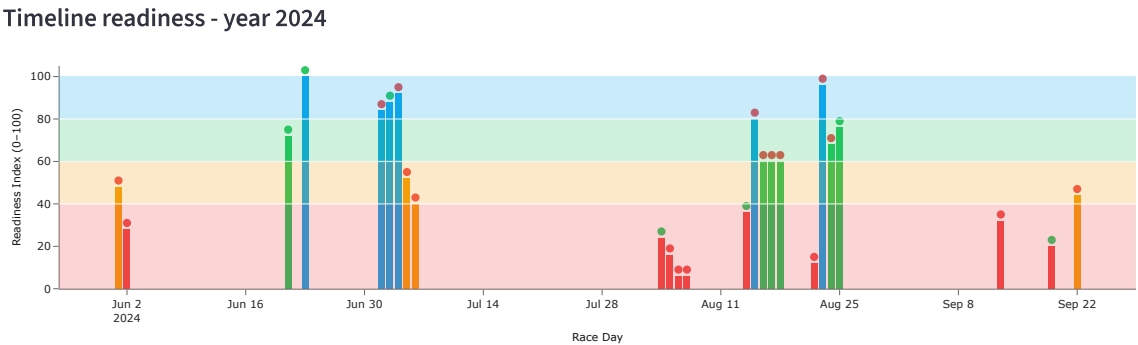


Figure 5.13: Timeline of the activity for Athlete A.

Each bar represents a race day, colored according to predefined readiness bands: red for Low readiness (0-40), orange for Neutral (40-60), green for Good (60-80), and blue for Excellent (80-100). Markers above the

bars indicate the actual race outcome: green markers for Top-20 results and red markers for Non Top-20 results. This view supports longitudinal analysis and helps identify patterns such as sustained high readiness, sudden drops, or inconsistencies between preparation and results.

The dashboard also includes a simplified confusion matrix (Figure 5.14), computed at the rider level, summarizing correct and incorrect predictions.

Confusion Matrix - Athlete A

Simple counts for the selected rider (Top-20 vs Non-Top-20 outcomes).

	Rider	Total Races	Correct Top20	Missed Top-20	Non-Top-20 predicted as Top-20	Correct Non-Top-20 predictions
0	Athlete A	25	4/7	3/7	4/18	14/18

Figure 5.14: Confusion Matrix for Athlete A .

This section provides a transparent and rider-specific evaluation of model behavior, supporting practical trust and interpretability for coaching staff.

5.6.3 Practical Usage in a Real-World Scenario

In a real-world professional cycling context, the Readiness & Performance Dashboard can be used as a pre-race decision-support tool integrated into the team's daily workflow. Its primary objective is to translate model-based probabilistic predictions into intuitive, rider-specific insights that can support race preparation and performance management.

In an operational setting, the dashboard would be deployed as a web-based application, automatically updated as new training and physiological data become available. Prior to each race, the system would generate predictions without requiring any technical interaction from coaches or athletes. Each rider would have a dedicated profile reflecting the personalized modelling approach described in Section 5.4. For every upcoming race, the dashboard presents a concise pre-race summary composed of three key elements:

- the *Readiness Index*, expressed on a 0-100 scale;
- the estimated probability of achieving a strong performance, derived directly from the predictive model;
- a qualitative readiness band (Low, Neutral, Good, Excellent), visualized through color-coded indicators and a gauge.

The Readiness Index (RI) acts as an interpretability layer on top of the raw probabilistic output. While the estimated probability provides a precise numerical measure of expected performance, the index enables a rapid qualitative assessment that can be easily communicated and interpreted during pre-race meetings. Importantly, in this phase the actual race outcome is not yet known: the dashboard is therefore used exclusively as a predictive and planning tool, supporting decisions such as rider role assignment, expectations management, and last-minute adjustments to tapering or recovery strategies.

The timeline view further supports longitudinal analysis by displaying readiness evolution across the season on a race-by-race basis. This allows coaches to contextualize the upcoming event within the athlete's broader preparation trajectory, identifying sustained high-readiness phases, warning signals such as sudden drops, or periods of unstable form. By selecting individual races, users can inspect the corresponding pre-race predictions and readiness states, reinforcing the exploratory and interactive nature of the tool.

After the race has taken place, the actual outcome can be incorporated into the dataset and visualized. While this post-event information does not influence pre-race decision-making, it enables ex-post evaluation of model behavior, highlighting discrepancies between predicted readiness and observed performance. Such cases may reveal the influence of unmodeled factors - such as race dynamics, tactical roles, or external conditions - and provide valuable feedback for both coaching staff and future model refinement.

Overall, the Readiness & Performance Dashboard is designed as a pre-race decision-support system that bridges the gap between data-driven modelling and practical performance management. By combining probabilistic predictions, individualized readiness indices, and intuitive visualizations, the application transforms model outputs into actionable insights for professional cycling teams. A complete overview of the dashboard interface is provided in Appendix C.

Chapter 6

Limitations and Future Work

6.1 Limitations

Despite the encouraging results obtained in this work, as reported in Chapter 5, several limitations should be acknowledged.

Small dataset

The most evident limitation is the size of the dataset. Only five riders and a limited number of race days were available, which naturally reduces the statistical power of the models and increases the risk of overfitting. More data - both in terms of seasons and number of athletes - would allow probably a more robust estimation and would improve the stability of rider-specific models.

Class imbalance

The number of *Top-20* results is very small compared to *Non-Top-20* outcomes. This class imbalance makes the learning task more difficult and often pushes the model toward conservative behavior, such as predicting the negative class (0) almost all the time. Techniques such as resampling or calibrated decision thresholds help mitigate the issue, but the imbalance remains a limitation of the problem.

Missing physiological and contextual variables

The dataset lacks variables related to sleep quality, recovery, wellness, illness, or stress, these KPIs were not available for all riders. Including these metrics only for a subset of athletes would have introduced inconsistencies, while removing riders without such information would have further reduced an already small dataset. *TrainingPeaks' Metrics* section could have provided richer detail, but it was not consistently filled in by all riders, as reported in Table 3.1 in Chapter 3. The absence of these physiological indicators limits the model's ability to capture deeper dynamics which are very important for describing the rider's psychophysical condition outside of the training session [62].

Lack of race profile and course characteristics

The modelling framework does not include information about race profiles and course characteristics, such as elevation gain, number and length of climbs, terrain type (flat, hilly, mountainous), or sprint likelihood. In professional cycling, riders have highly specialized physiological and tactical roles in a team: climbers, sprinters, time trial specialists, or punchers. A poor performance (e.g., a *Non-Top-20* result) may therefore not necessarily reflect inadequate preparation or low readiness, but rather a misalignment between the rider's characteristics and the demands of the specific race or stage. Ignoring this context risks attributing suboptimal results solely to training-related factors, while the true explanation may lie in the race profile itself.

Limited textual and contextual information

Although the raw data included fields such as *Title*, *CoachComments*, and *WorkoutDescription*, these were not systematically exploited. Free-text comments from riders and coaches could contain highly valuable contextual insights (e.g., fatigue complaints, illness, technical issues), but extracting insights from unstructured text is more challenging and would have required much more work.

Block construction heuristic

The definition of training blocks is defined on an heuristic of seven consecutive non-race days. While practical, this choice is somewhat arbitrary. Different block segmentation strategies (e.g., variable lengths, performance-driven segmentation, adaptive clustering) might better reflect how riders structure their training periods.

Time-series structure not fully exploited

The models were trained on aggregated block-level features, which simplifies how training evolves over time. The sequential patterns of training were only captured in an indirect way in the metadata file. Using proper time-series models in the future could help predict performance more accurately by modelling these dynamics directly with second-by-second data.

Exclusion of non-cycling activities

Some riders perform additional training modalities such as running, gym sessions, or cross-country skiing. These activities were removed. Incorporating all training modalities could provide a more complete picture of the athlete's overall load and physiological state.

Examples from elite sport make this point clearer: Primož Roglič transitioned from ski jumping to world-class cycling [63], showing how diverse athletic backgrounds can build a powerful and transferable endurance engine. Cross-country skiing champion Johannes Høsflot Klæbo has openly expressed interest in switching to professional cycling in the future [64], further illustrating the natural crossover between endurance sports with amazing Maximum Oxygen Consumption (VO_2max) improvements in short periods of time with specific training interventions [65]. To give an example, VO_2max represents the maximum amount of oxygen the body can use to produce movements. It has been proved that higher VO_2max values are strongly associated with better cycling performance [30, 66]. To conclude, another example, Sebastian Lipowitz, who comes from biathlon, demonstrates how athletes with mixed endurance backgrounds can successfully transition into high-level cycling [67].

These cases underline that non-cycling activities can have a real impact on performance. Including these multi-sport contributions would make future models more complete and accurate.

Simplified race outcome definition

Only *Top-20* vs *Non-Top-20* outcomes were considered. Other outcomes such as DNF, DNS, or OTL were ignored, despite the fact that a cluster of consecutive non-finishes often signals underlying issues (e.g., illness, poor preparation, mechanical problems). Ignoring these patterns may hide important signals.

Physiological performance proxies not used

Performance-relevant markers such as Vertical Ascent in Meters (VAM), VO_2max [30, 66], or lactate threshold [68] were not included due to data unavailability. Integrating these physiological estimators would likely strengthen the analysis, but collecting them would require laboratory testing or specialized protocols, which makes their inclusion very difficult [69].

Overall, these limitations highlight that performance prediction in professional cycling is a multi-factorial problem, where training load alone can not fully explain race outcomes. This motivates the future research directions outlined in the following section, which aim to enrich the modelling framework.

6.2 Future Work

Building on the limitations above, several promising research directions emerge.

Expanding the dataset

Collecting more data across more seasons and more riders would make the modelling approach much stronger. A larger dataset would reduce the risk of overfitting, allow more stable estimation of model parameters, and enable a clearer separation between rider-specific effects and general performance patterns. From an applied perspective, this would enhance the model's robustness for long-term decision support.

Improved block definitions

Exploring alternative strategies for defining training blocks could lead to a more accurate representation of how athletes structure their preparation, for example:

- adaptive blocks based on changes in training load or monotony may better capture transitions between build-up, overload, and recovery phases;
- variable-length windows linked to tapering or peak-form cycles could also improve the model's sensitivity

to performance-critical periods, changing the parameter of the 7 consecutive non-race days in the blocks construction phase;

- clustering techniques to group similar training periods, increasing both predictive accuracy and interpretability.

Overall, more flexible and data-driven block definitions would allow the model to better align with real training practices, improving its ability to capture meaningful preparation phases and their relationship with race performance.

Incorporation of physiological and recovery metrics

Adding subjective and objective wellness information (sleep, Heart Rate Variability (HRV), soreness, stress) would allow the model to move beyond external load and better approximate the athlete's internal state. If the team adopts a consistent routine for filling out *TrainingPeaks' Metrics*, these features could significantly enhance predictive performance: in fact, these variables are particularly important for readiness assessment, as two riders with similar training loads may respond very differently depending on recovery and stress levels. Including such information would therefore enhance the model's ability to distinguish productive training adaptations from early signs of overreaching.

Incorporation of race profile and rider specialization

Future work could integrate structured descriptors of race courses, such as total elevation gain, climb categorization, stage type, or terrain classification. These features could then be combined with rider-specific characteristics to better model performance expectations. This integration would allow the model to distinguish between poor readiness and contextual mismatch, improving interpretability and fairness of predictions. From a practical perspective, such information would be highly valuable for coaching staff, supporting decisions related to rider selection, role assignment, and training focus for specific races.

Text mining of rider and coach comments

A natural extension is applying Natural Language Processing (NLP) techniques to *Title*, *CoachComments*, and *WorkoutDescription*. Even simple sentiment analysis or keyword extraction could capture qualitative information related to motivation, perceived effort, illness, or technical issues. This would complement quantitative metrics and provide a more holistic view of the athlete's condition, improving both explanatory power and practical relevance.

Time-series or sequence models

More advanced time-series or sequence-based models could explicitly leverage the temporal structure of the data. Unlike aggregated block-level features, these approaches can capture trends, momentum, and sudden changes in training or performance. Such models may be particularly useful for detecting early warning signs of declining form or insufficient recovery, offering actionable insights for coaches before poor race outcomes or non-finishes occur.

Integration of non-cycling sports

Future datasets could also include structured information about running, swimming, gym sessions, or cross-country skiing. As demonstrated by several elite athletes with multi-sport backgrounds, endurance adaptations can transfer across disciplines. Accounting for these contributions would improve load quantification and reduce the risk of underestimating physiological stress, leading to more accurate readiness modelling.

Inclusion of DNF, DNS, and OTL dynamics

Analyzing sequences of DNF, DNS, or OTL events could be highly informative. For instance, repeated non-finishes after a block of training may indicate overreaching or inadequate tapering. Detecting patterns preceding such outcomes may offer valuable insights for coaches. Incorporating these outcomes would allow the model to detect such patterns earlier and to distinguish between competitive under performance and more serious physiological or contextual problems.

Physiological *engine* estimators

Future work could explore integrating:

- VAM-based power proxies,
- VO₂max estimators from training data,
- ML-based lactate threshold estimation [69].

These indicators would help anchor the modelling framework to well-established physiological concepts, improving interpretability and facilitating communication of results to coaches and athletes.

Overall, this future work aims to progressively move the proposed framework from a purely data-driven approach toward a more context-aware and physiologically grounded model. By integrating richer data sources, temporal dynamics, and race-specific information, the system could better distinguish between training-related readiness and contextual performance constraints. Such developments would not only improve predictive accuracy, but also enhance the practical usefulness of the model for coaches and performance staff.

Chapter 7

Conclusions

The problem addressed in this thesis investigated how machine learning and training-load can be used to model and predict race performance in professional road cycling. Working with two seasons of data from five riders of a Norwegian UCI Continental team, the study combined daily *TrainingPeaks* exports, time-series workout files, race metadata, and scraped PCS information into a multi-year dataset.

Predicting race outcomes is difficult: results depend not only on physiological condition, but also on tactics, team roles, race dynamics, environmental factors, and random variability. The aim was therefore not to forecast exact results, but to quantify, in a probabilistic and interpretable way, how prepared an athlete is to perform well on a given day.

Two main modelling families were explored: a baseline Logistic Regression model using training-load descriptors and a more expressive XGBoost model. The final focus moved from a metrics-based configuration toward a readiness-based framework, in which the model predicts the probability that a rider will finish in the Top-20 in a given race. These probabilities are then mapped to a personalized Readiness Index (RI), scaled between 0 and 100 for each athlete. Finally, the outputs were integrated into an interactive dashboard designed to support coaches in pre-race planning.

The results show that the readiness-based model offers clear improvements over the baseline. While the linear model struggled to detect successful performances, the readiness approach captured the majority of Top-20 results and showed meaningful discriminative ability for several riders. Although performance remains constrained by data limitations - small sample size, class imbalance, and missing physiological variables - the probabilistic formulation provides an interpretable view of each rider's pre-race condition. It is important to note that the RI is not a deterministic prediction, but an estimate of performance potential: high readiness increases the likelihood of a strong result, without guaranteeing it. Furthermore, an estimated poor performance (e.g., a *Non-Top-20* result) may therefore not necessarily reflect inadequate preparation or low readiness, but rather a misalignment between the rider's characteristics and the demands of the specific race or stage.

Beyond the numerical results, the practical value of this work lies in translating complex training histories into something simple and usable for coaches and staff. The Readiness Index, together with the interactive dashboard, provides an easy way to keep track of high or low performance, and support decisions on leadership and race planning. It is not meant to replace expert judgment, but to offer a clear, data-driven perspective that complements day-to-day coaching experience.

Overall, this thesis demonstrates that readiness modelling in professional cycling is both feasible and useful. By integrating machine learning with sport-science principles, the proposed approach provides a structured way to link training patterns with expected race performance. Hopefully, this work can serve as a basis for future research and inspire further developments in performance prediction. The ideas and tools presented here may also be useful to others working in data-driven decision support, both in cycling and in other endurance sports.

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Appendix A

A.1 Column grouping

```
GROUP_COLUMNS = {
    # --- Core Identification ---
    'rider_name': ['rider'],

    # --- Temporal Data ---
    'time_data': ['year', 'WorkoutDay', 'time'],

    # --- Movement Metrics ---
    'distance_metrics': ['distance'],
    'speed_metrics': ['speed', 'enhanced_speed'],

    # --- Physiological Data ---
    'cardiac_metrics': ['heart_rate'],

    # --- Performance Metrics ---
    'power_metrics': ['power', 'accumulated_power'],
    'cadence_metrics': ['cadence', 'fractional_cadence'],
    'pedal_metrics': ['left_right_balance'],

    # --- Environmental/Positional ---
    'altitude_metrics': ['altitude', 'enhanced_altitude'],
}
```

A.2 Imputation strategy for each group

```
GROUP_STRATEGIES = {
    # --- Metrics imputed by linear interpolation ---
    'speed_metrics': {'method': 'linear'},
    'cardiac_metrics': {'method': 'linear'},
    'power_metrics': {'method': 'linear'},
    'cadence_metrics': {'method': 'linear'},
    'distance_metrics': {'method': 'linear'},

    # --- Metrics imputed by forward/backward filling ---
    'pedal_metrics': {'method': 'ffill'},
    'altitude_metrics': {'method': 'ffill'},

    # --- Columns constant within a file ---
    'rider_name': {'method': 'first'},
    'time_data': {'method': 'first'}
}
```

A.3 Complete feature overview

It is provided the full list of features of the feature engineering process in Section 3.5. There are three groups: core features shared across all models, metrics-based features (used in the metrics model family), and readiness-based features (used in the readiness model).

Category	Features	
Core Features (29)	rider	block_id
	race_day	result
	top20	block_days.length
	days_since_last_race	km_sum_block
	altitude_gain_sum_block	CAD_mean_block
	CAD_std_block	kJ_sum_block
	LuTRIMP_sum_block	TRIMP_sum_block
	load_monotony_time	HR_mean_block
	NP_mean_block	PWR_mean_block
	HR_mean_prev_blocks	NP_mean_prev_blocks
	PWR_mean_prev_blocks	top20_count_cum_before
	top20_rate_cum	mean_result_cum_before
	success_rate_last3	days_since_last_top20
	taper_distance_last7_over_block	taper_HR_high_last7_over_block
	ewm_top20_hl60d	
Metrics-only (15)	TSS_sum_block	tTSS_sum_block
	IF_mean_block	CTL_mean_block
	ATL_mean_block	TSB_mean_block
	TSB_last	ACWR_TSS_7_28
	monotony_TSS	strain_TSS
	tsb_mean_30d	atl_mean_30d
	tsb_mean_45d	atl_mean_45d
	tss_sum_45d	
Readiness-only (16)	readiness_before	readiness_mean_block
	readiness_std_block	readiness_trend
	readiness_mean_last7	readiness_ema5_last
	mean_readiness_last28	ACWR_readiness_7_28
	readiness_last7_over_rest	readiness_last7_minus_rest
	readiness_taper_last7_over_block	readiness_monotony
	readiness_ewm_hl14d	readiness_mean_prev_blocks
	mean_readiness_block_last3	delta_readiness_vs_prev_block

Table A.1: Summary of features used across models: core, metrics-specific, and readiness-specific.

A.4 The dataset

This table presents the full list of block-level feature columns used to construct the two datasets described in Section 3.6: data metrics blocks and data readiness blocks.

Dataset	Features		
Data metrics blocks	rider	block_id	race_day
	result	top20	block.days_length
	days_since_last_race	km_sum_block	altitude_gain_sum_block
	CAD_mean_block	CAD_std_block	kJ_sum_block
	LuTRIMP_sum_block	TRIMP_sum_block	load_monotony_time
	HR_mean_block	NP_mean_block	PWR_mean_block
	HR_mean_prev_blocks	NP_mean_prev_blocks	PWR_mean_prev_blocks
	top20_count_cum.before	top20_rate_cum	mean_result_cum.before
	success_rate_last3	days_since_last_top20	taper_distance_last7_over_block
	taper_HR_high_last7_over_block	TSS_sum_block	tTSS_sum_block
	IF_mean_block	CTL_mean_block	ATL_mean_block
	TSB_mean_block	TSB_last	ACWR_TSS_7_28
	monotony_TSS	strain_TSS	tsb_mean_30d
	atl_mean_30d	tsb_mean_45d	atl_mean_45d
	tss_sum_45d	ewm_top20_hl60d	
Data readiness blocks	rider	block_id	race_day
	result	top20	block.days_length
	days_since_last_race	km_sum_block	altitude_gain_sum_block
	CAD_mean_block	CAD_std_block	kJ_sum_block
	LuTRIMP_sum_block	TRIMP_sum_block	load_monotony_time
	HR_mean_block	NP_mean_block	PWR_mean_block
	HR_mean_prev_blocks	NP_mean_prev_blocks	PWR_mean_prev_blocks
	top20_count_cum.before	top20_rate_cum	mean_result_cum.before
	success_rate_last3	days_since_last_top20	taper_distance_last7_over_block
	taper_HR_high_last7_over_block	readiness.before	readiness_mean_block
	readiness_std_block	readiness_trend	readiness_mean_last7
	readiness_ema5_last	mean_readiness_last28	ACWR_readiness_7_28
	readiness_last7_over_rest	readiness_last7_minus_rest	readiness_taper_last7_over_block
	readiness_monotony	readiness_ewm_hl14d	readiness_mean_prev_blocks
	mean_readiness_block_last3	delta_readiness_vs_prev_block	ewm_top20_hl60d

Table A.2: Feature sets used for the metrics-based and readiness-based block datasets.

Appendix B

B.1 XGBoost Hyperparameter Search Space

The XGBoost models are tuned using a rider-specific randomized search over 24 configurations. Below, the hyperparameter search space used in the work is reported.

Listing B.1: Random search hyperparameter grid used for XGBoost tuning.

```
search_space = {
  "n_estimators": [100, 150, 200, 300, 400],
  "max_depth": [2, 3, 4, 5, 6],
  "learning_rate": [0.02, 0.05, 0.08, 0.10, 0.15, 0.20],
  "subsample": [0.6, 0.7, 0.8, 0.9, 1.0],
  "colsample_bytree": [0.6, 0.7, 0.8, 0.9, 1.0],
  "min_child_weight": [1, 3, 5, 7],
  "gamma": [0.0, 0.1, 0.3, 0.5],
  "reg_lambda": [0.0, 0.5, 1.0, 2.0, 5.0],
  "reg_alpha": [0.0, 0.1, 0.5, 1.0]
}
```

B.2 Best XGBoost-Metrics based Model Hyperparameters per Rider

All models for the metrics-based XGBoost configuration share the same training pipeline; hyperparameters are optimized separately for each rider via randomized search. Below, the selected hyperparameters are reported.

Listing B.2: Best hyperparameters for Athlete A

```
params = {
  "n_estimators": 200,
  "max_depth": 5,
  "learning_rate": 0.1000,
  "subsample": 0.6000,
  "colsample_bytree": 0.8000,
  "min_child_weight": 1,
  "gamma": 0.3000,
  "reg_lambda": 5.0000,
  "reg_alpha": 0.0000
}
```

Listing B.3: Best hyperparameters for Athlete B

```
params = {
  "n_estimators": 150,
  "max_depth": 3,
  "learning_rate": 0.1000,
  "subsample": 1.0000,
  "colsample_bytree": 0.6000,
  "min_child_weight": 1,
  "gamma": 0.3000,
  "reg_lambda": 0.5000,
  "reg_alpha": 0.0000
}
```

Listing B.4: Best hyperparameters for Athlete C

```

params = {
    "n_estimators": 200,
    "max_depth": 3,
    "learning_rate": 0.0800,
    "subsample": 0.6000,
    "colsample_bytree": 0.7000,
    "min_child_weight": 7,
    "gamma": 0.0000,
    "reg_lambda": 0.5000,
    "reg_alpha": 0.1000
}

```

Listing B.5: Best hyperparameters for Athlete D

```

params = {
    "n_estimators": 300,
    "max_depth": 5,
    "learning_rate": 0.2000,
    "subsample": 0.9000,
    "colsample_bytree": 0.8000,
    "min_child_weight": 1,
    "gamma": 0.5000,
    "reg_lambda": 1.0000,
    "reg_alpha": 0.1000
}

```

Listing B.6: Best hyperparameters for Athlete E

```

params = {
    "n_estimators": 200,
    "max_depth": 2,
    "learning_rate": 0.2000,
    "subsample": 0.7000,
    "colsample_bytree": 0.7000,
    "min_child_weight": 1,
    "gamma": 0.3000,
    "reg_lambda": 0.5000,
    "reg_alpha": 1.0000
}

```

B.3 Best XGBoost Readiness-based Model Hyperparameters per Rider

Below are the hyperparameter configurations selected for each athlete in the readiness-based XGBoost model. Hyperparameters are optimized via rider-specific randomized search, with the same grid of above in section B.1.

Listing B.7: Best hyperparameters for Athlete A (readiness-based)

```

params = {
    "n_estimators": 200,
    "max_depth": 5,
    "learning_rate": 0.1000,
    "subsample": 0.6000,
    "colsample_bytree": 0.8000,
    "min_child_weight": 1,
    "gamma": 0.3000,
    "reg_lambda": 5.0000,
    "reg_alpha": 0.0000
}

```

Listing B.8: Best hyperparameters for Athlete B (readiness-based)

```

params = {
    "n_estimators": 150,
    "max_depth": 3,
    "learning_rate": 0.1000,
    "subsample": 1.0000,
    "colsample_bytree": 0.6000,
    "min_child_weight": 1,
    "gamma": 0.3000,
    "reg_lambda": 0.5000,
    "reg_alpha": 0.0000
}

```

Listing B.9: Best hyperparameters for Athlete C (readiness-based)

```

params = {
    "n_estimators": 200,
    "max_depth": 3,
    "learning_rate": 0.0800,
    "subsample": 0.6000,
    "colsample_bytree": 0.7000,
    "min_child_weight": 7,
    "gamma": 0.0000,
    "reg_lambda": 0.5000,
    "reg_alpha": 0.1000
}

```

Listing B.10: Best hyperparameters for Athlete D (readiness-based)

```

params = {
    "n_estimators": 300,
    "max_depth": 5,
    "learning_rate": 0.2000,
    "subsample": 0.9000,
    "colsample_bytree": 0.8000,
    "min_child_weight": 1,
    "gamma": 0.5000,
    "reg_lambda": 1.0000,
    "reg_alpha": 0.1000
}

```

Listing B.11: Best hyperparameters for Athlete E (readiness-based)

```

params = {
    "n_estimators": 300,
    "max_depth": 4,
    "learning_rate": 0.2000,
    "subsample": 0.6000,
    "colsample_bytree": 0.8000,
    "min_child_weight": 5,
    "gamma": 0.3000,
    "reg_lambda": 5.0000,
    "reg_alpha": 1.0000
}

```

Appendix C

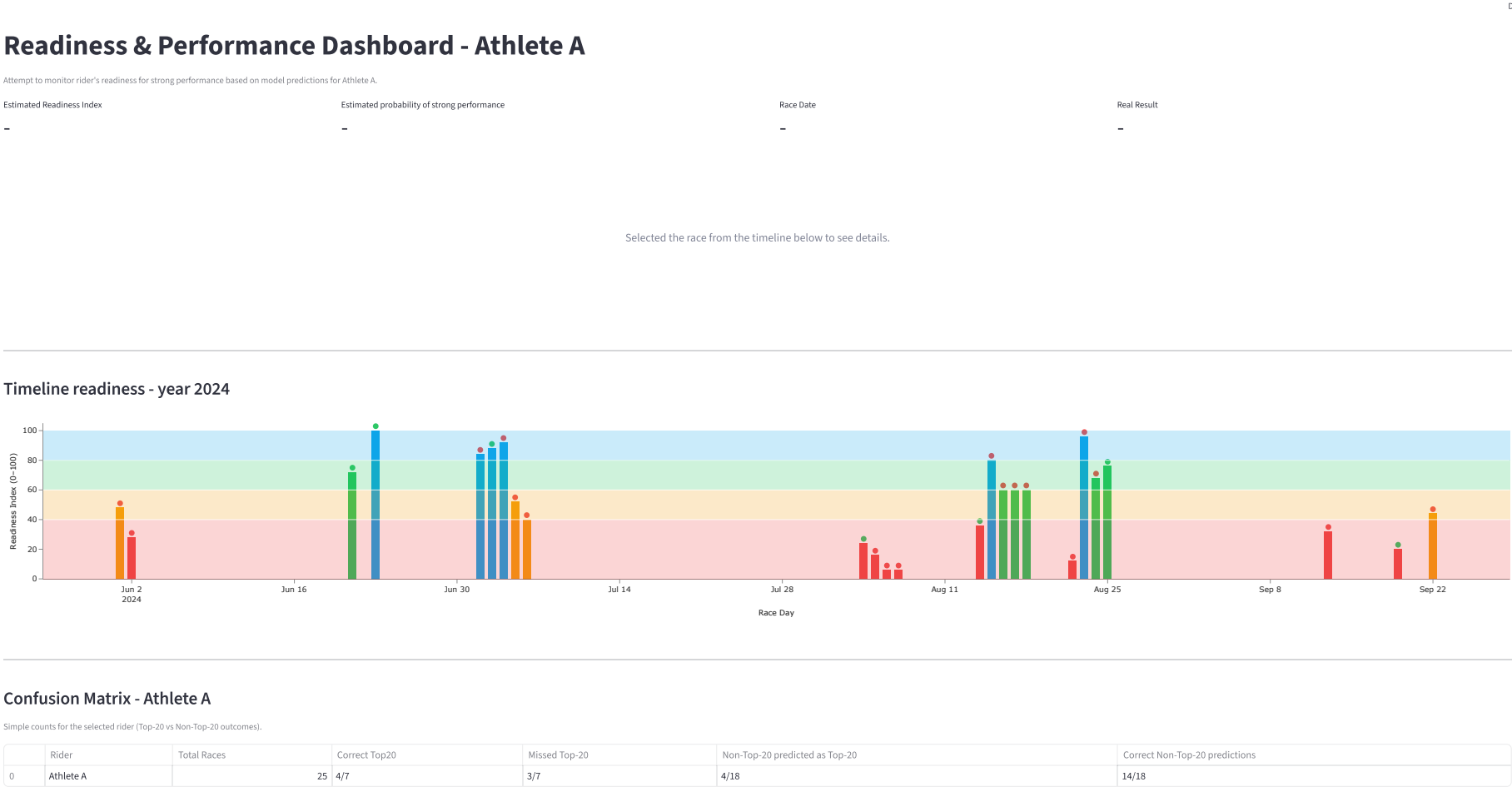


Figure C.1: Readiness & Performance Dashboard.